Optimization for Machine Learning

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Lecture 1: Introduction

An optimization problem asks for the minimum of a function f over a set $\mathcal{X}$:

$$\min_x f(x) \qquad \text{s.t. } x \in \mathcal{X}.$$

In machine learning f is usually a loss and $\mathcal{X}$ a constraint set. The course has three parts. Convex optimization, where f and $\mathcal{X}$ are convex, has a well-established theory. Nonconvex optimization is computationally hard in general but becomes feasible under specific structure. Minimax optimization,

$$\min_x \max_y f(x, y) \qquad \text{s.t. } (x, y) \in \mathcal{D},$$

is the two-player version of optimization; when f is convex in x and concave in y (the convex-concave setting) it again has a well-established theory.

1.1 Convexity

Definition 1.1 [Convex sets and functions]. A set $\mathcal{X} \subseteq \mathbb{R}^n$ is convex if $(1-\gamma)x + \gamma y \in \mathcal{X}$ for all $x, y \in \mathcal{X}$ and $\gamma \in [0, 1]$. A function $f : \mathcal{X} \to \mathbb{R}$ on a convex set is convex if for all $x, y \in \mathcal{X}$ and $\gamma \in [0, 1]$,

$$f\big((1-\gamma)x + \gamma y\big) \le (1-\gamma)f(x) + \gamma f(y).$$

So a set is convex if it contains every segment between two of its points, and a function is convex if every chord between $(x, f(x))$ and $(y, f(y))$ lies above its graph.

Proposition 1.2 [Examples of convex functions]. *Linear functions $f(x) = c^T x$ and the squared norm $f(x) = \|x\|^2 = x^T x$ on $\mathbb{R}^n$ are convex.*

Proposition 1.3 [Operations preserving convexity].

1. Nonnegative weighted sums. *If $f_1, \dots, f_m$ are convex and $\alpha_1, \dots, \alpha_m \ge 0$, then $\sum_{i=1}^m \alpha_i f_i$ is convex.*
2. Composition with an affine map. *If $f : \mathbb{R}^n \to \mathbb{R}$ is convex, $A \in \mathbb{R}^{n \times m}$ and $b \in \mathbb{R}^n$, then $g(x) = f(Ax + b)$ on $\mathbb{R}^m$ is convex.*
3. Pointwise supremum. *If $f_\theta : \mathcal{X} \to \mathbb{R}$ is convex for every $\theta \in \Theta$, then $f(x) = \sup_{\theta \in \Theta} f_\theta(x)$ is convex.*

Proof. All follow from the definition. A linear function satisfies it with equality, and $(1-\gamma)\|x\|^2 + \gamma\|y\|^2 - \|(1-\gamma)x + \gamma y\|^2 = \gamma(1-\gamma)\|x - y\|^2 \ge 0$. For sums, add the defining inequalities with weights α_i; for affine maps, apply the definition of f at $Ax + b$ and $Ay + b$.

For the supremum, $f_\theta((1-\gamma)x+\gamma y) \le (1-\gamma)f_\theta(x)+\gamma f_\theta(y) \le (1-\gamma)f(x)+\gamma f(y)$ for every θ; take the supremum on the left. □

1.2 Convex optimization

The problem $\min_{x\in\mathcal{X}} f(x)$ is a CONVEX OPTIMIZATION PROBLEM if both f and $\mathcal{X}$ are convex.

Example: *Linear regression.* Data $\{(x_i, y_i)\}_{i=1}^n$ come from the model $y_i = x_i^T w^\star + \epsilon_i$ with $\epsilon_i \sim N(0,1)$. Ordinary least squares,

$$\min_{w\in\mathbb{R}^d} \sum_{i=1}^n (y_i - x_i^T w)^2,$$

is convex: the objective is $\|\cdot\|^2$ composed with an affine map, and the constraint set $\mathbb{R}^d$ is convex.

Example: *Sparse linear regression.* If moreover $\|w^\star\|_0$ is known to be small, one would penalize $\|w\|_0$, which is not convex. Its convex relaxation $\|w\|_1$ gives the LASSO,

$$\min_{w\in\mathbb{R}^d} \sum_{i=1}^n (y_i - x_i^T w)^2 + \lambda\|w\|_1, \qquad \lambda \ge 0,$$

still convex as a sum of convex functions.

Convexity is special mainly because of the following fact.

Theorem 1.4 [LOCAL MINIMA ARE GLOBAL]. *If $f : \mathcal{X} \to \mathbb{R}$ is convex and x is a local minimum of f, then x is a global minimum.*

Proof. Fix $y \in \mathcal{X}$. For small enough $\gamma \in (0,1]$, local minimality and then convexity give $f(x) \le f((1-\gamma)x+\gamma y) \le (1-\gamma)f(x)+\gamma f(y)$, so $f(x) \le f(y)$. □

1.3 Nonconvex optimization

Most problems met in machine learning are nonconvex, and nonconvex optimization is NP-hard in general. Structure that is often present in practice can make it feasible.

Example: *Two-layer neural network.* With $W_2 \in \mathbb{R}^{d\times h}$, $w_1 \in \mathbb{R}^h$, and an activation $\sigma : \mathbb{R}^h \to \mathbb{R}^h$, the data follow $y_i = w_1^T \sigma(W_2^T x_i) + \epsilon_i$ with Gaussian noise: input $x_i \in \mathbb{R}^d$, a hidden layer of width h, and a scalar output. The square loss

$$\min_{w_1\in\mathbb{R}^h,\, W_2\in\mathbb{R}^{d\times h}} \sum_{i=1}^n \big(y_i - w_1^T \sigma(W_2^T x_i)\big)^2$$

is not convex, because of the nonlinear activation and because w_1 and W_2 multiply each other.

Example: *Matrix factorization.* Observations $M_{ij} = M^\star_{ij} + \epsilon_{ij}$, $(i,j) \in [d_1] \times [d_2]$, are noisy entries of a matrix with $\mathrm{rank}(M^\star) = r \ll \min\{d_1, d_2\}$; the observed matrix need not be low rank. Writing a rank-r matrix as UV^T with $U \in \mathbb{R}^{d_1 \times r}$, $V \in \mathbb{R}^{d_2 \times r}$, the square loss gives

$$\min_{U \in \mathbb{R}^{d_1 \times r},\, V \in \mathbb{R}^{d_2 \times r}} \|M - UV^T\|_F^2, \qquad \|A\|_F^2 = \sum_{ij} A_{ij}^2.$$

This is nonconvex, but every local minimum is a global minimum, so the problem reduces to escaping saddle points (revisited later in the course).

1.4 Minimax optimization

Example: *Adversarial training.* Data follow $y_i = f_{w^\star}(x_i) + \epsilon_i$, where f_w is parameterized by w (for example the weights of a network). A robust predictor should give the same output on x_i and on a slightly perturbed $x_i + \zeta_i$. This is the minimax problem

$$\min_{w \in \mathbb{R}^m} \max_{\zeta_1, \ldots, \zeta_n \in B_\infty} \sum_{i=1}^n \big(y_i - f_w(x_i + \zeta_i)\big)^2, \qquad B_\infty = \{\zeta \in \mathbb{R}^d : \|\zeta\|_\infty \le c\},$$

in which the max-player adds noise to break the predictor and the min-player seeks good performance despite it.

Example: *Generative adversarial networks.* A generator network tries to produce images that resemble natural images, and a discriminator network tries to decide whether an image is natural. As they compete, each improves; the task is a minimax problem between the two networks.

1.5 Oracle complexity and the black-box model

Statements about an algorithm A come in three kinds:

1. A converges;
2. A converges to an optimal point $x^\star$, or finds it;
3. a convergence rate: A finds an ϵ-approximation of $x^\star$ in, say, $O(1/\epsilon^2)$ steps, or its error decreases at rate, say, $O(1/\sqrt{t})$.

Only the third is a strong statement, so optimization theory focuses on rates. The first does not even guarantee correctness (an algorithm that always returns 0 converges), and correctness alone is easy to achieve, for example by grid search (Lecture 2), even for generic nonconvex functions.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 1 (Spring 2021), scribed by Seyed Sobhan Mir Yoosefi.

Lecture 2: Black-Box Model, Basic Properties of Convexity

Throughout the course the objective satisfies some regularity condition; the first is Lipschitz continuity.

Definition 2.1 [Lipschitz function]. $f : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz if $|f(x) - f(y)| \le L\|x - y\|$ for all x, y. Unless another norm is specified, $\|\cdot\|$ is the Euclidean (ℓ_2) norm.

2.1 Grid search

To minimize an L-Lipschitz f over $\mathcal{X} = [0,1]^d$, fix $\epsilon > 0$, let $G = \{0, \epsilon, 2\epsilon, \dots, \lfloor 1/\epsilon \rfloor \epsilon\}$, and evaluate f on all $O((1/\epsilon)^d)$ points of the ϵ-grid $\mathcal{X}_0 = G^d$.

Proposition 2.2 [Grid search]. $\min_{x \in \mathcal{X}_0} f(x) \le \min_{x \in \mathcal{X}} f(x) + \epsilon L \sqrt{d}$.

Proof. f is continuous on the compact set $\mathcal{X}$, so it has a minimizer $x^\star$. Rounding each coordinate of $x^\star$ down to G gives $x_0 \in \mathcal{X}_0$ with $\|x^\star - x_0\|_\infty < \epsilon$, so $\|x^\star - x_0\|_2 < \epsilon\sqrt{d}$ and $\min_{\mathcal{X}_0} f \le f(x_0) \le f(x^\star) + L\|x^\star - x_0\| \le f(x^\star) + \epsilon L\sqrt{d}$. □

If the global minimizer $x^\star$ is unique, letting $\epsilon \to 0$ makes the grid minimizer converge to $x^\star$, so grid search is a convergent algorithm. But it needs $O((1/\epsilon)^d)$ evaluations for an $\epsilon L\sqrt{d}$-optimal point, which is prohibitive in high dimension: a bad algorithm suffering the curse of dimensionality.

2.2 Measuring speed: oracle complexity

There are two rigorous ways to measure speed.

1. *Computational complexity*, counting Turing machine operations. Reductions to problems of known complexity are available, but fine-grained complexity is still young, and it is hard to pin down the optimal polynomial degree of a running time, which is exactly what optimization cares about.
2. *Oracle (query) complexity* in a black-box model, for $\min_{x \in \mathcal{X}} f(x)$.

In the black-box model a learner interacts with an oracle. The learner has unlimited computational power, knows the function class of f (convex, Lipschitz, smooth, and so on), and knows the domain $\mathcal{X}$, so it can compute projections onto $\mathcal{X}$ and separating hyperplanes. Only the oracle knows f. To find an ϵ-approximate minimizer, the learner queries the oracle at points $x \in \mathcal{X}$; we count the queries. An oracle is of nth order if it returns all partial derivatives of f up to order n at the query

point. A zeroth-order oracle returns $f(x)$; a first-order oracle also returns a gradient or subgradient. The course mostly uses first-order oracles.

The key question is how many queries suffice to find an ϵ-approximate minimum. Algorithms give upper bounds on the query complexity, and information-theoretic arguments give lower bounds; if they match for a function class, the algorithm is MINIMAX OPTIMAL.

The approach proves sharp upper and lower bounds on the number of queries, in a rigorous theory that develops independently of complexity theory. On the other hand, unlimited computation can permit algorithms with low query complexity that are still NP-hard to run (none appear in this course), and cheap side information about f, not tied to its derivatives, might permit faster algorithms.

2.3 Separating and supporting hyperplanes

Theorem 2.3 [HYPERPLANE SEPARATION]. *If $\mathcal{X}, \mathcal{Y} \subseteq \mathbb{R}^d$ are nonempty, disjoint, and convex, there are a nonzero $w \in \mathbb{R}^d$ and $c \in \mathbb{R}$ with $w^T x \geq c$ for all $x \in \mathcal{X}$ and $w^T y \leq c$ for all $y \in \mathcal{Y}$.*

Lemma 2.4 [MINIMUM-NORM POINT]. *A nonempty closed convex set $K \subseteq \mathbb{R}^d$ has a unique point of minimum Euclidean norm.*

Proof. A minimizer exists because K is closed (minimize over the compact set $K \cap \{\|v\| \leq \|v_0\|\}$ for any $v_0 \in K$). If $v_1 \neq v_2$ were both minimizers with norm m, their midpoint would lie in K with $\|\frac{v_1+v_2}{2}\|^2 = \frac{1}{2}\|v_1\|^2 + \frac{1}{2}\|v_2\|^2 - \frac{1}{4}\|v_1 - v_2\|^2 < m^2$, a contradiction. □

Proof of Theorem 2.3 when $\operatorname{dist}(\mathcal{X}, \mathcal{Y}) > 0$. The difference $K = \mathcal{X} - \mathcal{Y} = \{x - y : x \in \mathcal{X}, y \in \mathcal{Y}\}$ is convex, and so is its closure $\bar{K}$; since $\operatorname{dist}(\mathcal{X}, \mathcal{Y}) > 0$, $0 \notin \bar{K}$. Let $v \neq 0$ be the minimum-norm point of $\bar{K}$. For $u \in \bar{K}$ and $\gamma \in [0, 1]$, $v + \gamma(u - v) \in \bar{K}$, so

$$\|v\|^2 \leq \|v + \gamma(u - v)\|^2 = \|v\|^2 + 2\gamma v^T(u - v) + \gamma^2\|u - v\|^2.$$

Letting $\gamma \to 0$ gives $v^T(u - v) \geq 0$, that is, $v^T x \geq v^T y + \|v\|^2$ for all $x \in \mathcal{X}$, $y \in \mathcal{Y}$. Then $w = v$ and any $c \in [\sup_{y \in \mathcal{Y}} v^T y,\ \inf_{x \in \mathcal{X}} v^T x]$ work. □

Corollary 2.5 [SUPPORTING HYPERPLANE]. *If $\mathcal{X} \subseteq \mathbb{R}^d$ is convex and $x_0 \in \partial\mathcal{X}$, there is a nonzero $w \in \mathbb{R}^d$ with $w^T x \geq w^T x_0$ for all $x \in \mathcal{X}$.*

Proof. If $\operatorname{int}(\mathcal{X}) \neq \emptyset$, apply Theorem 2.3 to the convex set $\operatorname{int}(\mathcal{X})$ and $\{x_0\}$ (the inequality extends to $\mathcal{X}$, which lies in the closure of its interior). If $\operatorname{int}(\mathcal{X}) = \emptyset$, then $\mathcal{X}$ lies in a $(d-1)$-dimensional hyperplane, which itself supports $\mathcal{X}$ at x_0. □

2.4 Subgradients

Convex functions need not be differentiable, so we use subgradients.

Definition 2.6 [Subgradient]. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex and $f : \mathcal{X} \to \mathbb{R}$. A vector $g_x \in \mathbb{R}^d$ is a subgradient of f at $x \in \mathcal{X}$ if

$$f(y) \geq f(x) + g_x^T(y - x) \qquad \text{for all } y \in \mathcal{X}.$$

The subdifferential $\partial f(x)$ is the set of all subgradients at x.

Definition 2.7 [Epigraph]. The epigraph of $f : \mathcal{X} \to \mathbb{R}$ is $\text{epi}(f) = \{(x, t) \in \mathcal{X} \times \mathbb{R} : t \geq f(x)\}$.

Lemma 2.8. *f is convex if and only if* $\text{epi}(f)$ *is a convex set.*

Proof. If f is convex and $(x_1, t_1), (x_2, t_2) \in \text{epi}(f)$, then $f(\alpha x_1 + (1-\alpha)x_2) \leq \alpha f(x_1) + (1-\alpha) f(x_2) \leq \alpha t_1 + (1-\alpha)t_2$. Conversely, convexity of $\text{epi}(f)$ puts $\alpha(x_1, f(x_1)) + (1-\alpha)(x_2, f(x_2))$ in $\text{epi}(f)$, which is the convexity inequality. □

Proposition 2.9 [Existence of subgradients]. *Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex and $f : \mathcal{X} \to \mathbb{R}$.*

1. *If $\partial f(x) \neq \emptyset$ for all $x \in \mathcal{X}$, then f is convex.*
2. *If f is convex, then $\partial f(x) \neq \emptyset$ for all $x \in \text{int}(\mathcal{X})$.*
3. *If f is convex and differentiable at x, then $\nabla f(x) \in \partial f(x)$.*

Proof. (1) Let $z = (1-\gamma)x + \gamma y$ and $g \in \partial f(z)$. Then $f(x) \geq f(z) + \gamma g^T(x - y)$ and $f(y) \geq f(z) + (1-\gamma) g^T(y - x)$; adding $(1-\gamma)$ times the first to γ times the second gives $(1-\gamma)f(x) + \gamma f(y) \geq f(z)$.
(2) The point $(x, f(x))$ lies on the boundary of the convex set $\text{epi}(f)$, so by Corollary 2.5 there is $(a, b) \neq 0$ with $a^T y + bt \geq a^T x + b f(x)$ for all $(y, t) \in \text{epi}(f)$. Since $x \in \text{int}(\mathcal{X})$, $y = x - \epsilon a \in \mathcal{X}$ for small $\epsilon > 0$, which gives $b(t - f(x)) \geq \epsilon \|a\|^2$ for all $t \geq f(y)$. Taking t large shows $b \geq 0$, and $b = 0$ would force $a = 0$; hence $b > 0$. Taking $t = f(y)$ for arbitrary $y \in \mathcal{X}$ gives $f(y) \geq f(x) + (-a/b)^T(y - x)$, so $-a/b \in \partial f(x)$.
(3) Convexity gives $f(y) \geq f(x) + \big(f(x + \gamma(y - x)) - f(x)\big)/\gamma$ for $\gamma \in (0, 1)$; letting $\gamma \to 0$ yields $f(y) \geq f(x) + \nabla f(x)^T(y - x)$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 2 (Spring 2021), scribed by Hristo Papazov.

Lecture 3: The Ellipsoid Method

The ellipsoid method is the analogue of binary search in $\mathbb{R}^d$. Binary search keeps a segment containing the optimum, splits it in half, and keeps the half containing the optimum, halving its length. The ellipsoid method keeps an ellipsoid containing the optimum, cuts it by a hyperplane through its center, and keeps the half containing the optimum. To recurse, the half ellipsoid is replaced by a circumscribed ellipsoid whose volume is smaller than the original by a constant factor (Figure 1).

3.1 The geometric lemma

Lemma 3.1 [Half-ellipsoid]. *Let $\mathcal{E}_0 = \{x \in \mathbb{R}^d : (x - c_0)^T H_0^{-1}(x - c_0) \le 1\}$ with H_0 symmetric positive definite, $d \ge 2$, and $w \in \mathbb{R}^d$ nonzero. The ellipsoid $\mathcal{E} = \{x : (x - c)^T H^{-1}(x - c) \le 1\}$ with*

$$c = c_0 - \frac{1}{d+1} \cdot \frac{H_0 w}{\sqrt{w^T H_0 w}}, \qquad H = \frac{d^2}{d^2-1}\Big(H_0 - \frac{2}{d+1} \cdot \frac{H_0 w w^T H_0}{w^T H_0 w}\Big) \tag{3.1}$$

satisfies $\mathcal{E}_0 \cap \{x : (x - c_0)^T w \le 0\} \subseteq \mathcal{E}$ and $\operatorname{vol}(\mathcal{E}) < \operatorname{vol}(\mathcal{E}_0) \exp\big(-\frac{1}{2d}\big)$.

For $d = 1$ the lemma is halving a segment. For $d \ge 2$ one can verify (3.1) directly; the following symmetry argument derives it.

Proof. Unit ball. Let $c_0 = 0$, $H_0 = I_d$, and $\hat{w} = w/\|w\|$. Only $\hat{w}$ is a special direction, so take $c = -t\hat{w}$ with $t \in [0, 1]$ and $H^{-1} = a\hat{w}\hat{w}^T + b(I_d - \hat{w}\hat{w}^T)$, spherical orthogonal to $\hat{w}$. Require that $\mathcal{E}$ pass through $-\hat{w}$ and through the unit vectors $y \perp \hat{w}$, and have minimum volume. The two contact conditions give

$$(t-1)^2 a = 1, \qquad b + t^2 a = 1.$$

With $h = 1/(1-t)$ this is $a = h^2$ and $b = 1 - (h-1)^2 = 2h - h^2$, so

$$\frac{\operatorname{vol}(\mathcal{E})}{\operatorname{vol}(B(0,1))} = \frac{1}{\sqrt{a}}\Big(\frac{1}{\sqrt{b}}\Big)^{d-1} = \frac{1}{\sqrt{\varphi(h)}}, \qquad \varphi(h) = h^2(2h - h^2)^{d-1}.$$

Setting $(\ln \varphi)' = 2/h - 2(d-1)(h-1)/(h(2-h)) = 0$ gives $h = 1 + \frac{1}{d}$, that is, $t = \frac{1}{d+1}$, $a = (1 + \frac{1}{d})^2$, and $b = 1 - \frac{1}{d^2}$:

$$c = -\frac{1}{d+1} \cdot \frac{w}{\|w\|}, \qquad H^{-1} = \Big(1 - \frac{1}{d^2}\Big) I_d + \frac{2(d+1)}{d^2} \cdot \frac{w w^T}{\|w\|^2}.$$

This $\mathcal{E}$ contains the half ball: a point $x = -s\hat{w} + y$ with $s \in [0,1]$, $y \perp \hat{w}$, $\|y\|^2 \le 1 - s^2$ has $(x-c)^T H^{-1}(x-c) \le q(s) = a(s-t)^2 + b(1-s^2)$, which is convex in s (as $a > b$) with $q(0) = q(1) = 1$. For the volume, with $u = 1/d$,

$$\ln \varphi(1+u) = (d+1)\ln(1+u) + (d-1)\ln(1-u) = d\big[(1+u)\ln(1+u) + (1-u)\ln(1-u)\big] > du^2 = \frac{1}{d},$$

because $\psi(u) = (1+u)\ln(1+u) + (1-u)\ln(1-u) - u^2$ has $\psi(0) = \psi'(0) = 0$ and $\psi''(u) = \frac{2}{1-u^2} - 2 > 0$ on $(0,1)$. Hence $\mathrm{vol}(\mathcal{E})/\mathrm{vol}(B(0,1)) < \exp(-\frac{1}{2d})$.

General ellipsoid. The affine map $\Phi(z) = c_0 + H_0^{1/2} z$ sends $B(0,1)$ onto $\mathcal{E}_0$ and preserves inclusions and volume ratios. Since $(\Phi(z) - c_0)^T w = z^T (H_0^{1/2} w)$, the inclusion $\mathcal{E}_0 \cap \{(x - c_0)^T w \le 0\} \subseteq \mathcal{E}$ is equivalent to $B(0,1) \cap \{z^T H_0^{1/2} w \le 0\} \subseteq \Phi^{-1}(\mathcal{E})$. So $\mathcal{E}$ is the image under Φ of the unit-ball ellipsoid for the direction $H_0^{1/2} w$; inverting H^{-1} above and mapping back gives (3.1). □

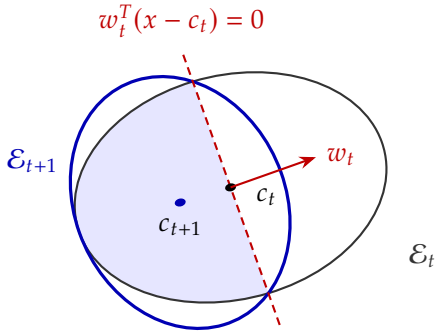


Figure 1: One step of the ellipsoid method. The hyperplane through the center c_t with normal w_t discards the half where $w_t^T(x - c_t) > 0$. The new ellipsoid $\mathcal{E}_{t+1}$, centered at c_{t+1}, contains the kept half of $\mathcal{E}_t$ (shaded).

3.2 The algorithm and its guarantee

The method uses a separation oracle for $\mathcal{X}$ and zeroth- and first-order oracles for the convex f. It starts from a ball $\mathcal{E}_0 = B(0,R)$ containing $\mathcal{X}$.

Algorithm 3.1: Ellipsoid method

Input: $\mathcal{E}_0 = B(0,R) \supseteq \mathcal{X}$ with center $c_0 = 0$, number of steps T

for $t = 0, 1, \ldots, T-1$ **do**
  if $c_t \notin \mathcal{X}$ **then**
    call the separation oracle for w_t with $\mathcal{X} \subseteq \{x : (x - c_t)^T w_t \le 0\}$
  else
    call the first-order oracle for $w_t \in \partial f(c_t)$
  let $\mathcal{E}_{t+1}$, with center c_{t+1}, be the ellipsoid of Lemma 3.1 for $\mathcal{E}_t$ and w_t
return $x_T = \arg\min_{c \in \{c_0, \ldots, c_{T-1}\} \cap \mathcal{X}} f(c)$

Theorem 3.2 [Ellipsoid method]. *Suppose $B(\tilde{x}, r) \subseteq \mathcal{X} \subseteq B(0,R)$ for some $\tilde{x}$ and $r > 0$, and $|f(x)| \le B$*

on $\mathcal{X}$. If $T > 2d^2 \ln \frac{R}{r}$, then $\{c_0, \dots, c_{T-1}\} \cap \mathcal{X} \neq \emptyset$ and

$$f(x_T) - \min_{x \in \mathcal{X}} f(x) \leq \frac{2BR}{r} \exp\Big(-\frac{T}{2d^2}\Big).$$

So $O\big(d^2 \ln \frac{BR}{r\epsilon}\big)$ iterations give an ϵ-optimal point.

Proof. By Lemma 3.1, $\mathrm{vol}(\mathcal{E}_T) < \mathrm{vol}(\mathcal{E}_0) e^{-T/(2d)}$.
Feasibility. If no center were in $\mathcal{X}$, every cut would come from the separation oracle and remove no point of $\mathcal{X}$, so $\mathcal{X} \subseteq \mathcal{E}_T$. But $\mathrm{vol}(\mathcal{E}_T) < \mathrm{vol}(\mathcal{E}_0) e^{-T/(2d)} < \mathrm{vol}(\mathcal{E}_0)(r/R)^d \leq \mathrm{vol}(\mathcal{X})$, a contradiction.
Only worse points are removed. If $c_t \in \mathcal{X}$, the removed part satisfies $(\mathcal{E}_t \setminus \mathcal{E}_{t+1}) \cap \mathcal{X} \subseteq \{x : w_t^T (x - c_t) > 0\}$, and on that set $f(x) \geq f(c_t) + w_t^T (x - c_t) > f(c_t)$ by convexity.
Accuracy. Let $x^\star \in \arg\min_{\mathcal{X}} f$ and $\mathcal{X}_\epsilon = \{(1-\epsilon)x^\star + \epsilon x : x \in \mathcal{X}\} \subseteq \mathcal{X}$. By convexity every $y \in \mathcal{X}_\epsilon$ has $f(y) \leq (1-\epsilon) f(x^\star) + \epsilon f(x) \leq f(x^\star) + 2\epsilon B$. Choose $\epsilon = \frac{R}{r} e^{-T/(2d^2)}$, which is less than 1 by the assumption on T. Then

$$\mathrm{vol}(\mathcal{E}_T) < \mathrm{vol}(\mathcal{E}_0) e^{-T/(2d)} \leq \mathrm{vol}(\mathcal{X}) \Big(\frac{R}{r}\Big)^d e^{-T/(2d)} = \mathrm{vol}(\mathcal{X}_\epsilon)\, \epsilon^{-d} \Big(\frac{R}{r}\Big)^d e^{-T/(2d)} = \mathrm{vol}(\mathcal{X}_\epsilon),$$

so some $y \in \mathcal{X}_\epsilon$ was removed at a step t with $c_t \in \mathcal{X}$. Hence $f(x_T) \leq f(c_t) < f(y) \leq f(x^\star) + 2\epsilon B = f(x^\star) + \frac{2BR}{r} e^{-T/(2d^2)}$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 3 (Spring 2021), scribed by Yuanhao Wang.

Lecture 4: Gradient Descent for Lipschitz Functions

4.1 Gradient descent

Gradient descent (GD) with step sizes (learning rates) $\{\eta_s\}_{s=1}^t$ iterates

$$x_{s+1} = x_s - \eta_s \nabla f(x_s), \qquad s = 1, \dots, t.$$

Each step minimizes the linearization of f at x_s plus a distance penalty weighted by $1/\eta_s$:

$$x_{s+1} = \arg\min_{x \in \mathbb{R}^d} \langle \nabla f(x_s), x - x_s \rangle + \frac{1}{2\eta_s} \|x - x_s\|^2,$$

since this objective is convex and its gradient $\nabla f(x_s) + (x - x_s)/\eta_s$ vanishes exactly at $x = x_s - \eta_s \nabla f(x_s)$.

The main advantage over the ellipsoid method, which needs $O(d^2 \log \frac{1}{\epsilon})$ iterations, is that the guarantees for GD do not depend on the dimension. With millions or billions of parameters, as in neural networks, a quadratic dependence on d is catastrophic.

4.2 Projected gradient descent

For a constrained problem, assume a projection oracle $\Pi_{\mathcal{X}}(y) = \arg\min_{x \in \mathcal{X}} \|x - y\|$.

Algorithm 4.1: Projected gradient descent

```
for s = 1, ..., t do
    y_{s+1} = x_s − η_s g_s, where g_s ∈ ∂f(x_s)
    x_{s+1} = Π_X(y_{s+1})
```

Lemma 4.1 [Projection]. *For $x \in \mathcal{X}$ and $y \in \mathbb{R}^d$, $\big(\Pi_{\mathcal{X}}(y) - x\big)^T \big(\Pi_{\mathcal{X}}(y) - y\big) \le 0$.*

Proof. Let $z = \Pi_{\mathcal{X}}(y)$. For $\lambda \in [0, 1]$ the point $(1 - \lambda)z + \lambda x$ lies in $\mathcal{X}$ by convexity, so

$$\|y - z\|^2 \le \|y - (1 - \lambda)z - \lambda x\|^2 = \|y - z\|^2 + 2\lambda (y - z)^T (z - x) + \lambda^2 \|z - x\|^2.$$

The right side minus the left is nonnegative on $[0, 1]$ and zero at $\lambda = 0$, so its derivative at 0 is nonnegative: $(z - x)^T (z - y) \le 0$. □

Corollary 4.2. *For $x \in \mathcal{X}$ and $y \in \mathbb{R}^d$, $\|\Pi_{\mathcal{X}}(y) - x\| \le \|y - x\|$.*

Proof. With $z = \Pi_{\mathcal{X}}(y)$, $\|y - x\|^2 = \|y - z\|^2 + 2\langle y - z, z - x\rangle + \|z - x\|^2 \geq \|z - x\|^2$, since the middle term is nonnegative by Lemma 4.1. □

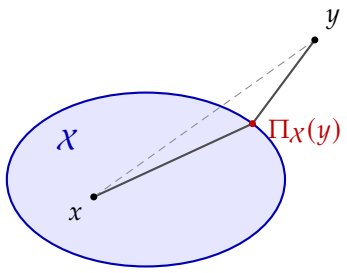


Figure 2: Lemma 4.1 and Corollary 4.2: the angle at $\Pi_{\mathcal{X}}(y)$ between y and any $x \in \mathcal{X}$ is at least 90°, so the projection is closer to x than y is.

4.3 Lipschitz convex functions

Definition 4.3 [Lipschitz on a set]. f is L-Lipschitz on $\mathcal{X}$ if $|f(x) - f(y)| \leq L\|x - y\|$ for all $x, y \in \mathcal{X}$.

Lemma 4.4. *If $\|g\| \leq L$ for all $x \in \mathcal{X}$ and $g \in \partial f(x)$, then f is L-Lipschitz. Conversely, if f is L-Lipschitz, then $\|g\| \leq L$ for all $x \in \operatorname{int}(\mathcal{X})$ and $g \in \partial f(x)$.*

Proof. For the first claim, the subgradient inequality at y and Cauchy–Schwarz give $f(y) - f(x) \leq g_y^T(y - x) \leq \|g_y\|\|y - x\| \leq L\|x - y\|$; swap x and y. Conversely, for $x \in \operatorname{int}(\mathcal{X})$, any v, and small $\lambda > 0$, $\lambda g_x^T v \leq f(x + \lambda v) - f(x) \leq L\lambda\|v\|$; take $v = g_x$. □

Theorem 4.5 [Lipschitz convex]. *Let f be convex and L-Lipschitz on $\mathcal{X}$ ($\|g_x\| \leq L$), with $\|x_1 - x^\star\| \leq R$. Projected GD with constant step $\eta_s = \eta = \frac{R}{L\sqrt{t}}$ satisfies*

$$f\Big(\frac{1}{t}\sum_{s=1}^{t} x_s\Big) - f(x^\star) \leq \frac{RL}{\sqrt{t}}.$$

Proof. By convexity, the update, and the identity $2a^Tb = \|a\|^2 + \|b\|^2 - \|a - b\|^2$,

$$f(x_s) - f(x^\star) \leq g_s^T(x_s - x^\star) = \frac{1}{\eta}(x_s - y_{s+1})^T(x_s - x^\star) = \frac{1}{2\eta}\big(\|x_s - x^\star\|^2 - \|y_{s+1} - x^\star\|^2\big) + \frac{\eta}{2}\|g_s\|^2.$$

By Corollary 4.2, $\|y_{s+1} - x^\star\| \geq \|x_{s+1} - x^\star\|$, and $\|g_s\| \leq L$, so

$$f(x_s) - f(x^\star) \leq \frac{1}{2\eta}\big(\|x_s - x^\star\|^2 - \|x_{s+1} - x^\star\|^2\big) + \frac{\eta L^2}{2}.$$

Summing telescopes:

$$\sum_{s=1}^{t}\big(f(x_s)-f(x^\star)\big)\le \frac{R^2}{2\eta}+\frac{\eta L^2 t}{2}=RL\sqrt{t}$$

for the stated η, which minimizes the bound. Divide by t and apply Jensen's inequality. □

Accuracy ϵ thus needs $t = R^2L^2/\epsilon^2 = O(1/\epsilon^2)$ iterations: much worse in ϵ than the ellipsoid method's $O(d^2 \log \frac{1}{\epsilon})$, but independent of d. The projection can be expensive (for example onto the feasible set of a linear program), and then the oracle model may not reflect the running time. The varying step $\eta_s = \frac{R}{L\sqrt{s}}$ avoids fixing t, at the cost of logarithmic factors.

4.4 Lipschitz strongly convex functions

Definition 4.6 [Strong convexity]. f is α-strongly convex if for all $x, y \in \mathcal{X}$ and $g_x \in \partial f(x)$,

$$f(y)\ge f(x)+g_x^T(y-x)+\frac{\alpha}{2}\|y-x\|^2.$$

Equivalently, $x \mapsto f(x) - \frac{\alpha}{2}\|x\|^2$ is convex; and if f is twice differentiable, f is α-strongly convex if and only if $\lambda_{\min}(\nabla^2 f(x)) \ge \alpha$ for all $x \in \mathcal{X}$.

Theorem 4.7 [Lipschitz strongly convex]. *If f is L-Lipschitz and α-strongly convex, projected GD with $\eta_s = \frac{2}{\alpha(s+1)}$ satisfies*

$$f\Big(\frac{2}{t(t+1)}\sum_{s=1}^{t} s\,x_s\Big)-f(x^\star)\le\frac{2L^2}{\alpha(t+1)}.$$

So $\frac{2L^2}{\alpha\epsilon}-1 = O(1/\epsilon)$ iterations suffice, still dimension-free and better than in the convex case.

Proof. Strong convexity at x_s toward $x^\star$ adds the term $-\frac{\alpha}{2}\|x_s-x^\star\|^2$ to the previous argument:

$$f(x_s)-f(x^\star)\le\frac{\eta_s L^2}{2}+\frac{1}{2\eta_s}\big(\|x_s-x^\star\|^2-\|x_{s+1}-x^\star\|^2\big)-\frac{\alpha}{2}\|x_s-x^\star\|^2.$$

Substituting $\eta_s = \frac{2}{\alpha(s+1)}$, so that $\frac{1}{2\eta_s}-\frac{\alpha}{2}=\frac{\alpha(s-1)}{4}$, and multiplying by s,

$$s\big(f(x_s)-f(x^\star)\big)\le\frac{L^2}{\alpha}+\frac{\alpha s(s-1)}{4}\|x_s-x^\star\|^2-\frac{\alpha s(s+1)}{4}\|x_{s+1}-x^\star\|^2,$$

which telescopes to $\sum_{s=1}^{t} s\big(f(x_s)-f(x^\star)\big)\le\frac{L^2 t}{\alpha}$; divide by $\frac{t(t+1)}{2}$ and apply convexity. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 4 (Spring 2021), scribed by Alex Damian.

Lecture 5: Gradient Descent for Smooth Functions

Definition 5.1 [Smoothness]. A differentiable f is ℓ-smooth if $\|\nabla f(x) - \nabla f(y)\| \le \ell \|x - y\|$ for all $x, y \in \mathcal{X}$.

For twice differentiable f, ℓ-smoothness is equivalent to $\|\nabla^2 f(x)\| \le \ell$ for all x; for convex f this reads $\lambda_{\max}(\nabla^2 f(x)) \le \ell$. For simplicity this lecture treats unconstrained problems; the same results hold with projections by a similar analysis.

5.1 The descent lemma

Lemma 5.2 [Quadratic upper bound]. *If f is ℓ-smooth, then for all x, y*

$$f(x) \le f(y) + \langle \nabla f(y), x - y \rangle + \frac{\ell}{2}\|x - y\|^2. \tag{5.1}$$

Proof. By the fundamental theorem of calculus and Cauchy–Schwarz,

$$\begin{aligned} f(x) - f(y) - \langle \nabla f(y), x - y \rangle &= \int_0^1 \left[\nabla f(y + t(x - y)) - \nabla f(y)\right]^T (x - y)\, dt \\ &\le \int_0^1 \ell t \|x - y\|^2 \, dt = \frac{\ell}{2}\|x - y\|^2. \end{aligned}$$

□

In fact f is ℓ-smooth if and only if $|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \le \frac{\ell}{2}\|x - y\|^2$ for all x, y.

GD, $x_{t+1} = x_t - \eta \nabla f(x_t)$, is equivalent to

$$x_{t+1} = \arg\min_{x \in \mathbb{R}^d} \; f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2\eta}\|x - x_t\|^2,$$

and when $\eta\ell \le 1$, (5.1) shows that this objective is at least $f(x)$ for every x. So GD iteratively minimizes an upper bound on f: the first-order Taylor expansion at the current iterate plus a quadratic regularizer.

Lemma 5.3 [Descent lemma]. *If f is ℓ-smooth and $\eta\ell \le 1$, GD satisfies*

$$f(x_{t+1}) \le f(x_t) - \frac{\eta}{2}\|\nabla f(x_t)\|^2.$$

Proof. By Lemma 5.2 and $\ell \le 1/\eta$, $f(x_{t+1}) \le f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{1}{2\eta}\|x_{t+1} - x_t\|^2 = f(x_t) - \eta\|\nabla f(x_t)\|^2 + \frac{\eta}{2}\|\nabla f(x_t)\|^2$. □

So with step size at most $1/\ell$, GD always decreases the function value of a smooth function, even a nonconvex one.

5.2 Smooth convex functions

Lemma 5.4 [Contraction]. *If f is convex and ℓ-smooth and $\eta\ell \le 1$, then $\|x_{t+1} - x^\star\| \le \|x_t - x^\star\|$.*

Proof. Convexity gives $\langle \nabla f(x_t), x_t - x^\star \rangle \ge f(x_t) - f(x^\star)$, and the descent lemma gives $\eta^2\|\nabla f(x_t)\|^2 \le 2\eta\big(f(x_t) - f(x_{t+1})\big)$. Hence

$$\begin{aligned}\|x_{t+1} - x^\star\|^2 &= \|x_t - x^\star\|^2 - 2\eta\langle \nabla f(x_t), x_t - x^\star \rangle + \eta^2\|\nabla f(x_t)\|^2 \\ &\le \|x_t - x^\star\|^2 - 2\eta\big(f(x_{t+1}) - f(x^\star)\big) \le \|x_t - x^\star\|^2.\end{aligned}$$

□

Theorem 5.5 [Smooth convex]. *If f is convex and ℓ-smooth and $\eta\ell = 1$, then*

$$f(x_t) - f(x^\star) \le \frac{2\ell\|x_1 - x^\star\|^2}{t-1}.$$

Proof. Let $\delta_s = f(x_s) - f(x^\star)$. The descent lemma gives $\delta_{s+1} \le \delta_s - \frac{1}{2\ell}\|\nabla f(x_s)\|^2$. Convexity, Cauchy–Schwarz, and the contraction lemma give $\delta_s \le \langle \nabla f(x_s), x_s - x^\star \rangle \le \|x_1 - x^\star\|\|\nabla f(x_s)\|$. Together,

$$\delta_{s+1} \le \delta_s - \frac{\delta_s^2}{2\ell\|x_1 - x^\star\|^2}.$$

Dividing by $\delta_s\delta_{s+1}$ and using $\delta_{s+1} \le \delta_s$,

$$\frac{1}{\delta_{s+1}} - \frac{1}{\delta_s} \ge \frac{1}{2\ell\|x_1 - x^\star\|^2} \cdot \frac{\delta_s}{\delta_{s+1}} \ge \frac{1}{2\ell\|x_1 - x^\star\|^2}.$$

Summing over $s = 1, \dots, t-1$ gives $1/\delta_t \ge (t-1)/(2\ell\|x_1 - x^\star\|^2)$. □

5.3 Smooth strongly convex functions

Write $\kappa = \ell/\alpha$ for the condition number.

Theorem 5.6 [Smooth strongly convex]. *If f is α-strongly convex and ℓ-smooth and $\eta\ell = 1$, then*

$$f(x_t) - f(x^\star) \le \frac{\ell}{2}\|x_1 - x^\star\|^2 \exp\Big(-\frac{t-1}{\kappa}\Big).$$

Proof. Strong convexity gives $f(x_t) - f(x^\star) + \frac{\alpha}{2}\|x_t - x^\star\|^2 \le \langle \nabla f(x_t), x_t - x^\star \rangle$. Combined with the descent lemma as in the contraction lemma,

$$\begin{aligned}\|x_{t+1} - x^\star\|^2 &\le (1 - \eta\alpha)\|x_t - x^\star\|^2 - 2\eta\big(f(x_{t+1}) - f(x^\star)\big) \\ &\le \Big(1 - \frac{1}{\kappa}\Big)\|x_t - x^\star\|^2 \le e^{-1/\kappa}\|x_t - x^\star\|^2.\end{aligned}$$

Finally (5.1) at $y = x^\star$, where $\nabla f(x^\star) = 0$, gives

$$f(x_t) - f(x^\star) \le \frac{\ell}{2}\|x_t - x^\star\|^2 \le \frac{\ell}{2}\|x_1 - x^\star\|^2 e^{-(t-1)/\kappa}. \qquad \square$$

5.4 Summary

With R bounding $\|x_1 - x^\star\|$, the rates of (projected) GD and the resulting iteration complexities for accuracy ϵ are, up to constants:

	Rate after t steps		Iterations for accuracy ϵ	
	L-Lipschitz	ℓ-smooth	L-Lipschitz	ℓ-smooth
Convex	$\frac{RL}{\sqrt{t}}$	$\frac{\ell R^2}{t}$	$\left(\frac{RL}{\epsilon}\right)^2$	$\frac{\ell R^2}{\epsilon}$
α-strongly convex	$\frac{L^2}{\alpha t}$	$\ell R^2 e^{-t/\kappa}$	$\frac{L^2}{\alpha\epsilon}$	$\kappa \log \frac{\ell R^2}{\epsilon}$

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 5 (Spring 2021), scribed by Qinghua Liu.

Lecture 6: Nesterov's Accelerated Gradient Descent

Consider $\min_{x\in\mathcal{X}} f(x)$ for a convex set $\mathcal{X}$ of diameter at most R and a convex f, with Lipschitz, strong convexity, and smoothness constants L, α, ℓ, and $\kappa = \ell/\alpha$. Lecture 5 gave the iteration complexities of GD in the four settings.

6.1 From strongly convex to convex rates

The strongly convex rates do not directly give the convex ones: $\alpha = 0$ makes $\kappa = \infty$. Regularization recovers them up to logarithmic factors.

Proposition 6.1 [Reduction by regularization]. *Let $\tilde{\alpha} = \epsilon/R^2$ and $\tilde{f}(x) = f(x) + \frac{\tilde{\alpha}}{2}\|x - x_1\|^2$, where x_1 is the starting point. Any x_t with $\tilde{f}(x_t) \le \min_{x\in\mathcal{X}} \tilde{f}(x) + \frac{\epsilon}{2}$ is ϵ-optimal for f. Running GD on $\tilde{f}$ therefore finds an ϵ-optimal point of f in $O\big(\frac{\ell R^2}{\epsilon}\log\frac{\ell R^2}{\epsilon}\big)$ steps if f is ℓ-smooth and $O\big(\frac{L^2R^2}{\epsilon^2}\big)$ steps if f is L-Lipschitz.*

Proof. Since $\tilde{f} \ge f$ and $\|x_1 - x^\star\| \le R$,

$$f(x_t) \le \tilde{f}(x_t) \le \min_{x\in\mathcal{X}} \tilde{f}(x) + \frac{\epsilon}{2} \le \tilde{f}(x^\star) + \frac{\epsilon}{2} = f(x^\star) + \frac{\epsilon}{2R^2}\|x_1 - x^\star\|^2 + \frac{\epsilon}{2} \le f(x^\star) + \epsilon.$$

The function $\tilde{f}$ is $\tilde{\alpha}$-strongly convex.

- If f is ℓ-smooth, $\tilde{f}$ is $(\ell + \tilde{\alpha})$-smooth. We may assume $\tilde{\alpha} \le \ell$; otherwise $\ell R^2 < \epsilon$ and, by smoothness, every point of $\mathcal{X}$ is already ϵ-optimal. So $\tilde{f}$ is 2ℓ-smooth, and the smooth strongly convex rate gives $O\big(\frac{2\ell}{\tilde{\alpha}}\log\frac{2\ell R^2}{\epsilon}\big) = O\big(\frac{\ell R^2}{\epsilon}\log\frac{\ell R^2}{\epsilon}\big)$.
- If f is L-Lipschitz, $\tilde{f}$ is $(L + \tilde{\alpha}R)$-Lipschitz on $\mathcal{X}$. We may assume $\tilde{\alpha}R \le L$; otherwise $LR < \epsilon$ and every point is ϵ-optimal. So $\tilde{f}$ is $2L$-Lipschitz, and the Lipschitz strongly convex rate gives $O\big(\frac{L^2}{\tilde{\alpha}\epsilon}\big) = O\big(\frac{L^2R^2}{\epsilon^2}\big)$. □

6.2 Momentum

Two accelerated methods are used in practice. The heavy ball method adds a momentum term,

$$x_{t+1} = x_t - \eta\nabla f(x_t) + \gamma(x_t - x_{t-1}),$$

with momentum parameter $\gamma \in [0, 1]$, usually $\gamma = 1 - \theta$ with $\theta \ll 1$. Nesterov's accelerated gradient descent (AGD) takes the gradient step at the extrapolated point:

Algorithm 6.1: Nesterov's accelerated gradient descent

Input: $x_0 = x_1$, step size η, momentum $\gamma \in [0, 1]$

for $t = 1, 2, \ldots$ **do**

$y_t = x_t + \gamma(x_t - x_{t-1})$

$x_{t+1} = y_t - \eta \nabla f(y_t)$

AGD improves the iteration complexity to $O(\epsilon^{-1/2})$ for smooth convex functions and to $O(\sqrt{\kappa} \log \frac{1}{\epsilon})$ for smooth strongly convex functions.

6.3 Intuition from quadratics

Let $f(x) = \frac{1}{2} x^T A x$ with $\alpha I \preceq A \preceq \ell I$, so the minimizer is 0 and $\nabla f(x) = Ax$. Write $A = V \Lambda V^T$ with V orthogonal.

Lemma 6.2 [GD on a quadratic]. *GD with $\eta = \frac{1}{\ell}$ satisfies $\|x_{t+1}\| \le (1 - \frac{1}{\kappa})^t \|x_1\| \le e^{-t/\kappa} \|x_1\|$, so it is ϵ-close to the optimum after $O(\kappa \log \frac{1}{\epsilon})$ steps.*

Proof. $x_{t+1} = (I - \eta A) x_t = V (I - \eta \Lambda)^t V^T x_1$, and $\frac{1}{\kappa} I \preceq \eta \Lambda \preceq I$ gives $0 \preceq I - \eta \Lambda \preceq (1 - \frac{1}{\kappa}) I$. □

Lemma 6.3 [AGD on a quadratic]. *AGD with $\eta = \frac{1}{\ell}$ and $\gamma = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}$ is ϵ-close to the optimum after $\tilde{O}(\sqrt{\kappa} \log \frac{1}{\epsilon})$ steps, a factor $\sqrt{\kappa}$ faster than GD.*

Proof. Substituting y_t, $x_{t+1} = (1+\gamma)(I - \eta A) x_t - \gamma (I - \eta A) x_{t-1}$. In the eigenbasis, the coordinates $v_t^{(i)}$ of $V^T x_t$ decouple: with $c_i = 1 - \eta \lambda_i \in [0, 1 - \frac{1}{\kappa}]$,

$$\begin{pmatrix} v_{t+1}^{(i)} \\ v_t^{(i)} \end{pmatrix} = T_i \begin{pmatrix} v_t^{(i)} \\ v_{t-1}^{(i)} \end{pmatrix}, \qquad T_i = \begin{pmatrix} (1+\gamma) c_i & -\gamma c_i \\ 1 & 0 \end{pmatrix}.$$

The eigenvalues of T_i solve $\mu^2 - (1 + \gamma) c_i \mu + \gamma c_i = 0$, whose discriminant is

$$\Delta = c_i \left[(1 - \gamma)^2 - (1 + \gamma)^2 \eta \lambda_i \right] = \frac{4 c_i}{(\sqrt{\kappa} + 1)^2} \left(1 - \kappa \eta \lambda_i \right) \le 0,$$

since $\eta \lambda_i \ge 1/\kappa$. So the roots are complex conjugates (or a double real root) with $|\mu|^2 = \mu_1 \mu_2 = \gamma c_i \le \gamma$, and the spectral radius is at most $\rho = \sqrt{\gamma} \le 1 - \frac{1}{\sqrt{\kappa} + 1} = 1 - \Theta(\frac{1}{\sqrt{\kappa}})$. By Cayley–Hamilton, $T_i^t = s_t T_i - \mu_1 \mu_2 s_{t-1} I$ with $s_t = \sum_{k=0}^{t-1} \mu_1^k \mu_2^{t-1-k}$, so $\|T_i^t\| = O(t \rho^{t-1})$. Since V is orthogonal, the same bound holds for $\|(x_{t+1}, x_t)\|$ relative to $\|(x_1, x_0)\|$, which gives the claim. □

6.4 Smooth strongly convex functions

Theorem 6.4 [AGD]. *If f is ℓ-smooth and α-strongly convex, AGD with $\eta = \frac{1}{\ell}$, $\gamma = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, and $x_0 = x_1$ satisfies*

$$f(x_t) - f(x^\star) \le \left(1 - \frac{1}{\sqrt{\kappa}}\right)^{t-1} \ell \|x_1 - x^\star\|^2.$$

The proof is a potential argument. Let $q = 1/\sqrt{\kappa}$, $v_t = y_t + \sqrt{\kappa}(y_t - x_t)$, and

$$\Phi_t = f(x_t) - f(x^\star) + \frac{\alpha}{2}\|v_t - x^\star\|^2.$$

The first term is a potential energy. In the second, $v_t - x^\star = (y_t - x^\star) + \sqrt{\kappa}(y_t - x_t)$ combines the distance of y_t to the optimum with a kinetic term. The key claim is

$$\Phi_{t+1} \le (1-q)\,\Phi_t. \tag{6.1}$$

Proof of the theorem from (6.1). Since $x_1 = x_0$, $y_1 = x_1$ and $v_1 = x_1$. Smoothness with $\nabla f(x^\star) = 0$ gives $f(x_1) - f(x^\star) \le \frac{\ell}{2}\|x_1 - x^\star\|^2$, and $\alpha \le \ell$, so $\Phi_1 \le \ell\|x_1 - x^\star\|^2$. Then $f(x_{t+1}) - f(x^\star) \le \Phi_{t+1} \le (1-q)^t \Phi_1$. □

Proof of (6.1). Write $g = \nabla f(y_t)$.
Function value. Smoothness and $x_{t+1} = y_t - \frac{1}{\ell} g$ give $f(x_{t+1}) \le f(y_t) - \frac{1}{2\ell}\|g\|^2$. Strong convexity at y_t gives

$$f(x_t) \ge f(y_t) + \langle g, x_t - y_t\rangle + \frac{\alpha}{2}\|x_t - y_t\|^2, \qquad f(x^\star) \ge f(y_t) + \langle g, x^\star - y_t\rangle + \frac{\alpha}{2}\|x^\star - y_t\|^2.$$

Subtracting $(1-q)$ times the first and q times the second from the smoothness bound,

$$\begin{aligned} f(x_{t+1}) - f(x^\star) \le (1-q)\big(f(x_t) - f(x^\star)\big) - \frac{1}{2\ell}\|g\|^2 + \langle g, y_t - (1-q)x_t - q x^\star\rangle \\ - \frac{q\alpha}{2}\|y_t - x^\star\|^2 - \frac{(1-q)\alpha}{2}\|x_t - y_t\|^2. \end{aligned} \tag{6.2}$$

The sequence v_t. From the definition, $x_t = (1+\sqrt{\kappa})q\, y_t - q\, v_t$. With $\gamma = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, so that $(1+\sqrt{\kappa})\gamma = \sqrt{\kappa} - 1$,

$$v_{t+1} = (1+\sqrt{\kappa})y_{t+1} - \sqrt{\kappa}x_{t+1} = x_{t+1} + (\sqrt{\kappa}-1)(x_{t+1} - x_t) = (1-q)v_t + q\, y_t - \frac{q}{\alpha}\, g,$$

using $x_{t+1} - y_t = -\frac{1}{\ell} g$ and $\sqrt{\kappa}/\ell = q/\alpha$ in the last step. Thus v_{t+1} mixes v_t with a long gradient step from y_t.
Distance. Let $a = (1-q)(v_t - x^\star) + q(y_t - x^\star)$. Then $\frac{\alpha}{2}\|v_{t+1} - x^\star\|^2 = \frac{\alpha}{2}\|a\|^2 - q\langle g, a\rangle + \frac{q^2}{2\alpha}\|g\|^2$,

where $\frac{q^2}{2\alpha} = \frac{1}{2\ell}$. By convexity of $\|\cdot\|^2$, $\|a\|^2 \le (1-q)\|v_t - x^\star\|^2 + q\|y_t - x^\star\|^2$, and the formula for x_t gives $q\big[(1-q)v_t + qy_t\big] = y_t - (1-q)x_t$. Hence

$$\frac{\alpha}{2}\|v_{t+1} - x^\star\|^2 \le (1-q)\frac{\alpha}{2}\|v_t - x^\star\|^2 + \frac{q\alpha}{2}\|y_t - x^\star\|^2 + \frac{1}{2\ell}\|g\|^2 - \langle g, y_t - (1-q)x_t - qx^\star\rangle. \tag{6.3}$$

Adding (6.2) and (6.3), the gradient-norm, inner-product, and $\|y_t - x^\star\|^2$ terms cancel, leaving $\Phi_{t+1} \le (1-q)\Phi_t - \frac{(1-q)\alpha}{2}\|x_t - y_t\|^2 \le (1-q)\Phi_t$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 6 (Spring 2021), scribed by Abhishek Panigrahi; proof completed from Lecture 7, scribed by Ye Tian.

Lecture 7: Lower Bounds: Linear Span Algorithms

7.1 Are the rates sharp?

The rates so far, with AGD improving the smooth column, are:

	L-Lipschitz	ℓ-smooth (GD)	ℓ-smooth (AGD)
Convex	$\frac{RL}{\sqrt{t}}$	$\frac{\ell R^2}{t}$	$\frac{\ell R^2}{t^2}$
α-strongly convex	$\frac{L^2}{\alpha t}$	$\ell R^2 e^{-t/\kappa}$	$\ell R^2 e^{-t/\sqrt{\kappa}}$

Can any algorithm converge faster? An upper bound has the form

$$\exists \text{ algorithm (e.g. AGD) } \forall f \ \ell\text{-smooth and convex}: \quad f(x_t) - f(x^\star) \le O\Big(\frac{\ell R^2}{t^2}\Big).$$

A matching lower bound negates it:

$$\forall \text{ algorithm } \exists f \ \ell\text{-smooth and convex}: \quad \min_{1\le s\le t} f(x_s) - f(x^\star) \ge \Omega\Big(\frac{\ell R^2}{t^2}\Big).$$

The hard instance may depend on the algorithm, and must, as trivial examples show: an algorithm that always outputs 0 is optimal for x^2 but fails on $(x-1)^2$, and one that always outputs 1 fails on x^2.

A general black-box algorithm chooses

$$x_{t+1} = A_t(x_1, g_1, x_2, g_2, \dots, x_t, g_t), \qquad g_s \in \partial f(x_s),$$

where A_t is an arbitrary, possibly randomized, map of the history.

7.2 Span algorithms

The classical setting for lower bounds is the class of span algorithms:

$$x_1 = 0, \qquad x_{t+1} \in \operatorname{span}(g_1, \dots, g_t).$$

GD and AGD belong to it, and the constructions below extend to general algorithms (Lecture 8). The class is restrictive, though: an algorithm that draws x_{t+1} from a Gaussian, ignoring the history entirely, is a poor algorithm, but these lower bounds say nothing about it.

Theorem 7.1 [Lipschitz convex]. *For any t, L, R and any span algorithm, there are a convex L-Lipschitz f and a convex set $\mathcal{X}$ of diameter R with*

$$\min_{1\le s\le t} f(x_s) - \min_{x\in\mathcal{X}} f(x) \ge \frac{RL}{2\sqrt{t}}.$$

So GD is optimal among span algorithms for Lipschitz convex functions.

Proof. The statement is dimension-free, so take $d \ge t$ and

$$f(x) = L \max_{1\le i\le t} e_i^T x, \qquad \mathcal{X} = B(0, R/2).$$

Then $\partial f(x) = L\,\mathrm{conv}\{e_i : e_i^T x = \max_{j\le t} e_j^T x\}$, and the oracle returns Le_i for the *first* maximizing index i.

1. f is convex as a maximum of linear functions, and L-Lipschitz since every returned subgradient has norm L.
2. The point $x = -\rho\sum_{i=1}^t e_i$ with $\rho = \frac{R}{2\sqrt{t}}$ has $\|x\| = \frac{R}{2}$, so $\min_{\mathcal{X}} f \le f(x) = -L\rho = -\frac{LR}{2\sqrt{t}}$.
3. By induction, $x_s \in \mathrm{span}(e_1, \dots, e_{s-1})$ for $s \le t$. This holds for $x_1 = 0$. If it holds for x_s, then $e_s^T x_s = 0$, so the maximum over $i \le t$ is at least 0 and is first attained at some index $i \le s$; thus $g_s \in \mathrm{span}(e_s) + \mathrm{span}(e_1, \dots, e_{s-1})$ and $x_{s+1} \in \mathrm{span}(g_1, \dots, g_s) \subseteq \mathrm{span}(e_1, \dots, e_s)$.

Hence $e_t^T x_s = 0$ and $f(x_s) \ge L e_t^T x_s = 0$ for all $s \le t$, so $f(x_s) - \min_{\mathcal{X}} f \ge \frac{LR}{2\sqrt{t}}$. □

Theorem 7.2 [Lipschitz strongly convex]. *For any t, L, α and any span algorithm, there are an α-strongly convex, L-Lipschitz f and a convex set $\mathcal{X}$ with*

$$\min_{1\le s\le t} f(x_s) - \min_{x\in\mathcal{X}} f(x) \ge \frac{L^2}{8\alpha t}.$$

Proof. Take

$$f(x) = \frac{L}{2} \max_{1\le i\le t} e_i^T x + \frac{\alpha}{2}\|x\|^2, \qquad \mathcal{X} = B\Big(0, \frac{L}{2\alpha}\Big),$$

so $\partial f(x) = \alpha x + \frac{L}{2}\,\mathrm{conv}\{e_i : e_i^T x = \max_{j\le t} e_j^T x\}$, and the oracle returns $\alpha x + \frac{L}{2}e_i$ for the first maximizing i. It is α-strongly convex, and L-Lipschitz on $\mathcal{X}$ since $\|g\| \le \frac{L}{2} + \alpha\cdot\frac{L}{2\alpha} = L$. The same induction gives $x_s \in \mathrm{span}(e_1, \dots, e_{s-1})$, so $e_t^T x_s = 0$ and $f(x_s) \ge 0$ for $s \le t$. For $x = -\rho\sum_{i=1}^t e_i$, $f(x) = -\frac{L\rho}{2} + \frac{\alpha t\rho^2}{2}$, minimized at $\rho^\star = \frac{L}{2\alpha t}$ with value $-\frac{L^2}{8\alpha t}$; this point has $\|x\| = \sqrt{t}\rho^\star = \frac{L}{2\alpha\sqrt{t}} \le \frac{L}{2\alpha}$, so it lies in $\mathcal{X}$. Hence $\min_s f(x_s) - \min_{\mathcal{X}} f \ge 0 + \frac{L^2}{8\alpha t}$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 7 (Spring 2021), scribed by Ye Tian.

Lecture 8: Lower Bounds II: General Randomized Algorithms

8.1 Smooth functions and span algorithms

Here a span algorithm is a black-box algorithm with $x_1 = 0$ and $x_{t+1} \in \operatorname{span}(x_1, g_1, \dots, x_t, g_t)$, $g_s \in \partial f(x_s)$.

Theorem 8.1 [Smooth convex]. *For any $\ell > 0$, $R > 0$, t and any span algorithm, there is a convex ℓ-smooth f on $\mathbb{R}^d$, $d \geq t$, with $\|x_1 - x^\star\| = R$ for its minimizer $x^\star$ and*

$$\min_{1 \leq s \leq t} f(x_s) - f(x^\star) \geq \frac{\ell \|x_1 - x^\star\|^2}{8t^2}.$$

So AGD is optimal among span algorithms for smooth convex functions.

The proof bounds the Hessian with a classical fact.

Lemma 8.2 [Gershgorin circle theorem]. *Every eigenvalue of a complex $n \times n$ matrix $A = (a_{ij})$ lies in at least one of the discs $D(a_{ii}, R_i) \subseteq \mathbb{C}$ with radius $R_i = \sum_{j \neq i} |a_{ij}|$.*

So eigenvalues stay close to the diagonal entries when the off-diagonal entries are small.

Proof of Theorem 8.1. With the coordinates $x_i = e_i^T x$, take the chain function

$$f(x) = \frac{\ell}{4}\left[\frac{1}{2}\left(x_1 - \frac{R}{\sqrt{t}}\right)^2 + \frac{1}{2}\sum_{i=1}^{t-1}(x_i - x_{i+1})^2\right].$$

It is a convex quadratic. On the first t coordinates its Hessian is tridiagonal with diagonal entries $\frac{\ell}{2}$ (the last one $\frac{\ell}{4}$) and off-diagonal entries $-\frac{\ell}{4}$, and it vanishes elsewhere. Every Gershgorin disc lies in $[0, \ell]$, so f is ℓ-smooth. Since $f \geq 0$ and f vanishes at $x^\star = \frac{R}{\sqrt{t}} \sum_{i=1}^{t} e_i$, this is a minimizer, with $f(x^\star) = 0$ and $\|x_1 - x^\star\| = \|x^\star\| = R$.
The gradient is $\nabla f(x) = \frac{\ell}{4}\left[(x_1 - \frac{R}{\sqrt{t}})e_1 + \sum_{i=1}^{t-1}(x_i - x_{i+1})(e_i - e_{i+1})\right]$, so $x \in \operatorname{span}(e_1, \dots, e_{s-1})$ implies $\nabla f(x) \in \operatorname{span}(e_1, \dots, e_s)$. Starting from $x_1 = 0$, induction gives $x_s \in \operatorname{span}(e_1, \dots, e_{s-1})$ and $g_s \in \operatorname{span}(e_1, \dots, e_s)$, so $e_j^T x_s = 0$ for $j \geq s$. Then

$$f(x_s) = \frac{\ell}{8}\left[\left(\frac{R}{\sqrt{t}} - x_1\right)^2 + (x_1 - x_2)^2 + \cdots + (x_{s-2} - x_{s-1})^2 + (x_{s-1} - 0)^2\right].$$

The s increments sum to $R/\sqrt{t}$, so by Cauchy–Schwarz the sum of their squares is at least $\frac{R^2}{ts}$, with equality at $x_s = \frac{R}{\sqrt{t}}\left(\frac{s-1}{s}, \frac{s-2}{s}, \dots, \frac{1}{s}, 0, \dots, 0\right)$. Hence $f(x_s) \geq \frac{\ell R^2}{8ts} \geq \frac{\ell R^2}{8t^2}$ for every $s \leq t$. □

The analogous lower bound for α-strongly convex, ℓ-smooth functions, left as an exercise, uses the regularized chain

$$f(x) = \frac{1}{2}\Big(x_1 - \frac{R}{\sqrt{t}}\Big)^2 + \frac{1}{2}\sum_{i=1}^{t-1}(x_i - x_{i+1})^2 + \frac{\alpha}{2}\|x\|^2.$$

8.2 Randomized algorithms

A general randomized algorithm chooses $x_{t+1} = A(x_1, g_1, \ldots, x_t, g_t, r_t)$ with independent random r_t. With $\mathrm{Gap}(A, t) = \min_{1\le s\le t} f(x_s) - f(x^\star)$, the goal is to lower bound

$$\min_{A \text{ randomized}} \max_{f\in\mathcal{F}} \mathbb{E}_A \,\mathrm{Gap}(A, t). \tag{8.1}$$

A single fixed hard instance cannot work: for $f(x) = \max_{i\le t} e_i^T x$ of Lecture 7, the non-span algorithm that always outputs the minimizer $-\frac{1}{\sqrt{t}}\sum_{i=1}^t e_i$ breaks it. Instead, put a prior $\mathcal{D}_\mathcal{F}$ on $\mathcal{F}$:

$$\min_{A \text{ rand.}} \max_{f\in\mathcal{F}} \mathbb{E}_A \,\mathrm{Gap}(A, t) \ge \min_{A \text{ rand.}} \mathbb{E}_{f\sim\mathcal{D}_\mathcal{F}} \mathbb{E}_A \,\mathrm{Gap}(A, t) = \min_{A \text{ det.}} \mathbb{E}_{f\sim\mathcal{D}_\mathcal{F}} \,\mathrm{Gap}(A, t).$$

The last equality holds because a randomized algorithm is a mixture of deterministic ones (fix the r_t), so the minimum is attained by a deterministic algorithm. It therefore suffices to lower bound the expected gap of deterministic algorithms on a random instance.

Theorem 8.3 [Randomized lower bound]. *For any L, R, t and any randomized algorithm querying points of $\mathcal{X}$, there are a convex L-Lipschitz f and a convex set $\mathcal{X}$ of diameter R with*

$$\mathbb{E}_A\Big[\min_{1\le s\le t} f(x_s) - \min_{x\in\mathcal{X}} f(x)\Big] \ge \Omega\Big(\frac{RL}{\sqrt{t}}\Big).$$

By scaling, take $L = 1$ and $\mathcal{X}$ the unit ball (diameter 2). The random instance is

$$f(x) = \max_{1\le i\le t}\big(v_i^T x - i\delta\big),$$

where $\delta > 0$ and $v_1, \ldots, v_t$ are the first t vectors of a uniformly random orthonormal basis of $\mathbb{R}^d$; the oracle returns v_i for the first maximizing index i. With e_i in place of v_i, the span structure guaranteed that e_i is revealed no earlier than step i, which fails for general algorithms. Random directions restore it with high probability: a query x chosen without knowledge of v_j has $|v_j^T x| = O(1/\sqrt{d})$, and the offsets $i\delta$ then make the indices appear in order.

Proof. f is convex and 1-Lipschitz, and $\min_\mathcal{X} f \le f\big(-\frac{1}{\sqrt{t}}\sum_i v_i\big) = -\frac{1}{\sqrt{t}} - \delta$.

Gap on a good event. On $\tilde{E} = \{|v_t^T x_s| \le \frac{\delta}{2} \text{ for all } s \le t\}$ we have $f(x_s) \ge v_t^T x_s - t\delta \ge -(t+\frac{1}{2})\delta$, so

$$\min_{s\le t} f(x_s) - \min_X f \ge \frac{1}{\sqrt{t}} - \Big(t - \frac{1}{2}\Big)\delta \ge \frac{1}{2\sqrt{t}} \qquad \text{for } \delta = \frac{1}{2t\sqrt{t}}.$$

Since the gap is always nonnegative, it suffices to show $\mathbb{P}[\tilde{E}] \ge \frac{1}{2}$. Consider the smaller event $E = \{|v_j^T x_s| \le \frac{\delta}{2} \text{ for all } 1 \le s \le j \le t\} \subseteq \tilde{E}$.

Simulation. Fix a deterministic algorithm and run it against the oracle that, at step r, answers with the first maximizer over $i \le r$ only. Its queries $\hat{x}_1, \dots, \hat{x}_t$ satisfy: $\hat{x}_r$ is a function of $v_1, \dots, v_{r-1}$. Let $F = \{|v_j^T \hat{x}_r| \le \frac{\delta}{2} \text{ for all } 1 \le r \le j \le t\}$. On F, for $j > r$, $v_j^T \hat{x}_r - j\delta \le \frac{\delta}{2} - (r+1)\delta \le v_r^T \hat{x}_r - r\delta$, so the true oracle gives the same answer at $\hat{x}_r$. By induction the true run coincides with the simulated one, $x_r = \hat{x}_r$, and so $F \subseteq E$.

Probability. Conditionally on $v_1, \dots, v_{r-1}$, each v_j with $j \ge r$ is uniform on the unit sphere of the $(d-r+1)$-dimensional orthogonal complement of $\operatorname{span}(v_1, \dots, v_{r-1})$, while $\hat{x}_r$ is fixed with $\|\hat{x}_r\| \le 1$. A uniform unit vector u in dimension m satisfies $\mathbb{P}[|u^T a| > \tau] \le 2e^{-m\tau^2/2}$ for $\|a\| \le 1$. A union bound over the at most t^2 pairs gives $\mathbb{P}[F^c] \le 2t^2 e^{-(d-t)\delta^2/8} \le \frac{1}{2}$ once $d \ge t + 8\delta^{-2}\ln(4t^2) = O(t^3 \log t)$. Hence $\mathbb{P}[\tilde{E}] \ge \mathbb{P}[F] \ge \frac{1}{2}$ and the expected gap is at least $\frac{1}{4\sqrt{t}}$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 8 (Spring 2021), scribed by Yuren Zhong.

Lecture 9: Mirror Descent I

9.1 Beyond the Euclidean geometry

The dimension-free rates of GD assume that f and $\mathcal{X}$ are well behaved in the Euclidean norm: $\|x\|_2$ for $x \in \mathcal{X}$ and $\|g\|_2$ for $g \in \partial f(x)$ do not depend on the dimension d. Otherwise GD can lose them. For example, if f is convex and differentiable on the Euclidean ball $B_{2,d}$ with $\|\nabla f(x)\|_\infty \le 1$, then only $\|\nabla f(x)\|_2 \le \sqrt{d}$ is guaranteed, and projected GD converges at rate $\sqrt{d/t}$. Mirror descent minimizes such functions over the ℓ_1-ball at the much faster rate $\sqrt{\log(d)/t}$.

The idea is clearest in a Banach space B whose norm does not come from an inner product (think of ℓ_1). The gradient $\nabla f(x)$ is an element of the dual space B^*, so the update $x - \eta \nabla f(x)$ does not even make sense; in a Hilbert space the problem does not arise because the dual is isometric to the space itself. Mirror descent instead maps x to the dual space, takes the gradient step there, and maps the result back to the primal space. The new point may lie outside $\mathcal{X}$, so it is then projected back. Both the primal-dual map and the projection come from a mirror map.

9.2 Norms and Bregman divergences

Fix an arbitrary norm $\|\cdot\|$ on $\mathbb{R}^d$ and a compact convex set $\mathcal{X} \subseteq \mathbb{R}^d$.

Definition 9.1 [Dual norm]. The dual norm is $\|g\|_* = \sup_{x \in \mathbb{R}^d : \|x\| \le 1} g^T x$.

For example, the dual of $\|\cdot\|_1$ is $\|\cdot\|_\infty$, and $g^T x \le \|g\|_* \|x\|$. A convex $f : \mathcal{X} \to \mathbb{R}$ is

- L-Lipschitz w.r.t. $\|\cdot\|$ if $\|g\|_* \le L$ for all $x \in \mathcal{X}$ and $g \in \partial f(x)$;
- ℓ-smooth w.r.t. $\|\cdot\|$ if $\|\nabla f(x) - \nabla f(y)\|_* \le \ell \|x - y\|$ for all $x, y \in \mathcal{X}$;
- α-strongly convex w.r.t. $\|\cdot\|$ if $f(x) - f(y) \le g^T(x - y) - \frac{\alpha}{2}\|x - y\|^2$ for all $x, y \in \mathcal{X}$ and $g \in \partial f(x)$.

Definition 9.2 [Bregman divergence]. The Bregman divergence of a differentiable convex f is

$$D_f(x, y) = f(x) - f(y) - \nabla f(y)^T (x - y) \ge 0,$$

the gap between $f(x)$ and its linearization at y.

Expanding the three divergences gives the three-point identity

$$\big(\nabla f(x) - \nabla f(y)\big)^T (x - z) = D_f(x, y) + D_f(z, x) - D_f(z, y) : \tag{9.1}$$

the function values cancel, and the remaining linear terms are $-\nabla f(y)^T(x-y) - \nabla f(x)^T(z-x) + \nabla f(y)^T(z-y)$, which equals the left side.

9.3 Mirror maps

Definition 9.3 [Mirror map]. Let $\mathcal{D} \subseteq \mathbb{R}^d$ be an open convex set whose closure contains $\mathcal{X}$, with $\mathcal{X} \cap \mathcal{D} \neq \emptyset$. A function $\Phi : \mathcal{D} \to \mathbb{R}$ is a mirror map if

1. Φ is strictly convex and differentiable;
2. its gradient takes all values: $\nabla\Phi(\mathcal{D}) = \mathbb{R}^d$;
3. its gradient diverges on the boundary: $\lim_{x \to \partial\mathcal{D}} \|\nabla\Phi(x)\| = +\infty$.

The gradient $\nabla\Phi$ maps points from the "primal" to the "dual" (all points lie in $\mathbb{R}^d$, so the two spaces are only an intuition). A point $x \in \mathcal{X} \cap \mathcal{D}$ is mapped to $\nabla\Phi(x)$, a gradient step gives $\nabla\Phi(x) - \eta\nabla f(x)$, and property 2 lets us write the result as $\nabla\Phi(y)$ for some $y \in \mathcal{D}$. If y lies outside $\mathcal{X}$, it is projected back using the Bregman divergence of Φ.

Definition 9.4 [Bregman projection]. The Bregman projection of $y \in \mathcal{D}$ onto $\mathcal{X}$ is

$$\Pi_{\mathcal{X}}^{\Phi}(y) = \operatorname*{arg\,min}_{x \in \mathcal{X} \cap \mathcal{D}} D_\Phi(x, y).$$

Properties 1 and 3 guarantee that the projection exists and is unique; in particular $x \mapsto D_\Phi(x, y)$ increases near the boundary of $\mathcal{D}$. Bregman projections behave like Euclidean projections (compare Lemma 4.1).

Lemma 9.5 [Bregman projection]. *For $x \in \mathcal{X} \cap \mathcal{D}$ and $y \in \mathcal{D}$,*

$$\big(\nabla\Phi(\Pi_{\mathcal{X}}^{\Phi}(y)) - \nabla\Phi(y)\big)^T \big(\Pi_{\mathcal{X}}^{\Phi}(y) - x\big) \leq 0,$$

which implies

$$D_\Phi\big(x, \Pi_{\mathcal{X}}^{\Phi}(y)\big) + D_\Phi\big(\Pi_{\mathcal{X}}^{\Phi}(y), y\big) \leq D_\Phi(x, y).$$

Proof. Let $z = \Pi_{\mathcal{X}}^{\Phi}(y)$. The function $g(x) = D_\Phi(x, y)$ is convex with gradient $\nabla\Phi(x) - \nabla\Phi(y)$, and z minimizes it over the convex set. By the first-order optimality condition, $\nabla g(z)^T(z - x) \leq 0$ for every feasible x: otherwise g would decrease along the segment from z toward x. This is the first inequality. By (9.1) with Φ, its left side equals $D_\Phi(z, y) + D_\Phi(x, z) - D_\Phi(x, y)$, which gives the second. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 9 (Spring 2021); written from S. Bubeck, Convex Optimization: Algorithms and Complexity, Chapter 4 introduction and Section 4.1.

Lecture 10: Mirror Descent II

10.1 The algorithm

With a mirror map Φ on $\mathcal{D}$ (Lecture 9), mirror descent alternates a gradient step in the dual with a Bregman projection in the primal (Figure 3).

Algorithm 10.1: Mirror descent

Input: mirror map Φ, step size η

$x_1 \in \arg\min_{x \in \mathcal{X} \cap \mathcal{D}} \Phi(x)$

for $t = 1, 2, \dots$ **do**

  choose $y_{t+1} \in \mathcal{D}$ with $\nabla\Phi(y_{t+1}) = \nabla\Phi(x_t) - \eta g_t$, where $g_t \in \partial f(x_t)$

  $x_{t+1} = \Pi^{\Phi}_{\mathcal{X}}(y_{t+1})$

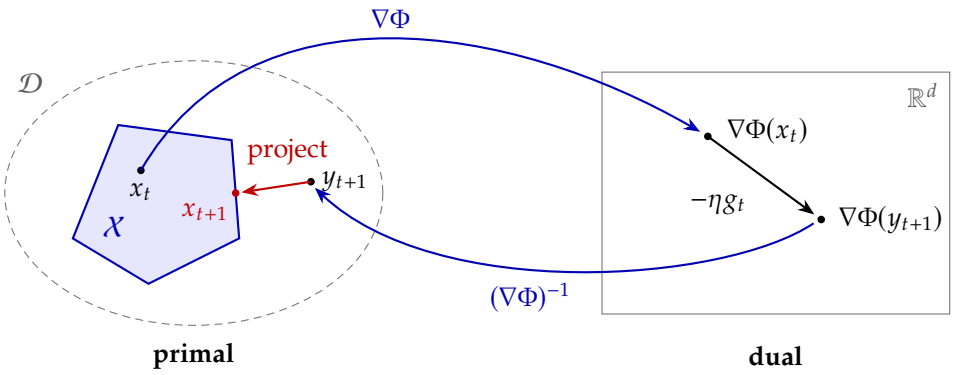


Figure 3: One step of mirror descent. The point x_t is mapped to the dual by $\nabla\Phi$, a gradient step is taken there, the result is mapped back to $y_{t+1} \in \mathcal{D}$, and y_{t+1} is projected onto $\mathcal{X}$ in the Bregman divergence of Φ. Schematic.

Theorem 10.1 [Mirror descent]. *Let Φ be a mirror map that is ρ-strongly convex on $\mathcal{X} \cap \mathcal{D}$ w.r.t. $\|\cdot\|$, let $R^2 = \sup_{x \in \mathcal{X} \cap \mathcal{D}} \Phi(x) - \Phi(x_1)$, and let f be convex and L-Lipschitz w.r.t. $\|\cdot\|$. Mirror descent with $\eta = \frac{R}{L}\sqrt{\frac{2\rho}{t}}$ satisfies*

$$f\Big(\frac{1}{t}\sum_{s=1}^{t} x_s\Big) - f(x^\star) \le RL\sqrt{\frac{2}{\rho t}}.$$

Proof. Fix $x \in \mathcal{X} \cap \mathcal{D}$; the bound follows by letting $x \to x^\star$. By convexity, the update, the

three-point identity, and Lemma 9.5,

$$\begin{aligned} f(x_s) - f(x) \le g_s^T(x_s - x) &= \frac{1}{\eta}\big(\nabla\Phi(x_s) - \nabla\Phi(y_{s+1})\big)^T(x_s - x) \\ &= \frac{1}{\eta}\big(D_\Phi(x, x_s) + D_\Phi(x_s, y_{s+1}) - D_\Phi(x, y_{s+1})\big) \\ &\le \frac{1}{\eta}\big(D_\Phi(x, x_s) - D_\Phi(x, x_{s+1}) + D_\Phi(x_s, y_{s+1}) - D_\Phi(x_{s+1}, y_{s+1})\big). \end{aligned}$$

The first difference telescopes over $s = 1, \dots, t$. For the second, use ρ-strong convexity of Φ and $az - bz^2 \le \frac{a^2}{4b}$:

$$\begin{aligned} D_\Phi(x_s, y_{s+1}) - D_\Phi(x_{s+1}, y_{s+1}) &= \Phi(x_s) - \Phi(x_{s+1}) - \nabla\Phi(y_{s+1})^T(x_s - x_{s+1}) \\ &\le \big(\nabla\Phi(x_s) - \nabla\Phi(y_{s+1})\big)^T(x_s - x_{s+1}) - \frac{\rho}{2}\|x_s - x_{s+1}\|^2 \\ &= \eta g_s^T(x_s - x_{s+1}) - \frac{\rho}{2}\|x_s - x_{s+1}\|^2 \\ &\le \eta L\|x_s - x_{s+1}\| - \frac{\rho}{2}\|x_s - x_{s+1}\|^2 \le \frac{(\eta L)^2}{2\rho}. \end{aligned}$$

Hence $\sum_{s=1}^t \big(f(x_s) - f(x)\big) \le \frac{D_\Phi(x, x_1)}{\eta} + \frac{\eta L^2 t}{2\rho}$. Since x_1 minimizes Φ over the feasible set, $\nabla\Phi(x_1)^T(x - x_1) \ge 0$ and $D_\Phi(x, x_1) \le \Phi(x) - \Phi(x_1) \le R^2$. The stated η balances the two terms, giving $\sum_s \big(f(x_s) - f(x)\big) \le RL\sqrt{2t/\rho}$; divide by t and apply convexity. □

Since $\nabla\Phi(y_{t+1}) = \nabla\Phi(x_t) - \eta g_t$, the projection step can be rewritten as

$$\begin{aligned} x_{t+1} &= \arg\min_{x \in \mathcal{X}\cap\mathcal{D}} D_\Phi(x, y_{t+1}) = \arg\min_{x \in \mathcal{X}\cap\mathcal{D}} \Phi(x) - \nabla\Phi(y_{t+1})^T x \\ &= \arg\min_{x \in \mathcal{X}\cap\mathcal{D}} \Phi(x) - \big(\nabla\Phi(x_t) - \eta g_t\big)^T x = \arg\min_{x \in \mathcal{X}\cap\mathcal{D}} \eta g_t^T x + D_\Phi(x, x_t), \end{aligned}$$

which is often taken as the definition. It gives a proximal view: mirror descent minimizes the linearization of f while not moving too far from the previous point, with distance measured by the Bregman divergence of the mirror map.

10.2 Standard setups

Ball setup. $\Phi(x) = \frac{1}{2}\|x\|_2^2$ on $\mathcal{D} = \mathbb{R}^d$ is a mirror map, 1-strongly convex w.r.t. $\|\cdot\|_2$, with $D_\Phi(x, y) = \frac{1}{2}\|x - y\|_2^2$. Mirror descent is then exactly projected subgradient descent, and Theorem 10.1 recovers the rate of Lecture 4.

Simplex setup. The negative entropy $\Phi(x) = \sum_{i=1}^{d} x(i) \log x(i)$ on $\mathcal{D} = \mathbb{R}^d_{++}$ gives the multiplicative update

$$y_{t+1}(i) = x_t(i) \exp\bigl(-\eta [\nabla f(x_t)](i)\bigr), \qquad i = 1, \dots, d,$$

and $D_\Phi(x, y) = \sum_i x(i) \log \frac{x(i)}{y(i)} - \sum_i x(i) + \sum_i y(i)$, the Kullback–Leibler divergence when x and y lie in the simplex $\Delta_d = \{x \in \mathbb{R}^d_+ : \sum_i x(i) = 1\}$. The Bregman projection onto Δ_d is the renormalization $y \mapsto y/\|y\|_1$. By Pinsker's inequality Φ is 1-strongly convex w.r.t. $\|\cdot\|_1$ on Δ_d, and for $\mathcal{X} = \Delta_d$, $x_1 = (\frac{1}{d}, \dots, \frac{1}{d})$ and $R^2 = \log d$. So for f with subgradients bounded in ℓ_∞ (the dual of ℓ_1), mirror descent with the negative entropy converges at rate of order $\sqrt{\log(d)/t}$, while subgradient descent achieves only order $\sqrt{d/t}$.

Spectrahedron setup. To minimize f over the SPECTRAHEDRON $\mathcal{S}_d = \{X \in \mathbb{S}^d_+ : \operatorname{tr}(X) = 1\}$ of positive semidefinite matrices with unit trace, use on $\mathcal{D} = \mathbb{S}^d_{++}$ the negative von Neumann entropy $\Phi(X) = \sum_{i=1}^{d} \lambda_i(X) \log \lambda_i(X)$, where $\lambda_1(X), \dots, \lambda_d(X)$ are the eigenvalues. The gradient update becomes

$$Y_{t+1} = \exp\bigl(\log X_t - \eta \nabla f(X_t)\bigr)$$

with the matrix exponential and logarithm, and the projection onto $\mathcal{S}_d$ is a trace renormalization. A nontrivial computation shows that Φ is $\frac{1}{2}$-strongly convex w.r.t. the Schatten 1-norm $\|X\|_1 = \sum_i |\lambda_i(X)|$. Here $X_1 = \frac{1}{d} I_d$ and $R^2 = \log d$, so the rate on the spectrahedron is the same as on the simplex.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 10 (Spring 2021); written from S. Bubeck, Convex Optimization: Algorithms and Complexity, Sections 4.2–4.3.

Lecture 11: Stochastic Gradient Descent

A stochastic gradient oracle returns, at any $x \in \mathcal{X}$, a random vector $\tilde{g}(x)$ with $\mathbb{E}[\tilde{g}(x)] \in \partial f(x)$.

Assumption 11.1 (Bounded variance)**.** $\mathrm{Var}(\tilde{g}(x)) = \mathbb{E}\|\tilde{g}(x) - \mathbb{E}\tilde{g}(x)\|^2 \le \sigma^2$ for all $x \in \mathcal{X}$.

Then $\mathbb{E}\|\tilde{g}(x)\|^2 = \|\mathbb{E}\tilde{g}(x)\|^2 + \mathrm{Var}(\tilde{g}(x))$, which is at most $L^2 + \sigma^2$ for L-Lipschitz f.

Algorithm 11.1: Projected stochastic gradient descent

```
for s = 1, ..., t do
    y_{s+1} = x_s − η g̃(x_s)
    x_{s+1} = Π_X(y_{s+1})
```

11.1 Lipschitz convex functions

Theorem 11.2 [SGD, Lipschitz convex]**.** *Let f be convex and L-Lipschitz, $R = \max_{x\in\mathcal{X}}\|x - x_1\|$, and $\eta = \frac{R}{\sqrt{(L^2+\sigma^2)t}}$. Then*

$$\mathbb{E}f\Big(\frac{1}{t}\sum_{s=1}^{t} x_s\Big) - f(x^\star) \le R\sqrt{\frac{L^2+\sigma^2}{t}} \lesssim \underbrace{\frac{RL}{\sqrt{t}}}_{\text{rate of GD}} + \underbrace{\frac{R\sigma}{\sqrt{t}}}_{\text{stochastic error}}.$$

Proof. Conditionally on x_s, $\tilde{g}(x_s)$ has mean in $\partial f(x_s)$. Following the deterministic proof (Lecture 4) inside the expectation,

$$\begin{aligned}
\mathbb{E}f(x_s) - f(x^\star) &\le \mathbb{E}\big[\tilde{g}(x_s)^T(x_s - x^\star)\big] = \frac{1}{\eta}\mathbb{E}\big[(x_s - y_{s+1})^T(x_s - x^\star)\big] \\
&= \frac{1}{2\eta}\mathbb{E}\big[\|x_s - y_{s+1}\|^2 + \|x_s - x^\star\|^2 - \|y_{s+1} - x^\star\|^2\big] \\
&\le \frac{1}{2\eta}\mathbb{E}\big[\|x_s - x^\star\|^2 - \|x_{s+1} - x^\star\|^2\big] + \frac{\eta}{2}\mathbb{E}\|\tilde{g}(x_s)\|^2 \\
&\le \frac{1}{2\eta}\mathbb{E}\big[\|x_s - x^\star\|^2 - \|x_{s+1} - x^\star\|^2\big] + \frac{\eta(L^2+\sigma^2)}{2},
\end{aligned}$$

using the projection inequality $\|y_{s+1} - x^\star\| \ge \|x_{s+1} - x^\star\|$. Telescoping over $s = 1, \dots, t$ and

applying convexity,

$$\mathbb{E} f\Big(\frac{1}{t}\sum_{s=1}^{t} x_s\Big) - f(x^\star) \le \frac{1}{t}\sum_{s=1}^{t} \mathbb{E}\big(f(x_s) - f(x^\star)\big) \le \frac{R^2}{2\eta t} + \frac{\eta(L^2+\sigma^2)}{2} = R\sqrt{\frac{L^2+\sigma^2}{t}}. \qquad \square$$

11.2 Smooth convex functions

Theorem 11.3 [SGD, SMOOTH CONVEX]. *Let f be convex and ℓ-smooth, $R = \|x^\star - x_1\|$, and $\eta = \min\{\frac{1}{\ell}, \frac{R}{\sigma\sqrt{2t}}\}$. Then*

$$\mathbb{E} f\Big(\frac{1}{t}\sum_{s=2}^{t+1} x_s\Big) - f(x^\star) \le \underbrace{\frac{\ell R^2}{2t}}_{\text{rate of GD}} + \underbrace{R\sigma\sqrt{\frac{2}{t}}}_{\text{stochastic error}}.$$

When σ is small, convergence has two phases: for small t the GD term dominates and the gap decreases at rate about $O(1/t)$; once t is large the noise term dominates and the rate is about $O(1/\sqrt{t})$.

Proof. Take $\mathcal{X} = \mathbb{R}^d$ and write $\tilde{g}_s = \tilde{g}(x_s)$, with $\mathbb{E}[\tilde{g}_s \mid x_s] = \nabla f(x_s)$ and $\mathbb{E}\|\tilde{g}_s\|^2 \le \mathbb{E}\|\nabla f(x_s)\|^2 + \sigma^2$. Smoothness and $\eta\ell \le 1$ give

$$\mathbb{E}\big[f(x_{s+1}) - f(x_s)\big] \le \mathbb{E}\Big[\langle \nabla f(x_s), x_{s+1} - x_s\rangle + \frac{1}{2\eta}\|x_{s+1} - x_s\|^2\Big] \le -\frac{\eta}{2}\mathbb{E}\|\nabla f(x_s)\|^2 + \frac{\eta\sigma^2}{2}.$$

As in Theorem 11.2, convexity gives

$$\mathbb{E}\big[f(x_s) - f(x^\star)\big] \le \frac{1}{2\eta}\mathbb{E}\big[\|x_s - x^\star\|^2 - \|x_{s+1} - x^\star\|^2\big] + \frac{\eta}{2}\mathbb{E}\|\nabla f(x_s)\|^2 + \frac{\eta}{2}\sigma^2.$$

Adding, the gradient terms cancel:

$$\mathbb{E}\big[f(x_{s+1}) - f(x^\star)\big] \le \frac{1}{2\eta}\mathbb{E}\big[\|x_s - x^\star\|^2 - \|x_{s+1} - x^\star\|^2\big] + \eta\sigma^2.$$

Telescoping over $s = 1, \dots, t$ and applying convexity bounds the left side of the theorem by $\frac{R^2}{2\eta t} + \eta\sigma^2$. With the stated η, $\frac{R^2}{2\eta t} \le \frac{\ell R^2}{2t} + \frac{R\sigma}{\sqrt{2t}}$ and $\eta\sigma^2 \le \frac{R\sigma}{\sqrt{2t}}$. $\square$

11.3 Summary

The strongly convex rows are stated without proof. Setting $\sigma = 0$ recovers the GD results.

Rate	L-Lipschitz	ℓ-smooth
Convex	$\frac{RL}{\sqrt{t}} + \frac{R\sigma}{\sqrt{t}}$	$\frac{\ell R^2}{t} + \frac{R\sigma}{\sqrt{t}}$
α-strongly convex	$\frac{L^2}{\alpha t} + \frac{\sigma^2}{\alpha t}$	$\ell R^2 e^{-t/\kappa} + \frac{\sigma^2}{\alpha t}$
Iterations		
Convex	$\left(\frac{RL}{\epsilon}\right)^2 + \left(\frac{R\sigma}{\epsilon}\right)^2$	$\frac{\ell R^2}{\epsilon} + \left(\frac{R\sigma}{\epsilon}\right)^2$
α-strongly convex	$\frac{L^2}{\alpha\epsilon} + \frac{\sigma^2}{\alpha\epsilon}$	$\kappa \log \frac{\ell R^2}{\epsilon} + \frac{\sigma^2}{\alpha\epsilon}$

11.4 Mini-batch SGD

Mini-batch SGD queries the stochastic oracle m times at the same point and averages. For independent queries this divides the variance by m,

$$\mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m} \tilde{g}_i(x)\right) = \frac{1}{m^2}\sum_{i=1}^{m} \mathrm{Var}(\tilde{g}_i(x)) \le \frac{\sigma^2}{m},$$

at the cost of m oracle calls per iteration.

Algorithm 11.2: Mini-batch SGD

```
for s = 1, ..., t do
    x_{s+1} = x_s − η · (1/m) Σ_{i=1}^{m} g̃_i(x_s)
```

Replacing σ^2 by σ^2/m and multiplying the iteration count by m gives the oracle complexity:

	L-Lipschitz	ℓ-smooth
Convex	$m\left(\frac{RL}{\epsilon}\right)^2 + \left(\frac{R\sigma}{\epsilon}\right)^2$	$m\frac{\ell R^2}{\epsilon} + \left(\frac{R\sigma}{\epsilon}\right)^2$
α-strongly convex	$m\frac{L^2}{\alpha\epsilon} + \frac{\sigma^2}{\alpha\epsilon}$	$m\kappa \log \frac{\ell R^2}{\epsilon} + \frac{\sigma^2}{\alpha\epsilon}$

The oracle complexity of mini-batch SGD is never smaller than that of SGD, but it is within a constant factor when $m \le \sigma^2/L^2$ (Lipschitz convex) or $m \le \sigma^2/(\ell\epsilon)$ (smooth convex). Mini-batch SGD can still be faster in wall-clock time, since the m queries of an iteration can run in parallel.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 11 (Spring 2021), scribed by Qinghua Liu.

Lecture 12: Stochastic Variance Reduced Gradient

12.1 The algorithm

Consider the finite-sum problem

$$\min_x F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x).$$

The stochastic variance reduced gradient (SVRG) method computes a full gradient at an anchor point once per outer loop and corrects cheap stochastic gradients with it.

Algorithm 12.1: SVRG

Input: x_0, step size η, inner loop length m

for $t = 0, 1, 2, \dots$ **do**
- $y_1 = x_t$; compute $\nabla F(y_1)$
- **for** $s = 1, \dots, m$ **do**
  - sample i_s uniformly from $[n]$
  - $y_{s+1} = y_s - \eta\big(\nabla f_{i_s}(y_s) - \nabla f_{i_s}(y_1) + \nabla F(y_1)\big)$
- $x_{t+1} = y_j$ with j uniform on $[m]$

Let $g_s = \nabla f_{i_s}(y_s) - \nabla f_{i_s}(y_1) + \nabla F(y_1)$. It is unbiased, $\mathbb{E}[g_s \mid y_s] = \nabla F(y_s)$, because $\mathbb{E}[\nabla f_{i_s}(y_1)] = \nabla F(y_1)$. When y_s is close to y_1, $\nabla f_{i_s}(y_s) \approx \nabla f_{i_s}(y_1)$, and $\nabla F(y_1)$ has no variance given y_1; in this sense the variance is reduced.

12.2 Convergence

Theorem 12.1 [SVRG]. *Let each f_i be convex and ℓ-smooth and let F be α-strongly convex, with $\kappa = \ell/\alpha$. SVRG with $\eta = \frac{1}{10\ell}$ and $m = 50\kappa$ satisfies*

$$\mathbb{E}\big[F(x_t) - F(x^\star)\big] \le \left(\frac{1}{2}\right)^t \big[F(x_0) - F(x^\star)\big].$$

Lemma 12.2. *If f is convex and ℓ-smooth, then for all x, y,*

$$\frac{1}{2\ell}\|\nabla f(x) - \nabla f(y)\|^2 \le f(y) - f(x) - \langle \nabla f(x), y - x\rangle.$$

Proof. Let $\Delta = \nabla f(y) - \nabla f(x)$ and $z = y - \frac{1}{\ell}\Delta$. Smoothness at y gives $f(z) \le f(y) + \langle \nabla f(y), z -$

$y\rangle + \frac{1}{2\ell}\|\Delta\|^2$, and convexity at x gives $f(z) \ge f(x) + \langle \nabla f(x), z - x\rangle$. Subtracting,

$$0 \le f(y) - f(x) - \langle \nabla f(x), y - x\rangle + \underbrace{\langle \Delta, z - y\rangle}_{=-\frac{1}{\ell}\|\Delta\|^2} + \frac{1}{2\ell}\|\Delta\|^2. \qquad \square$$

Lemma 12.3 [Variance bound]. $\mathbb{E}\|g_s\|^2 \le 4\ell\,\mathbb{E}\big[F(y_s) - F(x^\star) + F(y_1) - F(x^\star)\big]$.

Proof. Since $\nabla F(x^\star) = 0$, write $i = i_s$ and

$$g_s = \big[\nabla f_i(y_s) - \nabla f_i(x^\star)\big] - \big[Z - \mathbb{E}_i Z\big], \qquad Z = \nabla f_i(y_1) - \nabla f_i(x^\star), \quad \mathbb{E}_i Z = \nabla F(y_1) - \nabla F(x^\star).$$

By $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ and $\mathbb{E}\|Z - \mathbb{E}Z\|^2 \le \mathbb{E}\|Z\|^2$,

$$\mathbb{E}\|g_s\|^2 \le 2\mathbb{E}\|\nabla f_i(y_s) - \nabla f_i(x^\star)\|^2 + 2\mathbb{E}\|\nabla f_i(y_1) - \nabla f_i(x^\star)\|^2.$$

Lemma 12.2 applied to each f_i and averaged over i gives $\mathbb{E}_i\|\nabla f_i(x) - \nabla f_i(x^\star)\|^2 \le 2\ell\big(F(x) - F(x^\star) - \langle \nabla F(x^\star), x - x^\star\rangle\big) = 2\ell\big(F(x) - F(x^\star)\big)$. $\square$

Proof of Theorem 12.1. It suffices to show $\mathbb{E}[F(x_{t+1}) - F(x^\star)] \le \frac{1}{2}\mathbb{E}[F(x_t) - F(x^\star)]$ for one outer loop. By unbiasedness, convexity, and Lemma 12.3,

$$\begin{aligned}
\mathbb{E}\|y_{s+1} - x^\star\|^2 &= \mathbb{E}\big[\|y_s - x^\star\|^2 - 2\eta\langle \nabla F(y_s), y_s - x^\star\rangle + \eta^2\|g_s\|^2\big] \\
&\le \mathbb{E}\|y_s - x^\star\|^2 - 2\eta\mathbb{E}\big[F(y_s) - F(x^\star)\big] + 4\eta^2\ell\,\mathbb{E}\big[F(y_s) - F(x^\star) + F(y_1) - F(x^\star)\big] \\
&= \mathbb{E}\|y_s - x^\star\|^2 - 2\eta(1 - 2\eta\ell)\mathbb{E}\big[F(y_s) - F(x^\star)\big] + 4\eta^2\ell\,\mathbb{E}\big[F(y_1) - F(x^\star)\big].
\end{aligned}$$

Summing over $s = 1, \dots, m$,

$$2\eta(1 - 2\eta\ell)\sum_{s=1}^{m}\mathbb{E}\big[F(y_s) - F(x^\star)\big] \le \mathbb{E}\|y_1 - x^\star\|^2 + 4m\eta^2\ell\,\mathbb{E}\big[F(y_1) - F(x^\star)\big].$$

Since x_{t+1} is a uniformly random y_s, $y_1 = x_t$, and strong convexity gives $\|y_1 - x^\star\|^2 \le \frac{2}{\alpha}(F(y_1) - F(x^\star))$,

$$\mathbb{E}\big[F(x_{t+1}) - F(x^\star)\big] = \frac{1}{m}\sum_{s=1}^{m}\mathbb{E}\big[F(y_s) - F(x^\star)\big] \le \frac{1}{2m\eta(1 - 2\eta\ell)}\Big(\frac{2}{\alpha} + 4m\eta^2\ell\Big)\mathbb{E}\big[F(x_t) - F(x^\star)\big].$$

With $\eta = \frac{1}{10\ell}$ and $m = 50\kappa$, $2m\eta(1 - 2\eta\ell) = \frac{8}{\alpha}$, so the factor is $\frac{1}{4} + \frac{2\eta\ell}{1 - 2\eta\ell} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. $\square$

12.3 Complexity

With $\epsilon_0 = F(x_0) - F(x^\star)$, accuracy ϵ takes $\log \frac{\epsilon_0}{\epsilon}$ outer loops, each computing $n + 2m = O(n + \kappa)$ stochastic gradients, so SVRG needs

$$O\left((n + \kappa) \log \frac{\epsilon_0}{\epsilon}\right)$$

component gradients. GD needs $O(n\tilde{\kappa} \log \frac{\epsilon_0}{\epsilon})$, where $\tilde{\kappa} = \tilde{\ell}/\alpha$ uses the smoothness $\tilde{\ell} \le \ell$ of F itself, so $\tilde{\kappa} \le \kappa$. Unless the two condition numbers differ greatly, SVRG is a significant improvement, especially when both n and κ are large.

12.4 Optimal rates

For a stochastic gradient oracle with $\mathbb{E}\|\tilde{g}(x) - \nabla f(x)\|^2 \le \sigma^2$, the optimal oracle complexities are:

	Lipschitz	smooth
Convex	$\frac{R^2(L^2 + \sigma^2)}{\epsilon^2}$	$\frac{R^2\sigma^2}{\epsilon^2} + \sqrt{\frac{\ell R^2}{\epsilon}}$
Strongly convex	$\frac{L^2 + \sigma^2}{\alpha\epsilon}$	$\frac{\sigma^2}{\alpha\epsilon} + \sqrt{\kappa} \log \frac{\ell R^2}{\epsilon}$

Achieving the optimal smooth rates requires quite complicated algorithms. For finite sums $F = \frac{1}{n}\sum_i f_i$, with the assumptions on the f_i and strong convexity on F:

	Lipschitz (f_i)	smooth (f_i)
Convex	$\min\left\{\frac{R^2L^2}{\epsilon^2},\ n + \sqrt{n}\frac{LR}{\epsilon}\right\}$	$n + \sqrt{\frac{n\ell R^2}{\epsilon}}$
Strongly convex (F)	$\min\left\{\frac{L^2}{\alpha\epsilon},\ n + \sqrt{n}\frac{L}{\sqrt{\alpha\epsilon}}\right\}$	$(n + \sqrt{n\kappa}) \log \frac{\epsilon_0}{\epsilon}$

The smooth strongly convex optimal rate is achieved by the Catalyst framework and by Katyusha acceleration.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 12 (Spring 2021), scribed by Joseph Kim.

Lecture 13: Nonconvex Optimization Overview, Power Methods

Strong guarantees hold in the convex setting, but many problems in practice are nonconvex: deep learning, representation learning, and even reinforcement learning.

13.1 Difficulty of nonconvex optimization

A convex function has a single local minimum, which is global; a nonconvex function may have many local minima, which makes the global minimum hard to find. In the worst case, finding it takes exponentially many queries even for Lipschitz or smooth f.

Theorem 13.1 [Hardness of global minimization]. *For every $\ell > 0$ and every algorithm A, there is an ℓ-smooth f on $[-1,1]^d$ such that A needs at least $(\frac{\ell}{4\epsilon})^{\Omega(d)}$ gradient queries to output, with probability at least $\frac{1}{2}$, a point $\hat{x}$ with $f(\hat{x}) < f(x^\star) + \epsilon$, where $x^\star$ is the global minimizer.*

Proof. Grid $[-1,1]^d$ into blocks of side r. Choose one block at random to hold a bump of depth ϵ (the global minimum $-\epsilon$) and make $f \equiv 0$ on all other blocks (Figure 4). Then $f(\hat{x}) < f(x^\star) + \epsilon$ forces $\hat{x}$ into the special block, and queries outside it reveal nothing, so finding it takes $(\frac{2}{r})^{\Omega(d)}$ queries. For ℓ-smoothness, build the bump from quadratic pieces of curvature $\pm\ell$, each covering a depth $\frac{\epsilon}{2}$ over a distance $\frac{r}{4}$: $\frac{\ell(r/4)^2}{2} = \frac{\epsilon}{2}$, so $r = 4\sqrt{\epsilon/\ell}$ and the count is $(\frac{\ell}{4\epsilon})^{\Omega(d)}$. □

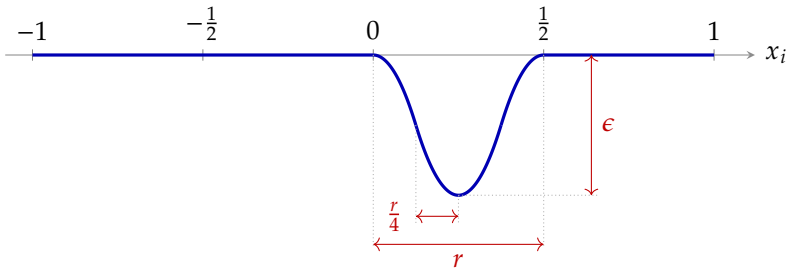


Figure 4: One coordinate of the hard instance with blocks of side $r = \frac{1}{2}$: zero on all blocks except one, where a C^1 bump of depth ϵ is built from a concave and a convex quadratic piece of curvature ℓ (here $\ell = 64$, $\epsilon = 1$).

Definition 13.2 [Stationary points]. For differentiable f, x is a stationary point if $\nabla f(x) = 0$. A stationary point x is

- a local minimum if there is $\delta > 0$ with $f(y) \geq f(x)$ for all $\|y - x\| \leq \delta$;
- a local maximum if there is $\delta > 0$ with $f(y) \leq f(x)$ for all $\|y - x\| \leq \delta$;
- a saddle point otherwise.

Stationarity is a necessary condition for a local minimum.

13.2 The eigenvector problem and the power method

The eigenvector problem is a special case that gives insight into general nonconvex optimization, and it is an important subroutine for large nonconvex problems. Given a positive semidefinite (symmetric) $A \in \mathbb{R}^{d\times d}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ and unit eigenvectors $v_1, \ldots, v_d$ ($Av_i = \lambda_i v_i$), find the top eigenvector v_1.

Algorithm 13.1: Power method

Input: PSD matrix A, number of steps t

draw $\tilde{w}_0 \sim N(0, I_d)$ and set $w_0 = \tilde{w}_0/\|\tilde{w}_0\|$

for $s = 0, \ldots, t-1$ **do**

  $\tilde{w}_{s+1} = Aw_s; \quad w_{s+1} = \tilde{w}_{s+1}/\|\tilde{w}_{s+1}\|$

Equivalently, $w_t = \bar{w}_t/\|\bar{w}_t\|$ with $\bar{w}_t = A^t w_0$.

Theorem 13.3 [Gap-dependent rate]. *If A is positive semidefinite with $\lambda_1 > \lambda_2$, the power method satisfies*

$$\langle v_1, w_t\rangle^2 \geq 1 - \Big(1 - \frac{\lambda_1 - \lambda_2}{\lambda_1}\Big)^{2t} \frac{1}{\langle v_1, w_0\rangle^2}.$$

Proof. Let P_1 project onto $\mathrm{span}(v_1)$ and $P_1^\perp$ onto its orthogonal complement. Then

$$1 - \langle v_1, w_t\rangle^2 = \|P_1^\perp w_t\|^2 = \frac{\|P_1^\perp \bar{w}_t\|^2}{\|\bar{w}_t\|^2} \leq \frac{\|P_1^\perp \bar{w}_t\|^2}{\|P_1 \bar{w}_t\|^2}.$$

Write $w_0 = \sum_i \alpha_i v_i$ with $\alpha_i = \langle v_i, w_0\rangle$ and $\sum_i \alpha_i^2 = 1$. Then $\bar{w}_t = \sum_i \lambda_i^t \alpha_i v_i$, so $\|P_1^\perp \bar{w}_t\|^2 = \sum_{i\geq 2} \lambda_i^{2t}\alpha_i^2 \leq \lambda_2^{2t}$ and $\|P_1\bar{w}_t\|^2 = \lambda_1^{2t}\alpha_1^2$. Hence $1 - \langle v_1, w_t\rangle^2 \leq (\lambda_2/\lambda_1)^{2t}/\alpha_1^2$. □

Corollary 13.4. *$\langle v_1, w_t\rangle^2 \geq 1 - \epsilon$ once $t \geq \frac{\lambda_1}{2(\lambda_1-\lambda_2)} \log \frac{1}{\epsilon\langle v_1, w_0\rangle^2}$, that is, after $O\big(\frac{\lambda_1}{\lambda_1-\lambda_2} \log \frac{1}{\epsilon\langle v_1, w_0\rangle^2}\big)$ steps.*

The guarantee depends on the gap between the top two eigenvalues and degrades when they are close; if $\lambda_1 = \lambda_2$ the power method may return any combination of v_1 and v_2. It also needs an initialization w_0 that is not too close to orthogonal to v_1, so that $\langle v_1, w_0\rangle$ is not too small.

13.3 Gap-free analysis

Definition 13.5 [Rayleigh quotient]. For symmetric A and $w \neq 0$, the Rayleigh quotient is $R(A, w) = \frac{w^T A w}{\|w\|^2}$.

Measuring progress by the Rayleigh quotient instead of the angle to v_1 removes the dependence on

the gap.

Theorem 13.6 [Gap-free rate]. *If A is positive semidefinite with top eigenvalue λ_1, the power method satisfies $w_t^T A w_t \geq (1-\epsilon)\lambda_1$ once $t \geq \frac{1}{\epsilon}\log\frac{2}{\epsilon\langle v_1, w_0\rangle^2}$, that is, after $O\big(\frac{1}{\epsilon}\log\frac{1}{\epsilon\langle v_1, w_0\rangle^2}\big)$ steps.*

Proof. Let P project onto the span of the eigenvectors with eigenvalue at least $(1-\frac{\epsilon}{2})\lambda_1$. Since $A \succeq 0$, if $\|Pw_t\|^2 \geq 1-\frac{\epsilon}{2}$ then

$$w_t^T A w_t \geq \Big(1-\frac{\epsilon}{2}\Big)\lambda_1\|Pw_t\|^2 \geq \Big(1-\frac{\epsilon}{2}\Big)^2\lambda_1 \geq (1-\epsilon)\lambda_1.$$

As in Theorem 13.3, the remaining eigenvalues are below $(1-\frac{\epsilon}{2})\lambda_1$, so

$$1-\|Pw_t\|^2 = \|P^\perp w_t\|^2 \leq \frac{\|P^\perp \bar{w}_t\|^2}{\|P_1\bar{w}_t\|^2} \leq \frac{\big[(1-\frac{\epsilon}{2})\lambda_1\big]^{2t}}{\lambda_1^{2t}\langle v_1, w_0\rangle^2} \leq \frac{e^{-\epsilon t}}{\langle v_1, w_0\rangle^2},$$

which is at most $\frac{\epsilon}{2}$ for the stated t. □

13.4 Geometry of the eigenvector problem

The top eigenvector solves

$$\max_w \; w^T A w \qquad \text{s.t. } \|w\|^2 \leq 1.$$

The objective and the constraint are both convex, but maximizing a convex function is a nonconvex problem, which makes this a simple nonconvex problem to study. Normalized gradient ascent with $\eta = \frac{1}{\|A\|} = \frac{1}{\lambda_1}$,

$$\tilde{w}_{s+1} = w_s + \eta A w_s, \qquad w_{s+1} = \frac{\tilde{w}_{s+1}}{\|\tilde{w}_{s+1}\|},$$

gives $w_t \propto (I + \frac{A}{\lambda_1})^t w_0$, the power method on $I + \frac{A}{\lambda_1}$. Its top two eigenvalues are 2 and $1 + \frac{\lambda_2}{\lambda_1}$, so the rate is $\tilde{O}\big(\frac{2}{2-(1+\lambda_2/\lambda_1)}\big) = \tilde{O}\big(\frac{2\lambda_1}{\lambda_1-\lambda_2}\big)$ iterations, about the same as the power method.

Example: For $A \in \mathbb{R}^{3\times 3}$ on the unit sphere, the Rayleigh quotient has stationary points $\pm v_1, \pm v_2, \pm v_3$. The point v_1 is a local (and global) maximum and v_3 a local (and global) minimum. The point v_2 is a saddle point: the objective increases along $v_2\cos\theta + v_1\sin\theta$ and decreases along $v_2\cos\theta + v_3\sin\theta$.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 13 (Spring 2021), scribed by Jack Ji.

Lecture 14: Lanczos Algorithm

The Lanczos algorithm improves the power method as AGD improves GD. With $\lambda_1 > \lambda_2$ the top eigenvalues of a positive semidefinite A, the iteration complexities for accuracy ϵ are:

	gap-dependent	gap-free
Power method	$\frac{\lambda_1}{\lambda_1 - \lambda_2} \log \frac{d}{\epsilon}$	$\frac{1}{\epsilon} \log \frac{d}{\epsilon}$
Lanczos	$\sqrt{\frac{\lambda_1}{\lambda_1 - \lambda_2}} \log \frac{d}{\epsilon}$	$\frac{1}{\sqrt{\epsilon}} \log \frac{d}{\epsilon}$

For intuition, consider $\max_{\|x\|^2 \le 1} x^T A x$ on the sphere with coordinates $v_1, v_2, \ldots$: moving away from v_1 decreases the objective at least quadratically, with curvature of order $\lambda_1 - \lambda_2$. The gap-dependent problem is then like maximizing a locally strongly concave function with condition number about $\frac{\lambda_1}{\lambda_1 - \lambda_2}$, and the gap-free problem like maximizing a locally concave one.

14.1 Krylov subspaces

The power method with $b = w_1$ has $w_t \propto A^{t-1} b$, so w_t lies in the Krylov subspace

$$K(A, b, t) = \text{span}\{b, Ab, A^2 b, \ldots, A^{t-1} b\}.$$

Assume these vectors are linearly independent, which is easy to satisfy when A has full rank and b is uniformly random on the sphere $\mathbb{S}^{d-1}$. Instead of the specific vector $A^{t-1} b$, the Lanczos algorithm takes the best vector of the subspace:

$$w_t = \operatorname*{arg\,max}_{w \in K(A,b,t)} \frac{w^T A w}{\|w\|^2}.$$

Three questions follow: how to compute this argmax fast, what it guarantees, and how it relates to AGD.

14.2 Fast computation by tridiagonalization

Let $q_1, \ldots, q_t$ be an orthonormal basis built incrementally, so that $q_1, \ldots, q_k$ spans $K(A, b, k)$ for each k, and let $Q_t = [q_1, \ldots, q_t] \in \mathbb{R}^{d \times t}$. (By orthogonality, $q_i \neq A^{i-1} b$ in general.)

Lemma 14.1 [Tridiagonal structure]. *$T = Q_t^T A Q_t$ is tridiagonal.*

Proof. T is symmetric, so it suffices to show $T_{ij} = q_i^T A q_j = 0$ for $i > j+1$. Since $q_j \in K(A,b,j)$, we have $Aq_j \in K(A,b,j+1) \subseteq K(A,b,i-1)$, and $q_i \perp \operatorname{span}\{q_1,\dots,q_{i-1}\} = K(A,b,i-1)$. □

Writing $w = Q_t v$ and using $Q_t^T Q_t = I$,

$$\max_{w \in K(A,b,t)} \frac{w^T A w}{\|w\|^2} = \max_{v \in \mathbb{R}^t} \frac{v^T Q_t^T A Q_t v}{v^T Q_t^T Q_t v} = \max_{v \in \mathbb{R}^t} \frac{v^T T v}{\|v\|^2}.$$

So if (μ_1, u_1) is the top eigenpair of T, then $w_t = Q_t u_1$ and $w_t^T A w_t = \mu_1$. The top eigenpair of a $k \times k$ tridiagonal matrix costs $O(k^2)$ with standard algorithms and $O(k \log k)$ with carefully designed ones, a lower-order term next to the matrix-vector products Aw.

It remains to compute Q_t and T. Since T is symmetric tridiagonal, only T_{ii} and $T_{(i-1)i} = T_{i(i-1)}$ are needed. Gram–Schmidt applied to the new vector Aq_t needs only the last two basis vectors, because $q_i^T A q_t = T_{it} = 0$ for $i < t-1$:

$$\tilde{q}_{t+1} = \Big(I - \sum_{i=1}^{t} q_i q_i^T\Big) A q_t = A q_t - T_{tt} q_t - T_{(t-1)t} q_{t-1}, \qquad q_{t+1} = \frac{\tilde{q}_{t+1}}{\|\tilde{q}_{t+1}\|}.$$

Algorithm 14.1: Lanczos algorithm

Input: PSD matrix A, number of steps m
draw q_1 uniformly from $\mathbb{S}^{d-1}$; $T_{11} = q_1^T A q_1$
for $k = 1, \dots, m-1$ **do**
  $\tilde{q}_{k+1} = Aq_k - T_{kk} q_k - T_{(k-1)k} q_{k-1}$ (with $q_0 = 0$)
  $q_{k+1} = \tilde{q}_{k+1} / \|\tilde{q}_{k+1}\|$
  $T_{(k+1)k} = T_{k(k+1)} = q_{k+1}^T A q_k$; $T_{(k+1)(k+1)} = q_{k+1}^T A q_{k+1}$
compute the top eigenpair (μ_1, u_1) of the tridiagonal $T \in \mathbb{R}^{m \times m}$
return $(\mu_1,\ Q_m u_1)$ as an approximation of (λ_1, v_1)

14.3 Guarantee via Chebyshev polynomials

Definition 14.2 [Chebyshev polynomials]. The Chebyshev polynomials are $T_0(x) = 1$, $T_1(x) = x$, and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$. Equivalently,

$$T_n(x) = \begin{cases} \cos(n \arccos x), & |x| \le 1, \\ \cosh(n \operatorname{arccosh} x), & x \ge 1, \\ (-1)^n \cosh(n \operatorname{arccosh}(-x)), & x \le -1, \end{cases}$$

and for $|x| \ge 1$, $T_n(x) = \frac{1}{2}\big[(x - \sqrt{x^2-1})^n + (x + \sqrt{x^2-1})^n\big]$.

So $|T_n(x)| \le 1$ on $[-1,1]$, while $|T_n|$ grows rapidly outside.

Theorem 14.3 [Lanczos]. *Let A be positive semidefinite with $\lambda_1 > \lambda_2$, and let $g = \frac{\lambda_1 - \lambda_2}{\lambda_1}$. After t steps the output $(\mu_1^{(t)}, u_1^{(t)})$ satisfies*

$$\lambda_1 \geq \mu_1^{(t)} \geq \Big[1 - \frac{4(1-\sqrt{g})^{2(t-1)}}{\langle v_1, q_1\rangle^2}\Big]\lambda_1, \qquad \langle v_1, Q_t u_1^{(t)}\rangle^2 \geq 1 - \frac{4(1-\sqrt{g})^{2(t-1)}}{\langle v_1, q_1\rangle^2} \cdot \frac{\lambda_1}{\lambda_1 - \lambda_2},$$

so $O\big(\sqrt{\frac{\lambda_1}{\lambda_1-\lambda_2}}\log\frac{d}{\epsilon}\big)$ iterations suffice when $\langle v_1, q_1\rangle^2 \gtrsim 1/d$.

Proof. $\lambda_1 \geq \mu_1^{(t)}$ because λ_1 maximizes the Rayleigh quotient over all of $\mathbb{R}^d$. Every $w \in K(A, b, t)$ with $b = q_1$ is $w = p(A)b$ for a polynomial p of degree at most $t-1$. With $b = \sum_i \alpha_i v_i$,

$$\mu_1^{(t)} = \max_p \frac{b^T p(A) A p(A) b}{b^T p(A)^2 b} = \max_p \frac{\sum_i \alpha_i^2 p(\lambda_i)^2 \lambda_i}{\sum_i \alpha_i^2 p(\lambda_i)^2} \geq \max_p \frac{\alpha_1^2 p(\lambda_1)^2 \lambda_1}{\sum_i \alpha_i^2 p(\lambda_i)^2}$$
$$= \max_p \lambda_1 \Big[1 - \frac{\sum_{i\geq 2} \alpha_i^2 p(\lambda_i)^2}{\sum_i \alpha_i^2 p(\lambda_i)^2}\Big] \geq \max_p \lambda_1 \Big[1 - \frac{\max_{i\geq 2} p(\lambda_i)^2}{\alpha_1^2 p(\lambda_1)^2}\Big],$$

and any specific polynomial gives a lower bound. The power method uses $p(z) = z^{t-1}$. Lanczos achieves the Chebyshev choice $p(\lambda) = T_{t-1}(\lambda/\lambda_2)$, which is small on $[0, \lambda_2] \ni \lambda_2, \ldots, \lambda_d$ and large at λ_1: there $\max_{i\geq 2} p(\lambda_i)^2 \leq 1$, while $x = \lambda_1/\lambda_2 = \frac{1}{1-g}$ satisfies

$$x + \sqrt{x^2 - 1} = \frac{1 + \sqrt{2g - g^2}}{1 - g} \geq \frac{1 + \sqrt{g}}{1 - g} = \frac{1}{1 - \sqrt{g}},$$

so $p(\lambda_1) \geq \frac{1}{2}(1-\sqrt{g})^{-(t-1)}$ and $\frac{1}{p(\lambda_1)^2} \leq 4(1-\sqrt{g})^{2(t-1)}$. For the eigenvector, a unit w with $c_1 = \langle v_1, w\rangle$ has $w^T A w \leq c_1^2 \lambda_1 + (1 - c_1^2)\lambda_2$, so $1 - c_1^2 \leq \frac{\lambda_1 - w^T A w}{\lambda_1 - \lambda_2}$; apply this to $w = Q_t u_1^{(t)}$. □

14.4 Connection to AGD

Consider the quadratic $f(x) = \frac{1}{2}x^T A x - b^T x$ with $\alpha I \preceq A \preceq \ell I$, so f is ℓ-smooth and α-strongly convex with $\nabla f(x) = Ax - b$. A linear-span algorithm such as GD or AGD has $x_t \in K(A, b, t)$, so $x_t = q_{t-1}(A)b$ for a polynomial q_{t-1} of degree $t-1$, and

$$\|\nabla f(x_t)\| = \|Ax_t - b\| = \|(I - A q_{t-1}(A))b\| = \|p_t(A)b\|,$$

where $p_t(z) = 1 - z q_{t-1}(z)$ has degree t and $p_t(0) = 1$. For GD, $\|\nabla f(x_t)\|$ and $\|x_t - x^\star\|$ decay like $e^{-t/\kappa}$. A better polynomial gives a better rate, since convergence is governed by $|p_t(\lambda_i)|$ on the eigenvalues.

Lemma 14.4 [Shifted Chebyshev polynomial]. *For every $\epsilon > 0$ there is a polynomial p of degree $O(\sqrt{\ell/\alpha}\log\frac{1}{\epsilon})$ with $p(0) = 1$ and $|p(z)| \leq \epsilon$ for all $z \in [\alpha, \ell]$.*

Proof. Take $p(z) = T_k\big(\frac{\ell+\alpha-2z}{\ell-\alpha}\big)/T_k\big(\frac{\ell+\alpha}{\ell-\alpha}\big)$, so $p(0) = 1$. On $[\alpha, \ell]$ the argument lies in $[-1, 1]$, so the numerator is at most 1 in absolute value. With $x = \frac{\kappa+1}{\kappa-1}$, $x + \sqrt{x^2-1} = \frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}$, so the denominator is at least $\frac{1}{2}(\frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1})^k$ and $|p(z)| \le 2e^{-2k/(\sqrt{\kappa}+1)} \le \epsilon$ for $k \ge \frac{\sqrt{\kappa}+1}{2}\log\frac{2}{\epsilon}$. □

The degree $O(\sqrt{\kappa}\log\frac{1}{\epsilon})$ matches the rate of AGD.

14.5 Initialization

Proposition 14.5 [Random initialization]. *If w_0 is uniform on $\mathbb{S}^{d-1}$, then for any fixed direction v_1, with probability at least $1-\delta$,*

$$\langle v_1, w_0\rangle^2 \ge \Omega\Big(\frac{\delta^2}{d + \sqrt{d}\log\frac{1}{\delta}}\Big).$$

Proof sketch. Write $\langle v_1, w_0\rangle^2 = \langle v_1, z_0\rangle^2/\|z_0\|^2$ with $z_0 \sim N(0, I)$. The numerator is the square of a one-dimensional Gaussian, and the denominator is a sum of d sub-exponential variables. □

Plugging this bound into the guarantees for the power method and Lanczos gives the $\log\frac{d}{\epsilon}$ factors in the table above.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 14 (Spring 2021), scribed by Mona Ye; continued in Lecture 15, scribed by Hui Yuan.

Lecture 15: Finding Stationary Points

We now turn to smooth, not necessarily convex, unconstrained problems $\min_{x\in\mathbb{R}^d} f(x)$, where f is ℓ-gradient Lipschitz (ℓ-smooth): $\|\nabla f(x_1) - \nabla f(x_2)\| \le \ell\|x_1 - x_2\|$ for all $x_1, x_2 \in \mathbb{R}^d$.

15.1 First-order stationary points

Without convexity, finding a global minimum can be exponentially hard even for smooth functions (Lecture 13), so we settle for a surrogate.

Definition 15.1 [ϵ-stationary point]. x is a stationary point of a differentiable f if $\|\nabla f(x)\| = 0$, and an ϵ-stationary point if $\|\nabla f(x)\| \le \epsilon$.

Theorem 15.2 [GD finds stationary points]. *If f is ℓ-gradient Lipschitz, then for every $\epsilon > 0$, GD with $\eta = \frac{1}{\ell}$ visits an ϵ-stationary point within*

$$\frac{2\ell\big(f(x_0) - f^\star\big)}{\epsilon^2}$$

iterations, where x_0 is the initialization and $f^\star$ the optimal value.

Proof. Smoothness with $x_{t+1} = x_t - \frac{1}{\ell}\nabla f(x_t)$ gives, without convexity,

$$f(x_{t+1}) \le f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t\rangle + \frac{\ell}{2}\|x_{t+1} - x_t\|^2 = f(x_t) - \frac{1}{2\ell}\|\nabla f(x_t)\|^2.$$

If no iterate among $x_0, \dots, x_{T-1}$ is ϵ-stationary, each step decreases f by more than $\frac{\epsilon^2}{2\ell}$, so $f^\star \le f(x_T) < f(x_0) - \frac{T\epsilon^2}{2\ell}$ and $T < \frac{2\ell(f(x_0)-f^\star)}{\epsilon^2}$. □

This rate is optimal for smooth functions.

15.2 Second-order stationary points

A stationary point may be a local minimum, a local maximum, or a saddle point (Lecture 13), so first-order stationarity alone is not satisfactory. Finding a local minimum is NP-hard in general, because of high-order stationary points (later lectures), but it has useful relaxations. Stationarity is the first-order necessary condition for a local minimum; adding the second-order necessary condition gives the following.

Definition 15.3 [Second-order stationary point]. For twice differentiable f, x is a second-order

stationary point (SOSP) if $\nabla f(x) = 0$ and $\nabla^2 f(x) \succeq 0$. It is an (ϵ_g, ϵ_H)-SOSP if $\|\nabla f(x)\| \le \epsilon_g$ and $\nabla^2 f(x) \succeq -\epsilon_H I$.

Finding second-order stationary points needs a higher-order assumption.

Definition 15.4 [Hessian Lipschitz]. A twice differentiable $f : \mathbb{R}^d \to \mathbb{R}$ is ρ-Hessian Lipschitz if $\|\nabla^2 f(x_1) - \nabla^2 f(x_2)\| \le \rho \|x_1 - x_2\|$ for all $x_1, x_2 \in \mathbb{R}^d$, in the spectral norm.

By Taylor's theorem this gives the cubic bound

$$\left| f(x_2) - f(x_1) - \langle \nabla f(x_1), x_2 - x_1 \rangle - \frac{1}{2}(x_2 - x_1)^T \nabla^2 f(x_1)(x_2 - x_1) \right| \le \frac{\rho}{6}\|x_1 - x_2\|^3.$$

The two error parameters are usually tied together.

Definition 15.5 [ϵ-SOSP]. For ρ-Hessian Lipschitz f, x is an ϵ-SOSP if $\|\nabla f(x)\| \le \epsilon$ and $\nabla^2 f(x) \succeq -\sqrt{\rho\epsilon}\, I$.

Algorithm 15.1: Cubic regularization

Input: initialization x_0

for $t = 0, 1, \ldots, T-1$ **do**

  $x_{t+1} = \arg\min_x \left\{ \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2}(x - x_t)^T \nabla^2 f(x_t)(x - x_t) + \frac{\rho}{6}\|x - x_t\|^3 \right\}$

Each step minimizes the second-order Taylor model plus a cubic penalty, which by the cubic bound is an upper bound on $f(x) - f(x_t)$.

Theorem 15.6 [Cubic regularization]. *If f is ρ-Hessian Lipschitz, cubic regularization visits an ϵ-SOSP within*

$$O\left(\frac{\sqrt{\rho}\,(f(x_0) - f^\star)}{\epsilon^{1.5}} \right)$$

iterations.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 15 (Spring 2021), scribed by Hui Yuan.

Lecture 16: Escaping Saddle Points

16.1 Preliminaries

Throughout, $f \in C^2(\mathbb{R}^d)$ is ℓ-gradient Lipschitz and ρ-Hessian Lipschitz, and the target is an ϵ-SOSP: $\|\nabla f(x)\| \le \epsilon$ and $\nabla^2 f(x) \succeq -\sqrt{\rho\epsilon}\, I$ (Lecture 15). A point x is *not* an ϵ-SOSP exactly when one of the following holds:

1. $\|\nabla f(x)\| > \epsilon$ (large gradient);
2. $\|\nabla f(x)\| \le \epsilon$ and $\lambda_{\min}(\nabla^2 f(x)) < -\sqrt{\rho\epsilon}$ (large negative curvature).

The second case approximates a strict saddle point, where $\nabla f(x) = 0$ and $\lambda_{\min}(\nabla^2 f(x)) < 0$.

The general scheme has two steps. First, guarantee that the function value decreases at every iterate that is not an ϵ-SOSP: by the descent lemma in case 1, and by moving approximately along the direction of most negative curvature in case 2. Second, conclude that an ϵ-SOSP must appear within a bounded number of iterations, since otherwise the function value would drop below its minimum.

16.2 Gradient descent with the power method

The simplest method runs GD in the large-gradient regime and, at an approximate saddle point, steps along the direction of most negative curvature, the bottom eigenvector of $\nabla^2 f(x)$. For $\eta \le 1/\ell$ the matrix $I - \eta\nabla^2 f(x)$ is positive semidefinite (the eigenvalues of $\nabla^2 f(x)$ lie in $[-\ell, \ell]$), and its top eigenvector is the bottom eigenvector of $\nabla^2 f(x)$, so the power method applies. Each power iteration needs only a Hessian-vector product, which two gradients approximate:

$$\nabla^2 f(x_t)\, v \approx \frac{\nabla f(x_t + \lambda v) - \nabla f(x_t)}{\lambda} \qquad (\lambda \to 0).$$

This is prone to numerical problems, but it shows that the method technically needs only a gradient oracle.

Algorithm 16.1: GD with the power method

Input: steps T, step size η, step length λ, power iterations $\mathcal{T}$, initial point x_1

for $t = 1, \dots, T$ **do**
  if $\|\nabla f(x_t)\| > \epsilon$ **then**
    $x_{t+1} = x_t - \eta \nabla f(x_t)$
  else
    $v_t \leftarrow \textsc{PowerMethod}\big(I - \eta \nabla^2 f(x_t), \mathcal{T}\big)$
    $x_{t+1} = \arg\min_{z \in \{x_t + \lambda v_t,\, x_t - \lambda v_t\}} f(z)$
return x_{T+1}

Theorem 16.1 [GD with the power method]. *For any $\epsilon, \delta > 0$, with $\eta = 1/\ell$, $\mathcal{T} = \tilde{\Theta}(\ell/\sqrt{\rho\epsilon})$, and $\lambda = \sqrt{\epsilon/\rho}$, with probability at least $1 - \delta$ one of the iterates is an ϵ-SOSP within $\tilde{O}\big(\ell(f(x_0) - f^\star)/\epsilon^2\big)$ gradient queries.*

The $\tilde{O}$ hides a factor logarithmic in d. So finding an ϵ-SOSP costs at most a logarithmic factor more gradient queries than finding an ϵ-stationary point.

Proof. Suppose no iterate is an ϵ-SOSP; each is in case 1 or case 2.
Case 1. The descent lemma gives $f(x_{t+1}) \le f(x_t) - \frac{\eta}{2}\|\nabla f(x_t)\|^2 \le f(x_t) - \frac{\epsilon^2}{2\ell}$.
Case 2. By the gap-free analysis of the power method (Lectures 13 and 14), for a PSD matrix A it returns w with $\langle w, Aw\rangle \ge (1 - \epsilon')\lambda_1(A)$ with probability $1 - \delta$ after $O(\frac{1}{\epsilon'} \log \frac{d}{\epsilon'\delta})$ iterations. Let $-\gamma = \lambda_{\min}(\nabla^2 f(x_t)) < -\sqrt{\rho\epsilon}$, so that $\lambda_1(I - \eta\nabla^2 f(x_t)) = 1 + \gamma/\ell$. With $\epsilon' = \gamma/(4\ell)$,

$$v_t^T (I - \eta \nabla^2 f(x_t)) v_t \ge (1 - \epsilon')\Big(1 + \frac{\gamma}{\ell}\Big) \ge 1 + \frac{\gamma}{\ell} - 2\epsilon' = 1 + \frac{\gamma}{2\ell},$$

that is, $v_t^T \nabla^2 f(x_t) v_t \le -\gamma/2 < -\sqrt{\rho\epsilon}/2$. This takes

$$\mathcal{T} = O\Big(\frac{\ell}{\gamma} \log \frac{\ell d}{\gamma \delta}\Big) \le O\Big(\frac{\ell}{\sqrt{\rho\epsilon}} \log \frac{\ell d}{\delta\sqrt{\rho\epsilon}}\Big) = \tilde{O}\Big(\frac{\ell}{\sqrt{\rho\epsilon}}\Big)$$

power iterations. Let $\tilde{x} \in \{x_t \pm \lambda v_t\}$ minimize $\langle \nabla f(x_t), x - x_t\rangle$, so this inner product is at most 0. By the cubic bound (Lecture 15) and $\|v_t\| = 1$,

$$\begin{aligned} f(x_{t+1}) \le f(\tilde{x}) &\le f(x_t) + \langle \nabla f(x_t), \tilde{x} - x_t\rangle + \frac{\lambda^2}{2} v_t^T \nabla^2 f(x_t) v_t + \frac{\rho}{6}\lambda^3 \\ &\le f(x_t) - \frac{\sqrt{\rho\epsilon}}{4}\lambda^2 + \frac{\rho}{6}\lambda^3 = f(x_t) - \frac{1}{12}\sqrt{\frac{\epsilon^3}{\rho}} \end{aligned}$$

for $\lambda = \sqrt{\epsilon/\rho}$. This decrease costs $\tilde{\Theta}(\ell/\sqrt{\rho\epsilon})$ queries, an average of $\tilde{\Omega}(\epsilon^2/\ell)$ per query, matching case 1 up to logarithmic factors.
Hence if no ϵ-SOSP appeared within $\tilde{O}(\ell(f(x_0) - f^\star)/\epsilon^2)$ queries, the function value would decrease by more than $f(x_0) - f^\star$, a contradiction. □

16.3 Intuition for the scaling

Nonconvex problems have little structure; only the smoothness conditions can be used, and each controls a ball in which the local picture is reliable.

- *Gradient Lipschitz.* If $\|\nabla f(x)\| \geq \epsilon$, then $\|\nabla f(y)\| \geq \epsilon/2$ on the ball of radius $R = O(\epsilon/\ell)$ around x. Nothing is known outside it, so a step should stay inside, and it decreases f by about $R \cdot \epsilon = O(\epsilon^2/\ell)$.
- *Hessian Lipschitz.* If $\lambda_{\min}(\nabla^2 f(x)) < -\sqrt{\rho\epsilon}$ with small gradient, then $\lambda_{\min}(\nabla^2 f(y)) < -\sqrt{\rho\epsilon}/2$ on the ball of radius $R = O(\sqrt{\rho\epsilon}/\rho) = O(\sqrt{\epsilon/\rho})$, and a step decreases f by about $\Omega(R^2\sqrt{\rho\epsilon}) = \Omega(\sqrt{\epsilon^3/\rho})$.

16.4 Perturbed gradient descent

Switching between two algorithms is not clean and can be noisy. The much simpler PERTURBED GRADIENT DESCENT (PGD) adds noise to escape saddle points where plain GD cannot:

$$x_{t+1} = x_t - \eta\big(\nabla f(x_t) + \zeta_t\big), \qquad \zeta_t \sim N\Big(0, \frac{r^2}{d} I\Big), \quad r = \tilde{\Theta}(\epsilon).$$

Its analysis is beyond the course; Lecture 17 analyzes a more deterministic variant that perturbs only at small gradients.

Algorithm 16.2: Perturbed gradient descent (variant)

Input: steps T, step size η, radius r, interval $\mathcal{T}$, initial point x_1

for $t = 1, \dots, T$ **do**
  if $\|\nabla f(x_t)\| \leq \epsilon$ **and** no perturbation was added in the last $\mathcal{T}$ steps **then**
    $x_t \leftarrow x_t + \eta\xi_t$, where $\xi_t \sim \text{Uniform}(B_0(r))$
  $x_{t+1} = x_t - \eta\nabla f(x_t)$

Theorem 16.2 [PGD]. *For any $\epsilon, \delta > 0$, with $\eta = 1/\ell$, $r = \tilde{\Theta}(\epsilon)$, and $\mathcal{T} = \tilde{\Theta}(\ell/\sqrt{\rho\epsilon})$, PGD run for $\tilde{O}\big(\ell(f(x_0) - f^\star)/\epsilon^2\big)$ iterations visits an ϵ-SOSP with probability at least $1 - \delta$.*

The logarithmic dependence is slightly worse than for GD with the power method ($\log^4$ versus $\log$), but the polynomial dependence is the same.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 16 (Spring 2021), scribed by Sabarish Sainathan.

Lecture 17: Escaping Saddle Points II

This lecture proves that perturbed gradient descent (Algorithm 16.2), which adds a random perturbation when the gradient is small, escapes saddle points efficiently.

Theorem 17.1 [PGD]. *Let f be ℓ-gradient Lipschitz and ρ-Hessian Lipschitz. For any $\epsilon > 0$, PGD with $\eta = 1/\ell$, $r = \tilde{\Theta}(\epsilon)$, and $\mathcal{T} = \tilde{\Theta}(\ell/\sqrt{\rho\epsilon})$ finds an ϵ-SOSP in $\tilde{O}\big(\ell(f(x_0) - f^\star)/\epsilon^2\big)$ iterations with high probability.*

In $\mathcal{T}$, the numerator ℓ bounds $\|\nabla^2 f(x)\|$ and the denominator $\sqrt{\rho\epsilon}$ is the tolerance on the minimum eigenvalue. PGD thus finds second-order stationary points in almost the same time as GD finds first-order ones.

Proof strategy. If no iterate is an ϵ-SOSP, each has a large gradient or is an approximate saddle point.

1. If $\|\nabla f(x_t)\| > \epsilon$, then $f(x_{t+1}) - f(x_t) \le -\Omega(\epsilon^2/\ell)$, by the descent lemma.
2. If $\|\nabla f(x_t)\| \le \epsilon$ and $\lambda_{\min}(\nabla^2 f(x_t)) < -\sqrt{\rho\epsilon}$, then with high probability $f(x_{t+\mathcal{T}}) - f(x_t) \le -\tilde{\Omega}(\mathcal{T}\epsilon^2/\ell) = -\tilde{\Omega}(\sqrt{\epsilon^3/\rho})$.

Either way the function value decreases by $\tilde{\Omega}(\epsilon^2/\ell)$ per step on average. It cannot go below $f^\star$, so after $\tilde{O}(\ell(f(x_0) - f^\star)/\epsilon^2)$ steps some iterate is an ϵ-SOSP. It remains to prove the second claim.

17.1 Saddle point escaping

Lemma 17.2 [Saddle point escaping]. *In the setting of Theorem 17.1, let $\|\nabla f(\tilde{x})\| \le \epsilon$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \le -\sqrt{\rho\epsilon}$, let $x_0 = \tilde{x} + \eta\xi$ with $\xi \sim \mathrm{Uniform}(B_0(r))$, and run GD from x_0. With probability at least $1 - \delta$,*

$$f(x_{\mathcal{T}}) - f(\tilde{x}) \le -\tilde{\Omega}(\mathcal{T}\epsilon^2/\ell) = -\tilde{\Omega}\big(\sqrt{\epsilon^3/\rho}\big).$$

The subroutine of GD with the power method has the same $\mathcal{T}$ and the same decrease, but the power method uses one fixed Hessian $\nabla^2 f(\tilde{x})$, whereas the Hessian changes along GD, and the negative curvature need not persist after a few steps. The analysis therefore localizes, in two steps:

1. *Improve or localize:* if GD makes little progress for many iterations, all those iterates stay in a small ball.
2. *Stuck probability is small:* with significant negative curvature, after a random perturbation GD is unlikely to stay in a small ball for long.

17.2 Improve or localize

Lemma 17.3 [Improve or localize]. *If f is ℓ-gradient Lipschitz and GD runs with $\eta = 1/\ell$, then for all $t \geq \tau > 0$,*

$$\|x_\tau - x_0\| \leq \sqrt{2\eta t\big(f(x_0) - f(x_t)\big)}.$$

Proof. By the triangle inequality, Cauchy–Schwarz, the update, and the descent lemma $f(x_s) - f(x_{s-1}) \leq -\frac{\eta}{2}\|\nabla f(x_{s-1})\|^2$,

$$\begin{aligned}\|x_\tau - x_0\| &\leq \sum_{s=1}^{\tau}\|x_s - x_{s-1}\| \leq \Big(t\sum_{s=1}^{\tau}\|x_s - x_{s-1}\|^2\Big)^{1/2} \\ &\leq \Big(\eta^2 t\sum_{s=1}^{t}\|\nabla f(x_{s-1})\|^2\Big)^{1/2} \leq \sqrt{2\eta t\big(f(x_0) - f(x_t)\big)}.\end{aligned}$$

□

So if the decrease is small, all iterates stay close to x_0. In particular, if $f(x_{\mathcal{T}}) - f(x_0) \geq -\tilde{O}(\mathcal{T}\epsilon^2/\ell)$, then $\max_{\tau\leq\mathcal{T}}\|x_\tau - x_0\| \leq \tilde{O}\big(\sqrt{2\eta\mathcal{T}\cdot\mathcal{T}\epsilon^2/\ell}\big) = \tilde{O}(\epsilon\mathcal{T}/\ell) = \tilde{O}(\sqrt{\epsilon/\rho})$.

17.3 Stuck probability is small near saddle points

Here $x_0 \sim \mathrm{Uniform}(B_{\tilde{x}}(\eta r))$. Define the stuck region

$$\mathcal{X}_{\text{stuck}} = \big\{x \in B_{\tilde{x}}(\eta r) :\ \text{GD from } x_0 = x \text{ has } \|x_t - x_0\| \leq \tilde{O}(\epsilon\mathcal{T}/\ell) \text{ for all } t \in [\mathcal{T}]\big\}.$$

If $x_0 \notin \mathcal{X}_{\text{stuck}}$, Lemma 17.3 forces a large decrease, and smoothness bounds the cost of the perturbation:

$$\begin{aligned}f(x_{\mathcal{T}}) - f(\tilde{x}) &= f(x_{\mathcal{T}}) - f(x_0) + f(x_0) - f(\tilde{x}) \\ &\leq -\tilde{\Omega}\big(\sqrt{\epsilon^3/\rho}\big) + \tilde{O}\big(\epsilon\cdot\eta r + \ell(\eta r)^2\big) \leq -\tilde{\Omega}\big(\sqrt{\epsilon^3/\rho}\big),\end{aligned}$$

because with $r = O(\epsilon)$ and $\eta = 1/\ell$, $\tilde{O}(\epsilon\eta r + \ell(\eta r)^2) = \tilde{O}(\epsilon^2/\ell) = \tilde{O}\big(\sqrt{\epsilon^3/\rho}\cdot\sqrt{\rho\epsilon}/\ell\big)$ and $\sqrt{\rho\epsilon}/\ell \leq 1$. To prove Lemma 17.2 it therefore suffices to show

$$\mathbb{P}(x_0 \in \mathcal{X}_{\text{stuck}}) = \frac{\mathrm{vol}(\mathcal{X}_{\text{stuck}})}{\mathrm{vol}(B_{\tilde{x}}(\eta r))} \leq \delta.$$

The shape of $\mathcal{X}_{\text{stuck}}$ is complicated, but it is thin (Figure 5). If every line in the direction e_1 meets it in a segment of length at most ηw, then

$$\mathbb{P}(x_0 \in \mathcal{X}_{\text{stuck}}) \leq \frac{\eta w\cdot\mathrm{vol}(B_0^{d-1}(\eta r))}{\mathrm{vol}(B_{\tilde{x}}^d(\eta r))} = \frac{w\,\Gamma(d/2+1)}{r\sqrt{\pi}\,\Gamma(d/2+1/2)} \leq \frac{w}{r}\sqrt{\frac{d}{\pi}},$$

so width $w \le O(\delta r/\sqrt{d})$ suffices.

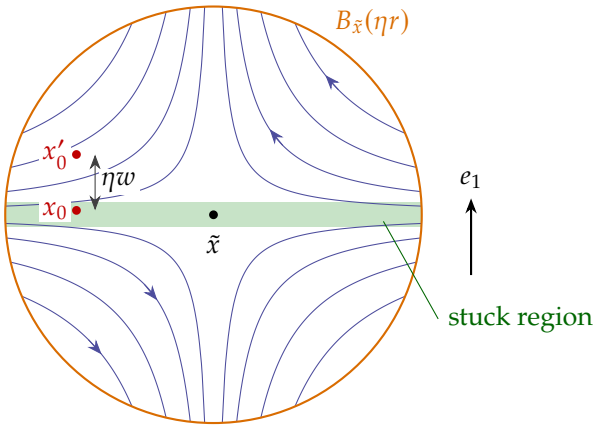


Figure 5: The perturbation ball $B_{\tilde{x}}(\eta r)$ around a saddle point, with gradient-flow lines of a model saddle, and the thin stuck region (shaded) around the directions of nonnegative curvature. Of two starting points separated by ηw along the negative curvature direction e_1, at least one escapes. Schematic.

The width is bounded by coupling two GD sequences $\{x_t\}$ and $\{x_t'\}$ with

1. $\max\{\|x_0 - \tilde{x}\|, \|x_0' - \tilde{x}\|\} \le \eta r$;
2. $x_0 - x_0' = \eta w e_1$, where e_1 is the bottom eigenvector of $\nabla^2 f(\tilde{x})$ and $w \ge w_0$ for a threshold w_0.

Lemma 17.4 [Coupling]. *For any $w_0 \in (0, \epsilon]$ and coupling sequences as above, if $\mathcal{T} \ge \Omega(\kappa \log \frac{\epsilon\kappa}{w_0})$ with $\kappa = \ell/\sqrt{\rho\epsilon}$, then for some $t \in [\mathcal{T}]$,*

$$\max\{\|x_t - x_0\|, \|x_t' - x_0'\|\} \ge \tilde{\Omega}(\epsilon\mathcal{T}/\ell).$$

So of any two starting points separated by at least ηw_0 along e_1, at least one lies outside $\mathcal{X}_{\text{stuck}}$: the width of the stuck region along e_1 is at most ηw_0, and w_0 can be made arbitrarily small at the price of only logarithmic factors in $\mathcal{T}$.

Proof. Suppose instead that both sequences stay within $\tilde{O}(\epsilon\mathcal{T}/\ell)$ of their starting points for all $t \in [\mathcal{T}]$. Let $\hat{x}_t = x_t - x_t'$, $H = \nabla^2 f(\tilde{x})$, $-\gamma = \lambda_{\min}(H) \le -\sqrt{\rho\epsilon}$, and

$$\Delta_t = \int_0^1 \left[\nabla^2 f\left(x_t' + \theta(x_t - x_t')\right) - H\right] d\theta.$$

By Taylor expansion,

$$\hat{x}_{t+1} = \hat{x}_t - \eta\left[\nabla f(x_t) - \nabla f(x_t')\right] = (I - \eta H)\hat{x}_t - \eta\Delta_t\hat{x}_t = \underbrace{(I - \eta H)^{t+1}\hat{x}_0}_{p(t+1)} - \underbrace{\eta\sum_{\tau=0}^{t}(I - \eta H)^{t-\tau}\Delta_\tau\hat{x}_\tau}_{q(t+1)}.$$

Since $\hat{x}_0 = \eta w e_1$, $p(t) = (1+\eta\gamma)^t \eta w e_1$ grows exponentially: $\|p(t)\| \geq (1+\sqrt{\rho\epsilon}/\ell)^t \eta w_0$. We show by induction that

$$\|q(t)\| \leq \frac{\|p(t)\|}{2} \qquad \text{for all } t \in [\mathcal{T}]. \tag{17.1}$$

Then $\|\hat{x}_{\mathcal{T}}\| \geq \|p(\mathcal{T})\|/2 \geq \tilde{\Omega}(\epsilon\mathcal{T}/\ell) = \tilde{\Omega}(\sqrt{\epsilon/\rho})$ once $\mathcal{T} \geq \Omega(\kappa \log \frac{\epsilon\kappa}{w_0})$, contradicting that both sequences stay in balls of radius $\tilde{O}(\epsilon\mathcal{T}/\ell)$.

For (17.1), $q(0) = 0$. Assume it up to t. Then for $\tau \leq t$, $\|\hat{x}_\tau\| \leq \|p(\tau)\| + \|q(\tau)\| \leq 2\|p(\tau)\| = 2(1+\eta\gamma)^\tau \eta w$. Hessian Lipschitzness and localization give $\|\Delta_t\| \leq \rho \max\{\|x_t - \tilde{x}\|, \|x'_t - \tilde{x}\|\} \leq \tilde{O}(\rho\epsilon\mathcal{T}/\ell) = \tilde{O}(\sqrt{\rho\epsilon})$, and $\|(I-\eta H)^{t-\tau}\| \leq (1+\eta\gamma)^{t-\tau}$. Hence

$$\begin{aligned}\|q(t+1)\| &\leq \eta \sum_{\tau=0}^{t} \|\Delta_\tau\| \|(I-\eta H)^{t-\tau}\| \|\hat{x}_\tau\| \leq \tilde{O}(\eta\sqrt{\rho\epsilon}) \sum_{\tau=0}^{t} (1+\eta\gamma)^t \eta w \\ &\leq \tilde{O}(\eta\sqrt{\rho\epsilon}\,\mathcal{T})(1+\eta\gamma)^t \eta w \leq \tilde{O}(1)\|p(t+1)\|.\end{aligned}$$

Choosing the constants and logarithmic factors carefully makes the $\tilde{O}(1)$ at most $\frac{1}{2}$. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 17 (Spring 2021), scribed by Yaqi Duan.

Lecture 18: Examples, Optimal Rates, High-Order Saddle Points

For unconstrained nonconvex problems we can efficiently find ϵ-SOSPs, approximations of points with $\nabla f(x) = 0$ and $\nabla^2 f(x) \succeq 0$. Is an SOSP what we want in practice? It is a global minimum whenever

- all local minima are global minima, and
- all saddle points are strict: $\lambda_{\min}(\nabla^2 f(x)) \le -\gamma$ for some $\gamma > 0$.

18.1 The top eigenvector problem

Let M be positive semidefinite with eigenvalues $\lambda_1 > \lambda_2 \ge \cdots \ge \lambda_d$ and eigenvectors $v_1, \dots, v_d$, and consider

$$\min_{x \in \mathbb{R}^d} f(x) = \frac{1}{2}\|xx^T - M\|_F^2,$$

whose global minima are $x^\star = \pm\sqrt{\lambda_1}\, v_1$. Expanding $f(x) = \frac{1}{2}\big(\|x\|^4 - 2x^T M x + \|M\|_F^2\big)$,

$$\nabla f(x) = 2\big(\|x\|^2 x - Mx\big), \qquad \nabla^2 f(x) = 2\big(\|x\|^2 I + 2xx^T - M\big).$$

Landscape. A stationary point satisfies $Mx = \|x\|^2 x$, so $x = 0$ or x is an eigenvector with eigenvalue $\|x\|^2$:

$$x = 0 \qquad \text{or} \qquad x = \pm\sqrt{\lambda_i}\, v_i \quad \text{for some } i.$$

Every stationary point other than $\pm\sqrt{\lambda_1}v_1$ is a strict saddle. At $\tilde{x} = \pm\sqrt{\lambda_i}v_i$ with $i > 1$, which is orthogonal to v_1,

$$v_1^T \nabla^2 f(\tilde{x}) v_1 = 2\big(\|\tilde{x}\|^2\|v_1\|^2 + 2(\tilde{x}^T v_1)^2 - v_1^T M v_1\big) = 2(\lambda_i - \lambda_1) < 0,$$

and at $x = 0$, $v_1^T \nabla^2 f(0) v_1 = -2\lambda_1 < 0$. So all saddle points are strict and all local minima are global. A robust version also holds: every ϵ-SOSP lies close to one of the global minima.

Smoothness. The algorithms also need the smoothness conditions. Since f is a polynomial, it is gradient Lipschitz and Hessian Lipschitz on any bounded set, and if $\|x_0\|$ is not too large, the GD iterates $\|x_t\|$ stay bounded, so the conditions hold where they are needed.

This is the general recipe for many nonconvex problems: to show that an algorithm finds a global minimum, it suffices to do landscape analysis, showing that all local minima are global and all saddle points are strict, and then apply the algorithmic results.

18.2 More problems where SOSPs are global minima

Example: *Matrix factorization.* For symmetric $M \in \mathbb{R}^{d\times d}$, $\min_{U\in\mathbb{R}^{d\times r}} \frac{1}{2}\|UU^T - M\|_F^2$ has a geometry very similar to the top eigenvector problem. For asymmetric $M \in \mathbb{R}^{d_1\times d_2}$,

$$\min_{U\in\mathbb{R}^{d_1\times r},\, V\in\mathbb{R}^{d_2\times r}} \frac{1}{2}\|UV^T - M\|_F^2 + \underbrace{\frac{1}{8}\|U^TU - V^TV\|_F^2}_{R(U,V)}.$$

The penalty R is needed because if $(U^\star, V^\star)$ minimizes $\frac{1}{2}\|UV^T - M\|_F^2$, so does $(U^\star R, V^\star(R^{-1})^T)$ for every invertible $R \in \mathbb{R}^{r\times r}$; $R(U, V)$ balances the two factors.

Example: *Matrix sensing.* Estimate a symmetric $M^\star$ of rank r from measurements $b_i = \langle A_i, M^\star\rangle = \mathrm{tr}(A_i^T M^\star)$, $i = 1, \dots, m$, by

$$\min_{U\in\mathbb{R}^{d\times r}} f(U) = \frac{1}{2m}\sum_{i=1}^m \big(\langle A_i, UU^T\rangle - b_i\big)^2 = \frac{1}{2m}\sum_{i=1}^m \langle A_i, UU^T - M^\star\rangle^2,$$

where $UU^T - M^\star$ has rank at most $2r$.

Definition 18.1 [Restricted isometry property]. The measurements $A_1, \dots, A_m$ satisfy the (r, δ)-RIP if for every symmetric M of rank at most r,

$$(1-\delta)\|M\|_F^2 \le \frac{1}{m}\sum_{i=1}^m \langle A_i, M\rangle^2 \le (1+\delta)\|M\|_F^2.$$

Under $(2r, \delta)$-RIP with small enough δ, the sensing objective is close to $\frac{1}{2}\|UU^T - M^\star\|_F^2$, so, as for matrix factorization, all SOSPs are global minima.

Example: *Matrix completion.* A rank-r matrix $M^\star$ is observed only on a set Ω of revealed entries, and

$$\min_{U\in\mathbb{R}^{d\times r}} f(U) = \frac{1}{2|\Omega|}\sum_{(i,j)\in\Omega} \big(e_i^T UU^T e_j - M^\star_{ij}\big)^2.$$

This is matrix sensing with the measurements $A_k = e_i e_j^T$, $k = 1, \dots, |\Omega|$.

18.3 Optimal rates

Gradient-query complexities for nonconvex optimization:

Condition	Target	Upper bound	Lower bound	Gap
ℓ-gradient Lipschitz	ϵ-FOSP	$O(\epsilon^{-2})$	$\Omega(\epsilon^{-2})$	$O(1)$
ℓ-gradient Lipschitz, ρ-Hessian Lipschitz	ϵ-FOSP and ϵ-SOSP	$\tilde{O}(\epsilon^{-7/4})$	$\Omega(\epsilon^{-12/7})$	$\tilde{O}(\epsilon^{-1/28})$
plus Lipschitz of order ≥ 3	ϵ-FOSP	$\tilde{O}(\epsilon^{-5/3})$	$\Omega(\epsilon^{-8/5})$	$\tilde{O}(\epsilon^{-1/15})$

GD achieves the first upper bound. In the second row, GD with noise or with the power method achieves $\tilde{O}(\epsilon^{-2})$, and AGD with noise or with Lanczos improves this to $\tilde{O}(\epsilon^{-7/4})$.

Intuition for AGD with Lanczos. If $\nabla^2 f \preceq -\sqrt{\rho\epsilon}\, I$ on a ball of radius $r = \sqrt{\epsilon/\rho}$, a negative curvature step decreases f by $\sqrt{\rho\epsilon}\, r^2 = \sqrt{\epsilon^3/\rho} \sim \epsilon^{1.5}$, so $\epsilon^{-1.5}$ such steps exhaust the decrease available. Finding the direction takes $\epsilon^{-0.5}$ power iterations, giving $\epsilon^{-1.5-0.5} = \epsilon^{-2}$ for GD with the power method, but only $\epsilon^{-0.25}$ Lanczos iterations, giving $\epsilon^{-1.75}$ for AGD with Lanczos. Closing the gaps in the last two rows and escaping high-order saddle points remain open problems.

18.4 Higher-order stationary points

Definition 18.2 [pTH-ORDER STATIONARY POINT]. x is a pTH-ORDER STATIONARY POINT of f if there is $\delta > 0$ such that $f(y) \geq f(x) - o(\|x - y\|^p)$ for all $\|y - x\| \leq \delta$.

Theorem 18.3 [HARDNESS FOR FOURTH ORDER]. *It is NP-hard to decide whether a point is a fourth-order stationary point.*

Proof sketch. Let $f(x) = \sum_{i,j=1}^{d} Q_{ij} x_i^2 x_j^2$. It is homogeneous of degree 4, $f(tx) = t^4 f(x)$, so 0 is a fourth-order stationary point if and only if $f(x) \geq 0$ for all x. With $u_i = x_i^2$, this says $u^T Q u \geq 0$ for all entrywise nonnegative u: Q is COPOSITIVE. Testing copositivity is as hard as INDEPENDENT SET, which is NP-hard. □

The hardness concerns computational complexity, not query complexity. It also shows that finding an SOSP under convex constraints is NP-hard: for $\min x^T Q x$ subject to $x_i \geq 0$ for all i, testing whether 0 is an SOSP or a strict saddle point is NP-hard. For $p = 3$, by contrast, a third-order stationary point can be found in polynomial time.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 18 (Spring 2021), scribed by Yihong Gu.

Lecture 19: Stochastic Nonconvex Optimization, SPIDER

19.1 Setup

Stochastic nonconvex optimization minimizes

$$\min_{x\in\mathbb{R}^d} f(x) = \mathbb{E}\big[F(x;\xi)\big]$$

over a random vector ξ; when ξ takes finitely many values this is the finite sum $f(x) = \frac{1}{n}\sum_{i=1}^n f_i(x)$. Assume:

1. bounded variance: $\mathbb{E}\|\nabla F(x;\xi) - \nabla f(x)\|^2 \le \sigma^2$;
2. mean-squared smoothness: $\mathbb{E}\|\nabla F(x;\xi) - \nabla F(y;\xi)\|^2 \le \ell^2\|x-y\|^2$ for all x, y, which implies $\|\nabla f(x) - \nabla f(y)\| \le \ell\|x-y\|$.

From x_0 with $f(x_0) - \inf_x f(x) \le \Delta$, the goal is an ϵ-FOSP, $\|\nabla f(x)\| \le \epsilon$, or an ϵ-SOSP, additionally with $\nabla^2 f(x) \succeq -\sqrt{\rho\epsilon}\, I$.

19.2 Warm-up: SGD

Algorithm 19.1: SGD for nonconvex objectives

```
for t = 0, ..., T − 1 do
    x_{t+1} = x_t − η_t ∇F(x_t; ξ_t)
return x̃ chosen uniformly at random from {x_t}_{t=0}^{T−1}
```

Theorem 19.1 [SGD]. *Under assumption 1 and ℓ-smoothness of f, with $\eta_t = \eta = \frac{\epsilon^2}{2\sigma^2\ell} \wedge \frac{1}{\ell}$ and $T = \max\big\{\frac{8\Delta\sigma^2\ell}{\epsilon^4}, \frac{4\Delta\ell}{\epsilon^2}\big\}$, SGD satisfies $\mathbb{E}\|\nabla f(\tilde{x})\|^2 \le \epsilon^2$.*

Proof. By smoothness, $f(x_{t+1}) \le f(x_t) - \eta\langle \nabla f(x_t), \nabla F(x_t;\xi_t)\rangle + \frac{\ell\eta^2}{2}\|\nabla F(x_t;\xi_t)\|^2$. Taking expectations and using $\mathbb{E}\|\nabla F\|^2 \le \mathbb{E}\|\nabla f\|^2 + \sigma^2$ and $\ell\eta \le 1$,

$$\mathbb{E}f(x_{t+1}) \le \mathbb{E}f(x_t) - \eta\Big(1 - \frac{\ell\eta}{2}\Big)\mathbb{E}\|\nabla f(x_t)\|^2 + \frac{\ell\eta^2\sigma^2}{2} \le \mathbb{E}f(x_t) - \frac{\eta}{2}\mathbb{E}\|\nabla f(x_t)\|^2 + \frac{\ell\eta^2\sigma^2}{2}.$$

Summing over $t = 0, \dots, T-1$,

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\|\nabla f(x_t)\|^2 \le \frac{2}{T\eta}\big(f(x_0) - \mathbb{E} f(x_T)\big) + \ell\eta\sigma^2 \le \frac{2\Delta}{T\eta} + \ell\eta\sigma^2,$$

and each term is at most $\frac{\epsilon^2}{2}$ for the stated η and T. □

So SGD needs $O(\epsilon^{-4})$ stochastic gradients. SVRG needs $O(\epsilon^{-10/3})$, and SPIDER improves this to $O(\epsilon^{-3})$.

19.3 SPIDER

The stochastic path-integrated differential estimator (SPIDER) tracks ∇f along the path of iterates. For a sample set B write $\nabla f_B(x) = \frac{1}{|B|}\sum_{\xi\in B}\nabla F(x;\xi)$.

Algorithm 19.2: SPIDER

Input: batch sizes S_1, S_2, epoch length q, steps T

for $t = 0, \dots, T-1$ **do**
  if $t \bmod q = 0$ **then**
    draw S_1 samples B_t; $v_t = \nabla f_{B_t}(x_t)$
  else
    draw S_2 samples B_t; $v_t = \nabla f_{B_t}(x_t) - \nabla f_{B_t}(x_{t-1}) + v_{t-1}$
  $x_{t+1} = x_t - \eta_t v_t$ with $\eta_t = \min\big(\frac{\epsilon}{\ell\|v_t\|}, \frac{1}{2\ell}\big)$
return $\tilde{x}$ chosen uniformly at random from $\{x_t\}_{t=0}^{T-1}$

The step size makes $\|x_{t+1} - x_t\| = \eta_t\|v_t\| \le \epsilon/\ell$.

Theorem 19.2 [SPIDER]. *Under assumptions 1 and 2, with $S_1 = \frac{2\sigma^2}{\epsilon^2}$, $S_2 = \frac{2\sigma}{\epsilon}$, $q = \frac{\sigma}{\epsilon}$, and $T = \frac{4\ell\Delta}{\epsilon^2} + 1$, SPIDER satisfies $\mathbb{E}\|\nabla f(\tilde{x})\| \le 5\epsilon$, using $\frac{T}{q}S_1 + 2TS_2 = O(\ell\Delta\sigma\epsilon^{-3})$ stochastic gradients.*

Lemma 19.3 [Descent]. *For every t, $f(x_{t+1}) \le f(x_t) - \frac{\epsilon\|v_t\|}{4\ell} + \frac{\epsilon^2}{2\ell} + \frac{\eta_t}{2}\|v_t - \nabla f(x_t)\|^2$.*

Proof. By smoothness and $-\langle a, b\rangle \le \frac{1}{2}\|a\|^2 + \frac{1}{2}\|b\|^2$,

$$\begin{aligned} f(x_{t+1}) &\le f(x_t) - \eta_t\langle \nabla f(x_t), v_t\rangle + \frac{\ell\eta_t^2}{2}\|v_t\|^2 = f(x_t) - \eta_t\Big(1 - \frac{\eta_t\ell}{2}\Big)\|v_t\|^2 - \eta_t\langle \nabla f(x_t) - v_t, v_t\rangle \\ &\le f(x_t) - \eta_t\Big(\frac{1}{2} - \frac{\eta_t\ell}{2}\Big)\|v_t\|^2 + \frac{\eta_t}{2}\|\nabla f(x_t) - v_t\|^2. \end{aligned}$$

Since $\eta_t\ell \le \frac{1}{2}$, the coefficient is at least $\frac{\eta_t}{4}$, and with $a = \|v_t\|/\epsilon$,

$$\frac{\eta_t}{4}\|v_t\|^2 = \frac{\epsilon^2}{8\ell}\min(2a, a^2) \ge \frac{\epsilon^2}{8\ell}(2a - 4) = \frac{\epsilon\|v_t\| - 2\epsilon^2}{4\ell},$$

because $a^2 - 2a + 4 > 0$. □

Lemma 19.4 [Estimator error]. *Under assumptions 1 and 2, $\mathbb{E}\|v_t - \nabla f(x_t)\|^2 \le \epsilon^2$ for all t.*

This variance control, impossible for SGD, is the point of SPIDER.

Proof. Let $k = \lfloor t/q \rfloor q$ be the start of the epoch. At $t = k$, $\mathbb{E}\|v_k - \nabla f(x_k)\|^2 \le \sigma^2/S_1 = \epsilon^2/2$. For $t \ne k$,

$$v_t - \nabla f(x_t) = \underbrace{\nabla f_{B_t}(x_t) - \nabla f_{B_t}(x_{t-1}) - \big(\nabla f(x_t) - \nabla f(x_{t-1})\big)}_{\text{mean zero given the past}} + \big(v_{t-1} - \nabla f(x_{t-1})\big),$$

so the cross term vanishes. The first term averages S_2 independent mean-zero terms, so by assumption 2 its second moment is at most $\frac{\ell^2}{S_2}\|x_t - x_{t-1}\|^2$. Unrolling to the start of the epoch and using $\|x_{\iota+1} - x_\iota\| \le \epsilon/\ell$,

$$\mathbb{E}\|v_t - \nabla f(x_t)\|^2 \le \sum_{\iota=k}^{t-1} \frac{\ell^2}{S_2}\mathbb{E}\|x_{\iota+1} - x_\iota\|^2 + \mathbb{E}\|v_k - \nabla f(x_k)\|^2 \le \frac{q\ell^2}{S_2} \cdot \frac{\epsilon^2}{\ell^2} + \frac{\epsilon^2}{2} = \epsilon^2. \qquad \square$$

Lemma 19.5. $\mathbb{E}\|\nabla f(x_t)\| \le \mathbb{E}\|v_t\| + \epsilon$.

Indeed, by Jensen's inequality and Lemma 19.4, $\mathbb{E}\|v_t - \nabla f(x_t)\| \le \epsilon$; apply the triangle inequality.

Proof of Theorem 19.2. Lemmas 19.3 and 19.4 with $\eta_t \le \frac{1}{2\ell}$ give

$$\mathbb{E}\big[f(x_{t+1}) - f(x_t)\big] \le -\frac{\epsilon}{4\ell}\mathbb{E}\|v_t\| + \frac{3\epsilon^2}{4\ell}.$$

Summing over $t = 0, \dots, T-1$ gives $\frac{1}{T}\sum_t \mathbb{E}\|v_t\| \le \frac{4\ell\Delta}{\epsilon T} + 3\epsilon \le 4\epsilon$ for the stated T, and Lemma 19.5 gives $\mathbb{E}\|\nabla f(\tilde{x})\| = \frac{1}{T}\sum_t \mathbb{E}\|\nabla f(x_t)\| \le 5\epsilon$. □

19.4 Summary

Stochastic gradient complexities for an ϵ-FOSP:

	SGD	SVRG	SPIDER
Online	ϵ^{-4}	$\epsilon^{-10/3}$	ϵ^{-3} (optimal)
Finite sum	$\epsilon^{-4} \wedge n\epsilon^{-2}$	$n + n^{2/3}\epsilon^{-2}$	$n + n^{1/2}\epsilon^{-2}$ (optimal)

Source. Cong Fang (guest lecturer), *ELE539/COS512: Optimization for Machine Learning*, Lecture 19 (Spring 2021), scribed by Ashwinee Panda.

Lecture 20: Nonsmooth Nonconvex Optimization

Smooth problems rely on gradient Lipschitzness; for nonsmooth problems L-Lipschitzness is the natural assumption. The nonsmooth nonconvex setting is challenging for three reasons.

1. Defining a "gradient" is hard. For $-|x|$ at $x = 0$, no linear function touching the graph at 0 lies below it, so the convex definition of a subgradient fails.
2. Tracking progress can be hard, even for Lipschitz convex functions.
3. The set of approximate stationary points can have measure zero. For $|x|$, the subgradient is 1 for $x > 0$, -1 for $x < 0$, and $[-1, 1]$ at $x = 0$, so no point other than 0 has a small subgradient.

20.1 Weakly convex functions

Definition 20.1 [Weak convexity]. f is ℓ-weakly convex if $f(\cdot) + \frac{\ell}{2}\|\cdot\|^2$ is convex.

For example, $|x|$ is weakly convex but $-|x|$ is not: sharp lower corners are allowed, sharp upper corners are not.

Example: For $\min_x \max_y f(x, y) = \min_x \phi(x)$ with f ℓ-smooth in (x, y), the function $\phi(x) = \max_y f(x, y)$ is ℓ-weakly convex: each $f(\cdot, y) + \frac{\ell}{2}\|\cdot\|^2$ is convex, and so is their maximum.

Definition 20.2 [Subgradient of a weakly convex function]. For ℓ-weakly convex f, v is a subgradient of f at x if $v + \ell x \in \partial\left[f + \frac{\ell}{2}\|\cdot\|^2\right](x)$; the subdifferential is $\partial f(x) = \partial\left[f + \frac{\ell}{2}\|\cdot\|^2\right](x) - \ell x$.

Proposition 20.3 [Characterizations]. *The following are equivalent:*

1. *f is ℓ-weakly convex;*
2. *$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y) + \frac{\ell\lambda(1-\lambda)}{2}\|x - y\|^2$ for all x, y and $\lambda \in [0, 1]$;*
3. *$f(y) \ge f(x) + \langle g_x, y - x\rangle - \frac{\ell}{2}\|y - x\|^2$ for all x, y and $g_x \in \partial f(x)$;*
4. *when f is twice differentiable, $\nabla^2 f(x) \succeq -\ell I$ for all x.*

Proof. Let $h = f + \frac{\ell}{2}\|\cdot\|^2$, which is convex exactly when f is ℓ-weakly convex. Item 2 is the convexity inequality for h, rewritten with $\lambda\|x\|^2 + (1-\lambda)\|y\|^2 - \|\lambda x + (1-\lambda)y\|^2 = \lambda(1-\lambda)\|x - y\|^2$. Item 3 is the subgradient inequality $h(y) \ge h(x) + \langle g_x + \ell x, y - x\rangle$, rewritten with $\frac{\ell}{2}\|x\|^2 - \frac{\ell}{2}\|y\|^2 + \ell\langle x, y - x\rangle = -\frac{\ell}{2}\|y - x\|^2$. Item 4 is $\nabla^2 h \succeq 0$. □

So ℓ-weak convexity requires the Hessian eigenvalues to be at least $-\ell$, while ℓ-smoothness requires them to lie in $[-\ell, \ell]$.

The algorithm is SUBGRADIENT DESCENT, $x_{t+1} = x_t - \eta g_t$ with $g_t \in \partial f(x_t)$.

20.2 Tracking progress: the Moreau envelope

Definition 20.4 [MOREAU ENVELOPE]. The MOREAU ENVELOPE of f with parameter λ is

$$\varphi_\lambda(x) = \min_z \Big(f(z) + \frac{1}{2\lambda}\|z - x\|^2 \Big),$$

and $\hat{x}$ denotes the minimizer.

If f is ℓ-weakly convex and $0 < \lambda < \frac{1}{\ell}$, the objective is strongly convex in z, so $\hat{x}$ is unique. The envelope is a lower bound, $\varphi_\lambda(x) \le f(x)$ (take $z = x$), it is smooth, and it has the same stationary points as f (Figure 6).

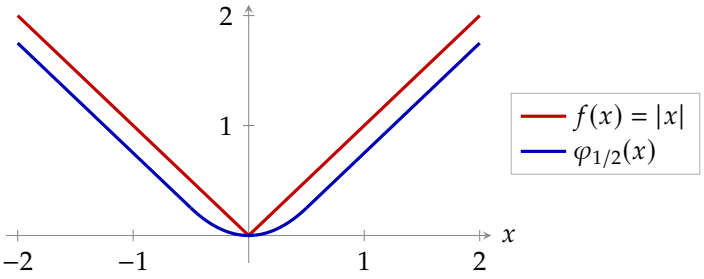


Figure 6: The function $|x|$ and its Moreau envelope with $\lambda = \frac{1}{2}$: $\varphi_{1/2}(x) = x^2$ for $|x| \le \frac{1}{2}$ and $|x| - \frac{1}{4}$ otherwise. It is a smooth lower bound with the same stationary point.

Lemma 20.5 [GRADIENT OF THE ENVELOPE]. *For ℓ-weakly convex f and $0 < \lambda < \frac{1}{\ell}$, φ_λ is smooth (gradient Lipschitz) with*

$$\nabla \varphi_\lambda(x) = \frac{1}{\lambda}(x - \hat{x}).$$

Proof idea. $\varphi_\lambda(x) = f(\hat{x}) + \frac{1}{2\lambda}\|x - \hat{x}\|^2 =: \mathcal{L}(x, \hat{x})$, where $\partial_x \mathcal{L} = \frac{1}{\lambda}(x - \hat{x})$ and the dependence through $\hat{x}$ contributes nothing because $\hat{x}$ is optimal ($0 \in \partial_{\hat{x}} \mathcal{L}$). □

Corollary 20.6.

1. $\|x - \hat{x}\| = \lambda \|\nabla \varphi_\lambda(x)\|$;
2. $f(\hat{x}) \le f(x)$;
3. $\operatorname{dist}\big(0, \partial f(\hat{x})\big) \le \|\nabla \varphi_\lambda(x)\|$.

Proof. (1) Take norms in Lemma 20.5. (2) $f(\hat{x}) \le f(\hat{x}) + \frac{1}{2\lambda}\|x-\hat{x}\|^2 \le f(x)$. (3) Optimality of $\hat{x}$ gives $0 \in \partial\big[f + \frac{1}{2\lambda}\|\cdot - x\|^2\big](\hat{x}) = \partial f(\hat{x}) + \frac{1}{\lambda}(\hat{x}-x)$, so $\nabla\varphi_\lambda(x) = \frac{1}{\lambda}(x-\hat{x}) \in \partial f(\hat{x})$, and $\operatorname{dist}(0,\partial f(\hat{x})) = \min_{v\in\partial f(\hat{x})}\|v\|$. □

20.3 Stationarity and subgradient descent

Definition 20.7 [ϵ-stationary point]. For ℓ-weakly convex f, x is an ϵ-stationary point if

$$\|\nabla\varphi_{1/(2\ell)}(x)\| \le \epsilon.$$

By the corollary, such an x is close to a point that is nearly stationary for f: there is $\hat{x}$ with

$$\|x-\hat{x}\| \le \frac{\epsilon}{2\ell}, \qquad \operatorname{dist}\big(0,\partial f(\hat{x})\big) \le \epsilon.$$

Theorem 20.8 [Subgradient descent]. *Let f be L-Lipschitz and ℓ-weakly convex, let $\Delta = \varphi_{1/(2\ell)}(x_1) - \min_x \varphi_{1/(2\ell)}(x)$, and run subgradient descent with $\eta = \gamma/\sqrt{T}$ for some $\gamma > 0$. Some iterate is an ϵ-stationary point once*

$$T = O\Big(\frac{\Delta^2 + \ell^2 L^4 \gamma^4}{\gamma^2}\cdot\frac{1}{\epsilon^4}\Big).$$

With $\gamma = \frac{1}{L}\sqrt{\Delta/\ell}$ this is $O(\ell L^2\Delta/\epsilon^4)$, a very slow ϵ^{-4} rate.

Proof. Write $\varphi = \varphi_{1/(2\ell)}$, so $\varphi(x) = \min_z f(z) + \ell\|z-x\|^2$, and let $\hat{x}_t$ be the minimizer for x_t. Using $\hat{x}_t$ as a candidate at x_{t+1} and $\|g_t\| \le L$,

$$\begin{aligned}
\varphi(x_{t+1}) &\le f(\hat{x}_t) + \ell\|\hat{x}_t - x_{t+1}\|^2 = f(\hat{x}_t) + \ell\|x_t - \eta g_t - \hat{x}_t\|^2 \\
&\le f(\hat{x}_t) + \ell\|x_t-\hat{x}_t\|^2 - 2\eta\ell\langle g_t, x_t-\hat{x}_t\rangle + \eta^2\ell L^2 \\
&\le \varphi(x_t) + 2\eta\ell\Big(f(\hat{x}_t) - f(x_t) + \frac{\ell}{2}\|x_t-\hat{x}_t\|^2\Big) + \eta^2\ell L^2,
\end{aligned}$$

where the last step uses item 3 of Proposition 20.3, $f(\hat{x}_t) \ge f(x_t) + \langle g_t, \hat{x}_t - x_t\rangle - \frac{\ell}{2}\|\hat{x}_t - x_t\|^2$. The function $\psi_t(x) = f(x) + \ell\|x-x_t\|^2$ is ℓ-strongly convex with minimizer $\hat{x}_t$, so $\psi_t(x_t) - \psi_t(\hat{x}_t) \ge \frac{\ell}{2}\|x_t-\hat{x}_t\|^2$ and

$$f(x_t) - f(\hat{x}_t) - \frac{\ell}{2}\|x_t-\hat{x}_t\|^2 = \psi_t(x_t) - \psi_t(\hat{x}_t) + \frac{\ell}{2}\|x_t-\hat{x}_t\|^2 \ge \ell\|x_t-\hat{x}_t\|^2 = \frac{1}{4\ell}\|\nabla\varphi(x_t)\|^2,$$

using $\nabla\varphi(x_t) = 2\ell(x_t-\hat{x}_t)$. Hence $\varphi(x_{t+1}) \le \varphi(x_t) - \frac{\eta}{2}\|\nabla\varphi(x_t)\|^2 + \eta^2\ell L^2$. Telescoping over

$t = 1, \dots, T$,

$$\min_t \|\nabla \varphi(x_t)\|^2 \le \frac{1}{T}\sum_{t=1}^{T} \|\nabla \varphi(x_t)\|^2 \le \frac{2\Delta}{\eta T} + 2\eta \ell L^2 = \frac{2}{\sqrt{T}}\left(\frac{\Delta}{\gamma} + \gamma \ell L^2\right),$$

which is at most ϵ^2 for the stated T. □

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 20 (Spring 2021), scribed by Mengxi Wang.

Lecture 21: Minimax Optimization, Gradient Descent Ascent for Lipschitz Functions

21.1 Examples

A two-player problem $\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x,y)$ is a minimax problem; GANs and adversarial training (Lecture 1) are examples.

Example: *Rock-paper-scissors.* With payoff matrix (rows and columns ordered rock, paper, scissors)

$$A = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix},$$

the game is the bilinear problem $\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} x^T A y$. With pure strategies, $\mathcal{X} = \mathcal{Y} = \{e_1, e_2, e_3\}$, which is not convex; with mixed strategies, $\mathcal{X} = \mathcal{Y} = \Delta$, the simplex in $\mathbb{R}^3$.

21.2 The minimax theorem

Proposition 21.1 [Weak duality]. $\max_{y\in\mathcal{Y}} \min_{x\in\mathcal{X}} f(x,y) \le \min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x,y)$.

Proof. With $y^\star \in \arg\max_y \min_x f(x,y)$ and $x^\star \in \arg\min_x \max_y f(x,y)$,

$$\max_y \min_x f(x,y) = \min_x f(x,y^\star) \le f(x^\star,y^\star) \le \max_y f(x^\star,y) = \min_x \max_y f(x,y). \qquad \square$$

Equality holds in two special cases.

Definition 21.2 [Convex-concave]. $f : \mathcal{X}\times\mathcal{Y} \to \mathbb{R}$ is convex-concave if $f(\cdot,y)$ is convex for every $y\in\mathcal{Y}$ and $f(x,\cdot)$ is concave for every $x\in\mathcal{X}$. It is μ-strongly-convex-strongly-concave if $f(\cdot,y)$ is μ-strongly convex and $f(x,\cdot)$ is μ-strongly concave.

Theorem 21.3 [Von Neumann's minimax theorem]. *If $\mathcal{X}\subseteq\mathbb{R}^{d_1}$ and $\mathcal{Y}\subseteq\mathbb{R}^{d_2}$ are compact convex sets and f is convex-concave, then* $\max_{y\in\mathcal{Y}} \min_{x\in\mathcal{X}} f(x,y) = \min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x,y)$.

For example, the bilinear game $\min_{x\in\Delta} \max_{y\in\Delta} x^T A y$ is convex-concave over compact convex sets.

Theorem 21.4 [Strongly-convex-strongly-concave]. *If f is strongly-convex-strongly-concave on $\mathbb{R}^{d_1}\times\mathbb{R}^{d_2}$ (the unconstrained case), then* $\max_y \min_x f(x,y) = \min_x \max_y f(x,y)$.

Proof. For fixed y, $f(\cdot, y)$ is strongly convex, so $\Phi(x) = \max_y f(x, y)$ is strongly convex and has a unique minimizer $x^\star$; let $y^\star$ be the unique maximizer of $f(x^\star, \cdot)$. Then $0 \in \partial_y(-f)(x^\star, y^\star)$. Since each maximizer $y^\star(x)$ is unique, $\partial\Phi(x)$ is given by the partial subgradients $\partial_x f(x, y^\star(x))$ (Danskin's theorem), so optimality of $x^\star$ gives $0 \in \partial_x f(x^\star, y^\star)$. Hence $x^\star$ minimizes $f(\cdot, y^\star)$ and $y^\star$ maximizes $f(x^\star, \cdot)$, and

$$\min_x \max_y f(x, y) \le \max_y f(x^\star, y) = f(x^\star, y^\star) = \min_x f(x, y^\star) \le \max_y \min_x f(x, y).$$

Weak duality gives the reverse inequality. □

21.3 Game-theoretic notions

A Stackelberg equilibrium is an optimal strategy when the opponent always plays the best response to the current strategy: $x^\star = \arg\min_x \max_y f(x, y)$ against best responses $\max_y f(x, y)$, and $y^\star = \arg\max_y \min_x f(x, y)$ against best responses $\min_x f(x, y)$. At a Nash equilibrium neither player gains by deviating while the other keeps playing it. For these Stackelberg points, the chain in the proof of Proposition 21.1 holds in general, and when the minimax theorem holds all its terms are equal: $\min_x f(x, y^\star) = f(x^\star, y^\star) = \max_y f(x^\star, y)$, so $(x^\star, y^\star)$ is a Nash equilibrium.

21.4 Convex-concave optimization

Optimality of a candidate $(\hat{x}, \hat{y})$ is measured by the duality gap

$$\operatorname{gap}(\hat{x}, \hat{y}) = \max_{y \in \mathcal{Y}} f(\hat{x}, y) - \min_{x \in \mathcal{X}} f(x, \hat{y}) \ge 0.$$

A point with gap 0 is a Nash equilibrium, and one with gap at most ϵ is an ϵ-Nash equilibrium. Writing $z = (x, y)$, f is L-Lipschitz if $|f(z_1) - f(z_2)| \le L\|z_1 - z_2\|$ and ℓ-smooth if $\|\nabla f(z_1) - \nabla f(z_2)\| \le \ell\|z_1 - z_2\|$.

Algorithm 21.1: Projected gradient descent ascent (GDA)

```
for s = 1, ..., t do
    x̃_{s+1} = x_s − η g_s with g_s ∈ ∂_x f(x_s, y_s);    x_{s+1} = P_X(x̃_{s+1})
    ỹ_{s+1} = y_s + η h_s with −h_s ∈ ∂_y(−f)(x_s, y_s);    y_{s+1} = P_Y(ỹ_{s+1})
```

Theorem 21.5 [GDA, Lipschitz convex-concave]. *Let f be L-Lipschitz and convex-concave, and let $\mathcal{X}$ and $\mathcal{Y}$ have diameter R. With $\eta = \frac{R}{L\sqrt{t}}$,*

$$\max_{y \in \mathcal{Y}} f\Big(\frac{1}{t}\sum_{s=1}^{t} x_s, y\Big) - \min_{x \in \mathcal{X}} f\Big(x, \frac{1}{t}\sum_{s=1}^{t} y_s\Big) \le \frac{2LR}{\sqrt{t}}.$$

Proof. Fix $(x, y) \in \mathcal{X} \times \mathcal{Y}$. By concavity in y and convexity in x,

$$f(x_s, y) - f(x, y_s) = f(x_s, y) - f(x_s, y_s) + f(x_s, y_s) - f(x, y_s) \le h_s^T (y - y_s) + g_s^T (x_s - x).$$

By the update, the identity $2a^T b = \|a\|^2 + \|b\|^2 - \|a-b\|^2$, $\|h_s\| \le L$, and the fact that projections do not increase distances to points of the sets,

$$\begin{aligned} h_s^T (y - y_s) &= \frac{1}{\eta} (\tilde{y}_{s+1} - y_s)^T (y - y_s) = \frac{1}{2\eta} \left[\|\tilde{y}_{s+1} - y_s\|^2 + \|y - y_s\|^2 - \|\tilde{y}_{s+1} - y\|^2 \right] \\ &\le \frac{1}{2\eta} \left[\eta^2 L^2 + \|y_s - y\|^2 - \|y_{s+1} - y\|^2 \right], \end{aligned}$$

and likewise $g_s^T (x_s - x) \le \frac{1}{2\eta} \left[\eta^2 L^2 + \|x_s - x\|^2 - \|x_{s+1} - x\|^2 \right]$. Averaging, telescoping, and convexity-concavity give, for every $(x, y) \in \mathcal{X} \times \mathcal{Y}$,

$$f\left(\frac{1}{t} \sum_s x_s, y \right) - f\left(x, \frac{1}{t} \sum_s y_s \right) \le \frac{1}{t} \sum_{s=1}^{t} \left(f(x_s, y) - f(x, y_s) \right) \le \eta L^2 + \frac{R^2}{\eta t} = \frac{2LR}{\sqrt{t}}. \qquad \square$$

A similar result holds for L-Lipschitz strongly-convex-strongly-concave functions. The smooth setting is very different: in smooth convex minimization the descent lemma gives monotone decrease and last-iterate guarantees, but minimax optimization has no such monotonicity. For $\min_{x \in [-1,1]} \max_{y \in [-1,1]} xy$, with Nash equilibrium $(0, 0)$, the gradient flow circles the equilibrium, and constant-step GDA, $(x_{t+1}, y_{t+1}) = (x_t - \eta y_t,\ y_t + \eta x_t)$, multiplies the distance to it by $\sqrt{1 + \eta^2}$ each step. It diverges without constraints and reaches the boundary with them (Figure 7). Only average-iterate guarantees are possible, at a slower rate.

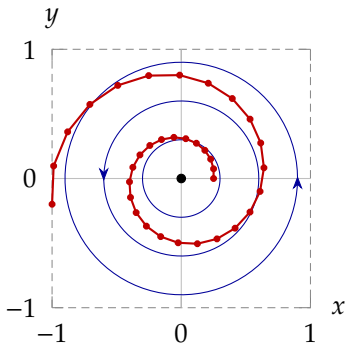


Figure 7: GDA on $f(x, y) = xy$ over $[-1, 1]^2$ (dashed box). Blue: gradient-flow circles around $(0, 0)$. Red: projected GDA with $\eta = 0.3$ from $(0.25, 0)$, spiraling out to the boundary.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 21 (Spring 2021), scribed by Chengzhuo Ni.

Lecture 22: Gradient Descent Ascent for Smooth Functions

22.1 Smooth convex-concave functions

Theorem 22.1 [GDA, smooth convex-concave]. *Let f be ℓ-smooth and convex-concave, let $\mathcal{X}$ and $\mathcal{Y}$ have diameter R, and let $L = 2\ell R + \|\nabla f(x_0, y_0)\|$. Projected GDA with $\eta = \frac{R}{L\sqrt{t}}$ satisfies*

$$\max_y f\left(\frac{1}{t}\sum_{s=1}^{t} x_s, y\right) - \min_x f\left(x, \frac{1}{t}\sum_{s=1}^{t} y_s\right) \le \frac{2RL}{\sqrt{t}}.$$

Proof. For $(x, y) \in \mathcal{X} \times \mathcal{Y}$, $\|\nabla f(x, y) - \nabla f(x_0, y_0)\| \le \ell\|x - x_0\| + \ell\|y - y_0\| \le 2\ell R$, so f is L-Lipschitz on $\mathcal{X} \times \mathcal{Y}$, and Theorem 21.5 applies. □

Three remarks:

- The rate $1/\sqrt{t}$ is slow (and tight for GDA in the worst case), essentially the Lipschitz rate, whereas GD on smooth convex functions achieves $1/t$.
- The step size is no longer $\frac{1}{\ell}$ but much smaller.
- There is no monotonicity, only an average-iterate guarantee. For $\min_{x\in[-1,1]} \max_{y\in[-1,1]} xy$, with equilibrium $(0, 0)$ and $F(z) = (y, -x)$ for $z = (x, y)$, the last iterate diverges without constraints, and even with them it does not converge to $(0, 0)$ (Figure 7).

22.2 Smooth strongly-convex-strongly-concave functions

Write $z = (x, y)$, the operator $F(z) = \begin{pmatrix} \nabla_x f(x, y) \\ -\nabla_y f(x, y) \end{pmatrix}$, so that GDA is $z_{t+1} = z_t - \eta F(z_t)$, and $\kappa = \ell/\mu$. In this section the problem is unconstrained, so the saddle point $z^\star$ has $F(z^\star) = 0$.

Theorem 22.2 [GDA, strongly-convex-strongly-concave]. *If f is ℓ-smooth and μ-strongly-convex-strongly-concave, GDA with $\eta = \mu/\ell^2$ satisfies*

$$\|z_t - z^\star\|^2 \le \left(1 - \frac{1}{\kappa^2}\right)^t \|z_0 - z^\star\|^2, \qquad \max_y f(x_t, y) - \min_x f(x, y_t) \le \kappa\ell\left(1 - \frac{1}{\kappa^2}\right)^t \|z_0 - z^\star\|^2.$$

So GDA converges in the last iterate here, but its step $\eta = \frac{1}{\kappa\ell}$ gives iteration complexity $O(\kappa^2 \log \frac{1}{\epsilon})$, slower than the $O(\kappa \log \frac{1}{\epsilon})$ of GD with $\eta = \frac{1}{\ell}$ on ℓ-smooth, μ-strongly convex functions.

Lemma 22.3 [Strong monotonicity]. *If f is μ-strongly-convex-strongly-concave, then F is* strongly

MONOTONE: $\langle F(z_1) - F(z_2), z_1 - z_2\rangle \geq \mu\|z_1 - z_2\|^2$ *for all* z_1, z_2.

Proof. For a single μ-strongly convex h, adding the strong convexity inequalities at x_1 and x_2 gives $\langle \nabla h(x_1) - \nabla h(x_2), x_1 - x_2\rangle \geq \mu\|x_1 - x_2\|^2$. For f, go from $z_1 = (x_1, y_1)$ to $z_2 = (x_2, y_2)$ along both paths $(x_1, y_1) \to (x_2, y_1) \to (x_2, y_2)$ and $(x_1, y_1) \to (x_1, y_2) \to (x_2, y_2)$:

$$\begin{aligned}
f(x_2, y_1) - f(x_1, y_1) &\geq \langle \nabla_x f(x_1, y_1), x_2 - x_1\rangle + \tfrac{\mu}{2}\|x_2 - x_1\|^2,\\
f(x_1, y_2) - f(x_2, y_2) &\geq \langle \nabla_x f(x_2, y_2), x_1 - x_2\rangle + \tfrac{\mu}{2}\|x_1 - x_2\|^2,\\
f(x_2, y_1) - f(x_2, y_2) &\leq \langle \nabla_y f(x_2, y_2), y_1 - y_2\rangle - \tfrac{\mu}{2}\|y_1 - y_2\|^2,\\
f(x_1, y_2) - f(x_1, y_1) &\leq \langle \nabla_y f(x_1, y_1), y_2 - y_1\rangle - \tfrac{\mu}{2}\|y_1 - y_2\|^2.
\end{aligned}$$

Subtracting the sum of the last two from the sum of the first two, the function values cancel and what remains is the claim. □

Lemma 22.4. *If f is ℓ-smooth, then $\|F(z_1) - F(z_2)\| \leq \ell\|z_1 - z_2\|$.*

Proof. $\|F(z_1)-F(z_2)\|^2 = \|\nabla_x f(z_1)-\nabla_x f(z_2)\|^2+\|\nabla_y f(z_1)-\nabla_y f(z_2)\|^2 = \|\nabla f(z_1)-\nabla f(z_2)\|^2$. □

Lemma 22.5 [SMOOTHNESS OF THE ENVELOPES]. *If f is ℓ-smooth and μ-strongly-convex-strongly-concave, then $\Phi(x) = \max_y f(x, y)$ and $\Psi(y) = \min_x f(x, y)$ are $(\ell + \kappa\ell)$-smooth.*

Proof. Consider Φ. By strong concavity the maximizer $y^\star(x)$ is unique, so by Danskin's theorem $\nabla\Phi(x) = \nabla_x f(x, y^\star(x))$. Write $y_i = y^\star(x_i)$. Strong concavity of $f(x_2, \cdot)$ and the optimality conditions $\nabla_y f(x_i, y_i) = 0$ give

$$\begin{aligned}
\mu\|y_1 - y_2\|^2 &\leq (y_1 - y_2)^T\big(\nabla_y f(x_2, y_2) - \nabla_y f(x_2, y_1)\big)\\
&= (y_1 - y_2)^T\big(\nabla_y f(x_1, y_1) - \nabla_y f(x_2, y_1)\big) \leq \|y_1 - y_2\|\,\ell\|x_1 - x_2\|,
\end{aligned}$$

so $y^\star$ is κ-Lipschitz. Then $\|\nabla\Phi(x_1) - \nabla\Phi(x_2)\| \leq \ell\|x_1 - x_2\| + \ell\|y_1 - y_2\| \leq (\ell + \kappa\ell)\|x_1 - x_2\|$. □

Proof of Theorem 22.2. By $F(z^\star) = 0$ and Lemmas 22.3 and 22.4,

$$\begin{aligned}
\|z_{t+1} - z^\star\|^2 &= \|z_t - z^\star\|^2 - 2\eta\langle F(z_t) - F(z^\star), z_t - z^\star\rangle + \eta^2\|F(z_t) - F(z^\star)\|^2\\
&\leq (1 - 2\eta\mu + \eta^2\ell^2)\|z_t - z^\star\|^2 = \Big(1 - \frac{1}{\kappa^2}\Big)\|z_t - z^\star\|^2
\end{aligned}$$

for $\eta = \mu/\ell^2$. For the gap, $\Phi(x^\star) = f(x^\star, y^\star) = \Psi(y^\star)$ and $\nabla\Phi(x^\star) = \nabla\Psi(y^\star) = 0$, so Lemma 22.5

gives

$$\max_y f(x_t, y) - \min_x f(x, y_t) = \big[\Phi(x_t) - \Phi(x^\star)\big] + \big[\Psi(y^\star) - \Psi(y_t)\big]$$
$$\le \frac{\ell + \kappa\ell}{2}\|z_t - z^\star\|^2 \le \kappa\ell\|z_t - z^\star\|^2. \qquad \square$$

The κ^2 is unavoidable for GDA. Take $f(x, y) = \ell xy + \frac{\mu}{2}x^2 - \frac{\mu}{2}y^2$ with $\mu \le \ell$, which is $(\ell+\mu)$-smooth and μ-strongly-convex-strongly-concave. GDA is linear:

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} - \eta \begin{pmatrix} \mu x_t + \ell y_t \\ -\ell x_t + \mu y_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1-\eta\mu & -\eta\ell \\ \eta\ell & 1-\eta\mu \end{pmatrix}}_{A} \begin{pmatrix} x_t \\ y_t \end{pmatrix},$$

so $z_t = A^t z_0$. The matrix A is not symmetric and has complex eigenvalues $1 - \eta\mu \pm i\eta\ell$, with $|\lambda|^2 = (1-\eta\mu)^2 + \eta^2\ell^2 = 1 - 2\eta\mu + \eta^2(\mu^2+\ell^2)$. Convergence requires $|\lambda| < 1$, that is, $\eta < \frac{2\mu}{\ell^2+\mu^2} = \Theta(\frac{1}{\kappa\ell})$, and then the spectral radius is

$$\rho(A) \ge 1 - \eta\mu \ge 1 - \frac{2\mu^2}{\ell^2 + \mu^2} = 1 - \Theta\Big(\frac{1}{\kappa^2}\Big).$$

The cross term ℓxy hurts the convergence rate.

22.3 The extragradient method

The EXTRAGRADIENT (EG) method does better than GDA. It takes a trial gradient step, evaluates F there, and uses that value to step from the original point (Figure 8).

Algorithm 22.1: Extragradient method

```
for t = 1, ..., T do
    z_{t+1/2} = z_t − ηF(z_t)
    z_{t+1} = z_t − ηF(z_{t+1/2})
```

Theorem 22.6 [EG, STRONGLY-CONVEX-STRONGLY-CONCAVE]. *If f is ℓ-smooth and μ-strongly-convex-strongly-concave, EG with $\eta = \frac{1}{4\ell}$ satisfies*

$$\|z_t - z^\star\|^2 \le \Big(1 - \frac{1}{4\kappa}\Big)^t \|z_0 - z^\star\|^2, \qquad \max_y f(x_t, y) - \min_x f(x, y_t) \le \kappa\ell\Big(1 - \frac{1}{4\kappa}\Big)^t \|z_0 - z^\star\|^2.$$

The proof is given in Lecture 23.

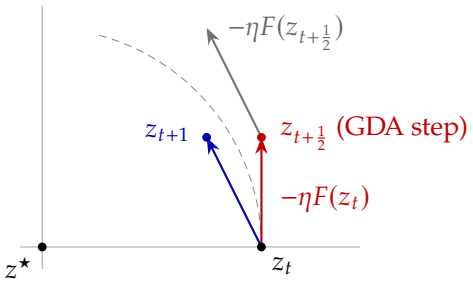


Figure 8: One step on $f(x, y) = xy$ with $\eta = \frac{1}{2}$ from $z_t = (1, 0)$. The GDA step (red) lands at $z_{t+\frac{1}{2}} = (1, \frac{1}{2})$, farther from the equilibrium $z^\star = 0$; EG instead steps from z_t along $-\eta F(z_{t+\frac{1}{2}})$ (gray) to $z_{t+1} = (\frac{3}{4}, \frac{1}{2})$ (blue), closer to $z^\star$. The dashed arc has radius $\|z_t - z^\star\| = 1$.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 22 (Spring 2021), scribed by Chang Yue.

Lecture 23: Extragradient Method, Last-Iterate Convergence, and Optimal Rates

23.1 Convergence of extragradient

Write $z = (x, y)$, $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, and $F(z) = \begin{pmatrix} \nabla_x f(x,y) \\ -\nabla_y f(x,y) \end{pmatrix}$. The projected extragradient method is

$$z_{t+\frac{1}{2}} = P_{\mathcal{Z}}\big(z_t - \eta F(z_t)\big), \qquad z_{t+1} = P_{\mathcal{Z}}\big(z_t - \eta F(z_{t+\frac{1}{2}})\big).$$

Theorem 23.1 [EG, smooth convex-concave**].** *If f is convex-concave and ℓ-smooth, $\mathcal{Z}$ has diameter R, and $\eta = 1/\ell$, then the averages $\bar{x}_T = \frac{1}{T}\sum_{t=0}^{T-1} x_{t+\frac{1}{2}}$ and $\bar{y}_T = \frac{1}{T}\sum_{t=0}^{T-1} y_{t+\frac{1}{2}}$ satisfy*

$$\max_{y \in \mathcal{Y}} f(\bar{x}_T, y) - \min_{x \in \mathcal{X}} f(x, \bar{y}_T) \le \frac{\ell R^2}{2T}.$$

The key is a one-step inequality. Let $\tilde{z}_{t+\frac{1}{2}} = z_t - \eta F(z_t)$ and $\tilde{z}_{t+1} = z_t - \eta F(z_{t+\frac{1}{2}})$ be the points before projection.

Lemma 23.2 [One step of EG**].** *If F is ℓ-Lipschitz, then for every $z \in \mathcal{Z}$,*

$$F(z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z) \le \frac{1}{2\eta}\Big[\|z_t - z\|^2 - \|z_{t+1} - z\|^2 + (\eta^2\ell^2 - 1)\|z_t - z_{t+\frac{1}{2}}\|^2\Big]. \tag{23.1}$$

Proof. Projections onto the convex set satisfy $(\tilde{z}_{t+\frac{1}{2}} - z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z) \ge 0$ and $(\tilde{z}_{t+1} - z_{t+1})^T (z_{t+1} - z) \ge 0$ for all $z \in \mathcal{Z}$ (Lecture 4). Since $\eta F(z_{t+\frac{1}{2}}) = z_t - \tilde{z}_{t+1}$, splitting $z_t - \tilde{z}_{t+1} = (z_t - \tilde{z}_{t+\frac{1}{2}}) + (\tilde{z}_{t+\frac{1}{2}} - \tilde{z}_{t+1})$ in the second term and applying the projection inequalities (the first with $z = z_{t+1}$),

$$\begin{aligned}
\eta F(z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z) &= (z_t - \tilde{z}_{t+1})^T (z_{t+1} - z) + (z_t - \tilde{z}_{t+1})^T (z_{t+\frac{1}{2}} - z_{t+1}) \\
&\le \underbrace{(z_t - z_{t+1})^T (z_{t+1} - z)}_{(a)} + \underbrace{(z_t - z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z_{t+1})}_{(b)} \\
&\quad + \underbrace{(\tilde{z}_{t+\frac{1}{2}} - \tilde{z}_{t+1})^T (z_{t+\frac{1}{2}} - z_{t+1})}_{(c)}.
\end{aligned}$$

By $a^T b = \frac{1}{2}(\|a+b\|^2 - \|a\|^2 - \|b\|^2)$,

$$(a) = \tfrac{1}{2}\left[\|z_t - z\|^2 - \|z_t - z_{t+1}\|^2 - \|z_{t+1} - z\|^2\right], \qquad (b) = \tfrac{1}{2}\left[\|z_t - z_{t+1}\|^2 - \|z_t - z_{t+\frac{1}{2}}\|^2 - \|z_{t+\frac{1}{2}} - z_{t+1}\|^2\right].$$

Since $\tilde{z}_{t+\frac{1}{2}} - \tilde{z}_{t+1} = \eta\big(F(z_{t+\frac{1}{2}}) - F(z_t)\big)$ and F is ℓ-Lipschitz,

$$(c) \le \eta\ell\|z_t - z_{t+\frac{1}{2}}\|\|z_{t+\frac{1}{2}} - z_{t+1}\| \le \tfrac{1}{2}\left[\eta^2\ell^2\|z_t - z_{t+\frac{1}{2}}\|^2 + \|z_{t+\frac{1}{2}} - z_{t+1}\|^2\right].$$

Adding (a), (b), and (c), the terms $\|z_t - z_{t+1}\|^2$ and $\|z_{t+\frac{1}{2}} - z_{t+1}\|^2$ cancel. □

Proof of Theorem 23.1. F is ℓ-Lipschitz (Lemma 22.4). For $(x, y) \in \mathcal{X} \times \mathcal{Y}$, concavity in y and convexity in x give

$$\begin{aligned} f(x_{t+\frac{1}{2}}, y) - f(x, y_{t+\frac{1}{2}}) &= \underbrace{f(x_{t+\frac{1}{2}}, y) - f(x_{t+\frac{1}{2}}, y_{t+\frac{1}{2}})}_{\text{concave in } y} + \underbrace{f(x_{t+\frac{1}{2}}, y_{t+\frac{1}{2}}) - f(x, y_{t+\frac{1}{2}})}_{\text{convex in } x} \\ &\le \nabla_y f(z_{t+\frac{1}{2}})^T (y - y_{t+\frac{1}{2}}) + \nabla_x f(z_{t+\frac{1}{2}})^T (x_{t+\frac{1}{2}} - x) = F(z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z). \end{aligned}$$

With $\eta = 1/\ell$ the last term of (23.1) vanishes, so the sum over $t = 0, \dots, T-1$ telescopes. By convexity-concavity,

$$\max_y f(\bar{x}_T, y) - \min_x f(x, \bar{y}_T) \le \max_{z \in \mathcal{Z}} \frac{1}{T} \sum_{t=0}^{T-1} F(z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z) \le \max_{z \in \mathcal{Z}} \frac{\|z_0 - z\|^2}{2\eta T} \le \frac{\ell R^2}{2T}. \qquad \square$$

For strongly-convex-strongly-concave f, EG converges linearly (Theorem 22.6).

Proof of Theorem 22.6. Apply (23.1) at the saddle point $z = z^\star$, where $F(z^\star)^T (z - z^\star) \ge 0$ for all $z \in \mathcal{Z}$ (equality to zero without constraints). By strong monotonicity (Lemma 22.3) and $\|a\|^2 \ge \frac{1}{2}\|b\|^2 - \|a - b\|^2$,

$$\begin{aligned} \|z_t - z^\star\|^2 - \|z_{t+1} - z^\star\|^2 + (\eta^2\ell^2 - 1)\|z_t - z_{t+\frac{1}{2}}\|^2 &\ge 2\eta F(z_{t+\frac{1}{2}})^T (z_{t+\frac{1}{2}} - z^\star) \ge 2\eta\mu\|z_{t+\frac{1}{2}} - z^\star\|^2 \\ &\ge \eta\mu\left[\|z_t - z^\star\|^2 - 2\|z_{t+\frac{1}{2}} - z_t\|^2\right]. \end{aligned}$$

Rearranging,

$$\|z_{t+1} - z^\star\|^2 \le (1 - \eta\mu)\|z_t - z^\star\|^2 + (\eta^2\ell^2 + 2\eta\mu - 1)\|z_t - z_{t+\frac{1}{2}}\|^2 \le \left(1 - \frac{1}{4\kappa}\right)\|z_t - z^\star\|^2,$$

since $\eta = \frac{1}{4\ell}$ makes $\eta\mu = \frac{1}{4\kappa}$ and $\eta^2\ell^2 + 2\eta\mu - 1 \le \frac{1}{16} + \frac{1}{2} - 1 < 0$. The gap bound follows from Lemma 22.5 as for GDA. □

23.2 Comparison and last iterates

Gradient complexities for smooth minimax problems (logarithmic factors omitted in the strongly-convex-strongly-concave row):

	GDA	EG	OGDA
Convex-concave	$1/\epsilon^2$	$1/\epsilon$	$1/\epsilon$
Strongly-convex-strongly-concave	κ^2	κ	κ

Optimistic GDA (OGDA) performs much like EG; without constraints its update is

$$z_{t+1} = z_t - 2\eta F(z_t) + \eta F(z_{t-1}).$$

In the convex-concave case these guarantees are for averaged iterates; last-iterate guarantees, easy in smooth convex minimization thanks to the descent lemma, are much harder. Known results:

Smooth convex-concave	GDA	EG	OGDA
Average iterate	$1/\epsilon^2$	$1/\epsilon$	$1/\epsilon$
Last iterate	none	$1/\epsilon^2$	$1/\epsilon^9$

Last-iterate guarantees come more easily with a small modification: the reduction from convex to strongly convex optimization (Lecture 6). For convex-concave f, consider the proxy

$$\tilde{f}(x,y) = f(x,y) + \frac{\mu}{2}\|x - x_0\|^2 - \frac{\mu}{2}\|y - y_0\|^2,$$

which is $(\ell+\mu)$-smooth and μ-strongly-convex-strongly-concave if f is ℓ-smooth and convex-concave. By Theorem 22.6, EG (or OGDA) on $\tilde{f}$ reaches $\max_y \tilde{f}(x_t, y) - \min_x \tilde{f}(x, y_t) \le \epsilon/2$ within $\tilde{\Theta}(\ell/\mu)$ iterations. With $\mu = \epsilon/(2R^2)$, $|f(x,y) - \tilde{f}(x,y)| \le \frac{\mu}{2}R^2 = \frac{\epsilon}{4}$ on $\mathcal{X} \times \mathcal{Y}$, so

$$\begin{aligned}\max_y f(x_t, y) - \min_x f(x, y_t) \le &\left[\max_y f(x_t, y) - \max_y \tilde{f}(x_t, y)\right] + \left[\max_y \tilde{f}(x_t, y) - \min_x \tilde{f}(x, y_t)\right] \\ &+ \left[\min_x \tilde{f}(x, y_t) - \min_x f(x, y_t)\right] \le \frac{\epsilon}{4} + \frac{\epsilon}{2} + \frac{\epsilon}{4} = \epsilon.\end{aligned}$$

So EG or OGDA on $\tilde{f}$ achieves duality gap ϵ on f in the last iterate with $\tilde{O}(\ell R^2/\epsilon)$ iterations.

23.3 The role of momentum

Minimization	GD	AGD	Minimax	GDA	EG/OGDA	Lower bound
Convex	$\frac{1}{\epsilon}$	$\frac{1}{\sqrt{\epsilon}}$	Convex-concave	$\frac{1}{\epsilon^2}$	$\frac{1}{\epsilon}$	$\Omega(\frac{1}{\epsilon})$
Strongly convex	$\kappa \ln \frac{1}{\epsilon}$	$\sqrt{\kappa} \ln \frac{1}{\epsilon}$	SC-SC	$\kappa^2 \ln \frac{1}{\epsilon}$	$\kappa \ln \frac{1}{\epsilon}$	$\Omega(\kappa \ln \frac{1}{\epsilon})$

Here SC-SC abbreviates strongly-convex-strongly-concave. EG and OGDA match the lower bounds, so in this sense momentum is not needed for minimax optimization, in contrast with convex minimization. But if $\mathcal{Y} = \{y_0\}$ is trivial, the problem is minimization in x, where momentum accelerates. A more refined description resolves this:

- if f is ℓ-smooth, μ_x-strongly convex in x, and μ_y-strongly concave in y, let $\kappa_x = \ell/\mu_x$ and $\kappa_y = \ell/\mu_y$;
- EG and OGDA need $O((\kappa_x + \kappa_y) \ln \frac{1}{\epsilon})$ iterations, which cannot be improved for them, while the lower bound is $\Omega(\sqrt{\kappa_x \kappa_y} \ln \frac{1}{\epsilon})$;
- a near-optimal algorithm achieves $O(\sqrt{\kappa_x \kappa_y} \ln^3 \frac{1}{\epsilon})$.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 23 (Spring 2021), scribed by Yuanhao Wang.

Lecture 24: Nonconvex Minimax Optimization

When $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is not convex-concave, the minimax theorem fails: in general

$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x,y) \neq \max_{y\in\mathcal{Y}} \min_{x\in\mathcal{X}} f(x,y),$$

so the order matters. Without loss of generality we solve

$$\min_{x\in\mathcal{X}} \max_{y\in\mathcal{Y}} f(x,y) = \min_{x\in\mathcal{X}} \Phi(x), \qquad \Phi(x) = \max_{y\in\mathcal{Y}} f(x,y).$$

Two scenarios differ: either the inner maximization can be solved for fixed x (for example, when f is nonconvex-concave), or it is intractable (for example, nonconvex-nonconcave).

24.1 Solvable inner maximization

Assume $\Phi(x)$ can be computed pointwise, that is, for fixed x an (approximate) maximizer of $f(x,\cdot)$ can be found. Minimizing over x is still hard because of nonconvexity, so we need Φ to be smooth, or nonsmooth and weakly convex (Lecture 20). The latter holds.

Proposition 24.1. *Let f be ℓ-smooth and $\Phi(x) = \max_{y\in\mathcal{Y}} f(x,y)$. Then*

1. *Φ can be nonsmooth;*
2. *Φ is ℓ-weakly convex, and $\nabla_x f(x, y^\star(x)) \in \partial\Phi(x)$ for any $y^\star(x) \in \arg\max_{y\in\mathcal{Y}} f(x,y)$.*

Proof. (1) $f(x,y) = xy$ on $\mathcal{X} = \mathcal{Y} = [-1,1]$ is 1-smooth, but $\Phi(x) = |x|$.
(2) For any x_1, x_2, let $y_2 = y^\star(x_2)$. By ℓ-smoothness of f,

$$\begin{aligned}\Phi(x_1) \geq f(x_1,y_2) &\geq f(x_2,y_2) + \langle \nabla_x f(x_2,y_2), x_1 - x_2\rangle - \frac{\ell}{2}\|x_1-x_2\|^2 \\ &= \Phi(x_2) + \langle \nabla_x f(x_2,y_2), x_1 - x_2\rangle - \frac{\ell}{2}\|x_1-x_2\|^2,\end{aligned}$$

which is item 3 of Proposition 20.3 with $g_{x_2} = \nabla_x f(x_2,y_2)$. □

GD-max. Solve the inner maximization up to error δ, then take a gradient step.

Algorithm 24.1: GD-max

for $t = 0, 1, \dots, T-1$ **do**
  find y_t with $f(x_t, y_t) \geq \max_{y \in \mathcal{Y}} f(x_t, y) - \delta$
  $x_{t+1} = x_t - \eta \nabla_x f(x_t, y_t)$

With $\delta = 0$, $y_t \in \arg\max_y f(x_t, y)$, and by Proposition 24.1 GD-max is subgradient descent on the weakly convex Φ. It therefore finds stationary points in the sense of the Moreau envelope $\varphi_\lambda(x) = \min_z \left\{\Phi(z) + \frac{1}{2\lambda}\|z - x\|^2\right\}$.

Theorem 24.2 [GD-max]. *If f is ℓ-smooth and L-Lipschitz, then for any $T, \gamma, \delta > 0$, GD-max with $\eta = \gamma/\sqrt{T}$ satisfies*

$$\min_{0 \leq t < T} \|\nabla \varphi_{1/(2\ell)}(x_t)\|^2 \leq 2 \cdot \frac{\left[\varphi_{1/(2\ell)}(x_0) - \min_x \Phi(x)\right] + \ell L^2 \gamma^2}{\gamma \sqrt{T}} + 4\ell\delta.$$

The first term is the standard rate for stationary points of weakly convex functions (Theorem 20.8); the second is the error from solving the inner maximization approximately. The proof combines that analysis with an error analysis for δ and is omitted.

For nonconvex-concave f and an ϵ-stationary point of Φ, each term must be $O(\epsilon^2)$: the outer loop needs $O(\epsilon^{-4})$ iterations, and with $\delta = \Theta(\epsilon^2)$ the concave inner problem takes $O(\epsilon^{-1})$ iterations of AGD, for a total of $O(\epsilon^{-5})$.

PPA-minimax. GD-based methods like GD-max cannot do much better, but a proximal point algorithm (PPA) in the outer loop can.

Algorithm 24.2: PPA-minimax

for $t = 0, 1, \dots, T-1$ **do**
  find x_{t+1} with $\Phi(x_{t+1}) + \ell\|x_{t+1} - x_t\|^2 \leq \min_x \left\{\Phi(x) + \ell\|x - x_t\|^2\right\} + \delta$

With $\delta = 0$, the outer loop is the PPA update $x_{t+1} = \arg\min_x \{\Phi(x) + \ell\|x - x_t\|^2\}$, which converges like GD in the smooth setting even though Φ is not smooth. Each step is the minimax problem $\min_x \max_y \; f(x, y) + \ell\|x - x_t\|^2$, which is ($\ell$-strongly convex)-concave, so the convex-concave results apply.

Theorem 24.3 [PPA-minimax]. *If f is ℓ-smooth, then for any $T, \delta > 0$, PPA-minimax satisfies*

$$\min_{0 \leq t < T} \|\nabla \varphi_{1/(2\ell)}(x_t)\|^2 \leq \frac{8\ell\left[\Phi(x_0) - \min_x \Phi(x)\right]}{T} + 24\ell\delta.$$

Proof. Let $\psi_t(x) = \Phi(x) + \ell\|x - x_t\|^2$ and $\hat{x}_t = \arg\min_x \psi_t(x)$, so that $\nabla\varphi_{1/(2\ell)}(x_t) = 2\ell(x_t - \hat{x}_t)$ (Lemma 20.5). Since Φ is ℓ-weakly convex, ψ_t is ℓ-strongly convex, so $\psi_t(x_{t+1}) \le \psi_t(\hat{x}_t) + \delta$ gives

$$\frac{\ell}{2}\|x_{t+1} - \hat{x}_t\|^2 \le \psi_t(x_{t+1}) - \psi_t(\hat{x}_t) \le \delta. \tag{24.1}$$

Comparing with the candidate $x = x_t$,

$$\Phi(x_{t+1}) + \ell\|x_{t+1} - x_t\|^2 \le \Phi(x_t) + \delta. \tag{24.2}$$

By $2a^2 \ge (a+b)^2 - 2b^2$ with $a = \|x_{t+1} - x_t\|$, $b = \|x_{t+1} - \hat{x}_t\|$, and $a + b \ge \|\hat{x}_t - x_t\|$, then (24.1),

$$\ell\|x_{t+1} - x_t\|^2 \ge \frac{\ell}{2}\|\hat{x}_t - x_t\|^2 - \ell\|x_{t+1} - \hat{x}_t\|^2 \ge \frac{1}{8\ell}\|\nabla\varphi_{1/(2\ell)}(x_t)\|^2 - 2\delta.$$

Combined with (24.2), $\|\nabla\varphi_{1/(2\ell)}(x_t)\|^2 \le 8\ell\big[\Phi(x_t) - \Phi(x_{t+1})\big] + 24\ell\delta$. Summing over $t = 0, \dots, T-1$, dividing by T, and bounding the minimum by the average completes the proof. □

For nonconvex-concave f, the outer loop needs $O(\epsilon^{-2})$ iterations and $\delta = \Theta(\epsilon^2)$. The inner problem has $\kappa_x = O(1)$, and adding a strongly concave term in y gives $\kappa_y = O(1/\delta)$, so the near-optimal $\tilde{O}(\sqrt{\kappa_x\kappa_y})$ minimax method (Lecture 23) solves it in $\tilde{O}(\epsilon^{-1})$ iterations. The total is $\tilde{O}(\epsilon^{-3})$, much faster than GD-max.

24.2 Intractable inner maximization

In the nonconvex-nonconcave setting most problems are open and few rigorous results exist.

- The simplest goal is a stationary point of f over $\mathcal{Z}$. Constraints require a KKT-type definition; such points are the fixed points of projected GDA. For general $\mathcal{Z}$, finding them is PPAD-complete; for decoupled constraints $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ the question is open.
- The main open question is whether structure in specific problems, such as adversarial training or GANs, can be exploited.

Source. Chi Jin, *ELE539/COS512: Optimization for Machine Learning*, Lecture 24 (Spring 2021), scribed by Seyed Sobhan Mir Yoosefi.